

Abstract

We study the topological properties of phonons in trapped Rydberg atom arrays, which arise from dipole-dipole interactions between the atoms. For various one-dimensional geometries, from zigzag to armchair configurations, we analyze the symmetries of the phononic Hamiltonian which give rise to topologically localized phonon excitations on the system boundaries. Because the phonon-phonon interactions here do not naturally respect chiral symmetry, which is crucial for realizing 1D topological phases, we show how an appropriate, geometry-dependent choice of the dipole moment orientation can restore this symmetry and enable topological characterization. The interplay between two phononic degrees of freedom in the harmonic trap potential, mediated by the dipole-dipole interactions, leads to interesting topological phases with winding numbers between zero and two. The corresponding edge states exhibit strong localization and robustness, making them potentially useful for applications in Rydberg-array-based quantum transport.

As an outlook we aim to explore how the phonon topology influences transport phenomena. Topological states in the electronic degrees of freedom in Rydberg arrays have already been shown to support topologically protected photon pumping [**meidanTopologicalClassificationInteractiondriven201**, 1–4]. Studying the effect of spin-phonon coupling can provide new control mechanisms in the realm of topological transport.

Hamiltonian

I am considering the interaction term H_I of a Hamiltonian of a system of N particles with positions $\{\bar{\mathbf{R}}_l\}_{l \in \{1, \dots, N\}}$. It shall consist of two-body interactions, which depend on the position of the involved particles. Thus H_I can be written in the form

$$H_I(\{\bar{\mathbf{R}}_l\}) = \sum_{i,j; i>j} V_{i,j}(\bar{\mathbf{R}}_i, \bar{\mathbf{R}}_j)$$

In order to being able to derive the second order expansion of H_I , that will be needed later, in a more formal way, I write the two-body interactions as dependent on all positions.

$$H_I(\{\bar{\mathbf{R}}_l\}) = \sum_{i,j; i>j} V_{i,j}(\{\bar{\mathbf{R}}_l\})$$

The $V_{i,j}$ are formally functions of all positions $\{\bar{\mathbf{R}}_l\}$, i.e. $V_{i,j}(\{\bar{\mathbf{R}}_l\})$. But of course they only depend on $\bar{\mathbf{R}}_i$ and $\bar{\mathbf{R}}_j$, i.e.

$$\frac{\partial V_{i,j}}{\partial \bar{\mathbf{R}}_l} = 0 \quad \text{if } i \neq l \neq j$$

Now I expand the interaction Hamiltonian in small displacements $\{\delta \mathbf{R}_l\}$ from the tweezer centers $\{\mathbf{R}_l\}$.

$$\begin{aligned} H_I(\{\mathbf{R}_l + \delta \mathbf{R}_l\}) &= H_I(\{\mathbf{R}_l\}) + \sum_n \frac{\partial H_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n} \delta \mathbf{R}_n + \frac{1}{2} \sum_{n,m} \delta \mathbf{R}_n \frac{\partial^2 H_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n \partial \mathbf{R}_m} \delta \mathbf{R}_m \\ &= H_I(\{\mathbf{R}_l\}) + \sum_n \frac{\partial H_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n} \delta \mathbf{R}_n + \frac{1}{2} \sum_{n \neq m} \delta \mathbf{R}_n \frac{\partial^2 H_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n \partial \mathbf{R}_m} \delta \mathbf{R}_m + \frac{1}{2} \sum_{n=m} \delta \mathbf{R}_n \frac{\partial^2 H_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n \partial \mathbf{R}_m} \delta \mathbf{R}_m \\ &= H_I(\{\mathbf{R}_l\}) + \sum_n \frac{\partial H_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n} \delta \mathbf{R}_n + \frac{1}{2} \sum_{n \neq m} \sum_{i,j; i>j} \delta \mathbf{R}_n \frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n \partial \mathbf{R}_m} \delta \mathbf{R}_m \\ &\quad + \frac{1}{2} \sum_n \sum_{i,j; i>j} \delta \mathbf{R}_n \frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n^2} \delta \mathbf{R}_n \\ &= H_I(\{\mathbf{R}_l\}) + \sum_n \frac{\partial H_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n} \delta \mathbf{R}_n + \frac{1}{2} \sum_{n \neq m} \sum_{i,j; i>j} \delta \mathbf{R}_n \frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n \partial \mathbf{R}_m} \delta \mathbf{R}_m (\delta_{n=i, m=j} + \delta_{n=j, m=i}) \\ &\quad + \frac{1}{2} \sum_n \sum_{i,j; i>j} \delta \mathbf{R}_n \frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n^2} \delta \mathbf{R}_n (\delta_{n=i} + \delta_{n=j}) \end{aligned}$$

Using the fact, that

$$\sum_{i,j; i>j} \left(\frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_i^2} \delta \mathbf{R}_i^2 + \frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_j^2} \delta \mathbf{R}_j^2 \right) = \sum_{i,j; i \neq j} \frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_i^2} \delta \mathbf{R}_i^2$$

and

$$\sum_{i,j; i>j} \left(\delta \mathbf{R}_i \frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_i \partial \mathbf{R}_j} \delta \mathbf{R}_j + \delta \mathbf{R}_j \frac{\partial^2 V_{j,i}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_j \partial \mathbf{R}_i} \delta \mathbf{R}_i \right) = \sum_{i,j; i \neq j} \delta \mathbf{R}_i \frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_i \partial \mathbf{R}_j} \delta \mathbf{R}_j$$

and similar for the mixed terms, the expansion can be written as

$$\begin{aligned} H_I(\{\mathbf{R}_l + \delta \mathbf{R}_l\}) &= H_I(\{\mathbf{R}_l\}) + \sum_i \frac{\partial H_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_i} \delta \mathbf{R}_i + \frac{1}{2} \sum_{i,j; i>j} \frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_i \partial \mathbf{R}_j} (\delta \mathbf{R}_i \delta \mathbf{R}_j + \delta \mathbf{R}_j \delta \mathbf{R}_i) \\ &\quad + \frac{1}{2} \sum_{i,j; i \neq j} \frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_i^2} \delta \mathbf{R}_i^2 \\ &= H_I(\{\mathbf{R}_l\}) + \sum_i \frac{\partial H_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_i} \delta \mathbf{R}_i + \frac{1}{2} \sum_{i,j; i \neq j} \frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_i \partial \mathbf{R}_j} \delta \mathbf{R}_i \delta \mathbf{R}_j \\ &\quad + \frac{1}{2} \sum_{i,j; i \neq j} \frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_i^2} \delta \mathbf{R}_i^2 \end{aligned}$$

If I introduce a matrix U the expanded Hamiltonian can be brought into the same form as in the paper.

$$H_I(\{\mathbf{R}_l + \delta \mathbf{R}_l\}) = H_I(\{\mathbf{R}_l\}) + \sum_n \frac{\partial H_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n} \delta \mathbf{R}_n + \sum_{n,m} U_{n,m} \delta \mathbf{R}_n \delta \mathbf{R}_m$$

with

$$\begin{aligned} U_{n,m} &= \frac{1}{2} \frac{\partial^2 V_{n,m}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n \partial \mathbf{R}_m} \quad \text{if } n \neq m \\ U_{n,n} &= \frac{1}{2} \sum_{j; j \neq n} \frac{\partial^2 V_{n,j}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n^2} \end{aligned}$$

Swithing again to the reduced writting, where V_{ij} is only written as a function of the variables of which it actually depends, reads

$$\begin{aligned} U_{n,m} &= \frac{1}{2} \frac{\partial^2 V_{n,m}(\mathbf{R}_n, \mathbf{R}_m)}{\partial \mathbf{R}_n \partial \mathbf{R}_m} \quad \text{if } n \neq m \\ U_{n,n} &= \frac{1}{2} \sum_{j; j \neq n} \frac{\partial^2 V_{n,j}(\mathbf{R}_n, \mathbf{R}_j)}{\partial \mathbf{R}_n^2} \end{aligned}$$

This result is a good point to further specify the potential that is used in the next sections. The potential will only depend on the relative vector between the two positions, i.e. $\mathbf{r}_{ij} = \mathbf{R}_i - \mathbf{R}_j$. Using the derivative

$$\frac{\partial \mathbf{r}_{ij}}{\partial \mathbf{R}_i} = \mathbb{1} = - \frac{\partial \mathbf{r}_{ij}}{\partial \mathbf{R}_j}$$

the derivatives of the U matrix can be written as

$$\begin{aligned} \frac{\partial^2 V_{n,m}(\mathbf{r}_{nm})}{\partial \mathbf{R}_n \partial \mathbf{R}_m} &= \frac{\partial}{\partial \mathbf{R}_n} \left(\frac{\partial V_{n,m}(\mathbf{r}_{nm})}{\partial \mathbf{r}_{nm}} \frac{\partial \mathbf{r}_{nm}}{\partial \mathbf{R}_m} \right) \\ &= \frac{\partial^2 V_{n,m}(\mathbf{r}_{nm})}{\partial \mathbf{r}_{nm}^2} \frac{\partial \mathbf{r}_{nm}}{\partial \mathbf{R}_m} \frac{\partial \mathbf{r}_{nm}}{\partial \mathbf{R}_n} + \frac{\partial V_{n,m}(\mathbf{r}_{nm})}{\partial \mathbf{r}_{nm}} \frac{\partial^2 \mathbf{r}_{nm}}{\partial \mathbf{R}_n \partial \mathbf{R}_m} \\ &= - \frac{\partial^2 V_{n,m}(\mathbf{r}_{nm})}{\partial \mathbf{r}_{nm}^2} \end{aligned}$$

and

$$\begin{aligned}\frac{\partial^2 V_{n,m}(\mathbf{r}_{nm})}{\partial \mathbf{R}_m^2} &= \frac{\partial^2 V_{n,m}(\mathbf{r}_{nm})}{\partial \mathbf{r}_{nm}^2} \frac{\partial \mathbf{r}_{nm}}{\partial \mathbf{R}_m} \frac{\partial \mathbf{r}_{nm}}{\partial \mathbf{R}_m} \\ &= \frac{\partial^2 V_{n,m}(\mathbf{r}_{nm})}{\partial \mathbf{r}_{nm}^2}\end{aligned}$$

Thus

$$\begin{aligned}U_{n,m} &= -\frac{1}{2} \frac{\partial^2 V_{n,m}(\mathbf{r}_{nm})}{\partial \mathbf{r}_{nm}^2} \quad \text{if } n \neq m \\ U_{n,n} &= \frac{1}{2} \sum_{j; j \neq n} \frac{\partial^2 V_{n,j}(\mathbf{r}_{nj})}{\partial \mathbf{r}_{nj}^2}\end{aligned}$$

At this point I can eliminate the linear term of the Hamiltonian by introducing the trap potential

$$H_0(\{\delta \mathbf{R}_l\}) = \sum_i \frac{\mathbf{P}_i^2}{2m} + \frac{1}{2} m \omega^2 \delta \mathbf{R}_i^2,$$

where $\mathbf{P}_i$ is the momentum of the i th atom. I can now shift the considered displacements from the trap centers by a constant $\{\delta \mathbf{R}_l\} = \{\delta \tilde{\mathbf{R}}_l + \mathbf{\Delta}_l\}$, where $\delta \tilde{\mathbf{R}}_l$ and $\delta \mathbf{R}_l$ are operators and $\mathbf{\Delta}_l$ are just constant vectors. Neglecting all constant terms the Hamiltonian minus the momentum term $\tilde{H} = H - \sum_i \mathbf{P}_i^2/2m$ up to second order reads

$$\begin{aligned}\tilde{H}(\{\delta \mathbf{R}_l\}) &= \sum_i \frac{1}{2} m \omega^2 (\delta \tilde{\mathbf{R}}_i + \mathbf{\Delta}_i)^2 + \sum_i \frac{\partial H_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_i} (\delta \tilde{\mathbf{R}}_i + \mathbf{\Delta}_i) + \sum_{i,j} (\delta \tilde{\mathbf{R}}_i + \mathbf{\Delta}_i) U_{i,j} (\delta \tilde{\mathbf{R}}_j + \mathbf{\Delta}_j) \\ &= \sum_i \frac{1}{2} m \omega^2 \delta \tilde{\mathbf{R}}_i^2 + \sum_{i,j} \delta \tilde{\mathbf{R}}_i U_{i,j} \delta \tilde{\mathbf{R}}_j + \sum_i \left(\frac{\partial H_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_i} + \sum_j (2 U_{j,i} + m \omega^2 \delta_{ij}) \mathbf{\Delta}_j \right) \delta \tilde{\mathbf{R}}_i,\end{aligned}$$

where it was used that U is symmetric. As the matrix $(2 U_{j,i} + m \omega^2 \delta_{ij})$ is invertible, a set of vectors $\{\mathbf{\Delta}_n\}$ can be found such that the linear term vanishes. Thus

$$H(\{\delta \tilde{\mathbf{R}}_l\}) = \sum_n \frac{\mathbf{P}_n^2}{2m} + \sum_n \frac{1}{2} m \omega^2 \delta \tilde{\mathbf{R}}_n^2 + \sum_{n,m} U_{n,m} \delta \tilde{\mathbf{R}}_n \delta \tilde{\mathbf{R}}_m$$

At this point I introduce the quantized displacements in form of the quadrature operators

$$\begin{aligned}\delta \tilde{\mathbf{R}}_n &= \sqrt{\frac{\hbar}{2m\omega}} (a_n + a_n^\dagger) \\ \mathbf{P}_n &= i \sqrt{\frac{\hbar m \omega}{2}} (a_n^\dagger - a_n)\end{aligned}$$

With that the Hamiltonian gets its final form

$$H = \sum_n \hbar \omega a_n^\dagger a_n + \frac{\hbar}{2m\omega} \sum_{n,m} U_{n,m} (a_n a_m + a_n^\dagger a_m + a_n a_m^\dagger + a_n^\dagger a_m^\dagger)$$

Physical implementation

As a physical implementation I consider a system of Rydberg atoms. The interactions between Rydberg atoms can be tuned in a wide range of magnitude. What is however restricted are the lifetimes of such Rydberg states. They can reach for conventional Rydberg states up to several hundred μs . For Circular Rydberg states the lifetimes can reach even a few ms. In both cases the trap depth (ω) can easily have a few MHz [5, 6]. In order to observe the physics induced by the Rydberg-Hamiltonian during the lifetime T , this Hamiltonian has to be on energy scales $|H| \gg 1/T \sim 5\text{-}10\text{ kHz}$. For the implementation of a Rice-Mele transfer one can estimate the duration T_{RM} of the process of $T_{RM} \gg \langle i|\dot{H}|f\rangle/\Delta^2 \sim 1/(T_{RM}\Delta^2) \rightarrow T_{RM} \gg 1/\Delta$, where Δ is the band gap of the topological insulator ($\Delta \sim |H|$). Thus for circular Rydberg states as well as conventional ones it is possible to fulfill this condition whilst at the same time satisfying $|H| \ll \omega$. The latter assumption leads to a large energy splitting of different motional excitation numbers and hence to an effective conservation of motional excitations. This allows one to analyze the topological properties of the system via a single-particle fermionic formalism.

One can, however, also consider a regime where the motional excitations are not conserved, i.e. $\omega \lesssim |H|$. For this investigation I assume dipole-dipole interactions between dressed Rydberg states. The corresponding potential reads

$$\begin{aligned} V_{dd}(\mathbf{r}) &= \frac{\langle d \rangle^2}{4\pi\epsilon_0 r^3} (1 - 3(\hat{\mathbf{r}} \cdot \hat{\mathbf{d}})^2) =: V_0(r) f(\hat{\mathbf{r}}) = V_0(a) \frac{a^3}{r^3} f(\hat{\mathbf{r}}) \\ \frac{\partial V_{dd}}{\partial r_\alpha} &= \frac{\langle d \rangle^2}{4\pi\epsilon_0 r^3} \frac{1}{r} \left(-3 \frac{r_\alpha}{r} + 15 \frac{r_\alpha}{r} (\hat{\mathbf{r}} \cdot \hat{\mathbf{d}})^2 - 6 \hat{d}_\alpha (\hat{\mathbf{r}} \cdot \hat{\mathbf{d}}) \right) =: \frac{V_0(r)}{r} g(\hat{\mathbf{r}}) = \frac{V_0(a)}{a} \frac{a^4}{r^4} g(\hat{\mathbf{r}}) \\ \frac{\partial^2 V_{dd}}{\partial r_\alpha \partial r_\beta} &= \frac{\langle d \rangle^2}{4\pi\epsilon_0 r^3} \frac{1}{r^2} \left[\delta_{\alpha\beta} (-3 + 15 (\hat{\mathbf{r}} \cdot \hat{\mathbf{d}})^2) - 6 \hat{d}_\alpha \hat{d}_\beta + 30 \frac{\hat{d}_\alpha r_\beta + \hat{d}_\beta r_\alpha}{r} (\hat{\mathbf{r}} \cdot \hat{\mathbf{d}}) + 15 \frac{r_\alpha r_\beta}{r^2} (1 - 7 (\hat{\mathbf{r}} \cdot \hat{\mathbf{d}})^2) \right] \\ &=: \frac{V_0(r)}{r^2} c(\hat{\mathbf{r}}) = \frac{V_0(a)}{a^2} \frac{a^5}{r^5} c(\hat{\mathbf{r}}) \end{aligned}$$

For a topological characterization of the system the Hamiltonian has to be in a quadratic form in the phonon operators. For this purpose I will now investigate for which parameter configurations it is possible to tune the first order terms in the displacement to zero. In the following I will consider a chain of atoms and will restrict the displacements of the atoms to the x direction, which is the direction in which the one dimensional chain extends. The chain consists of two sublattices A and B with a distance of $a \hat{\mathbf{e}}_x$ between neighboring atoms of the same sublattice and a distance of $b \hat{\mathbf{e}}_x + h \hat{\mathbf{e}}_y$ between neighboring atoms of different sublattices. Interactions shall only be relevant between nearest neighbors.

At this point let us rediscuss the shift of the displacements which allowed us to eliminate the first order term in the Hamiltonian. The necessary condition was

$$\left(\frac{\partial H_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_i} + \sum_j (2U_{j,i} + m\omega^2 \delta_{ij}) \Delta_j \right) = 0 \quad \forall i$$

whereas

$$\begin{aligned} \frac{\partial H_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_i} &= \sum_{n,m; n < m} \frac{\partial V_{n,m}(\mathbf{R}_n, \mathbf{R}_m)}{\partial \mathbf{R}_i} (\delta_{i=n} + \delta_{i=m}) \\ &= \sum_{m; i < m} \frac{\partial V_{i,m}(\mathbf{R}_m - \mathbf{R}_i)}{\partial \mathbf{R}_i} + \sum_{n; n < i} \frac{\partial V_{n,i}(\mathbf{R}_i - \mathbf{R}_n)}{\partial \mathbf{R}_i} \\ &= \sum_{m; i < m} \frac{\partial V_{i,m}(\mathbf{R}_m - \mathbf{R}_i)}{\partial \mathbf{r}_{i,m}} - \sum_{n; n < i} \frac{\partial V_{n,i}(\mathbf{R}_i - \mathbf{R}_n)}{\partial \mathbf{r}_{n,i}} \end{aligned}$$

with $\mathbf{r}_{i,j} = \mathbf{R}_j - \mathbf{R}_i$. By defining

$$\begin{aligned} \tilde{g}_i &:= \frac{a}{V_0(a)} \frac{\partial H_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_i} \\ \tilde{U}_{i,j} &:= \frac{a^2}{V_0(a)} 2U_{j,i} \end{aligned}$$

The condition for the shift can be written as

$$\frac{V_0(a)}{a} G_i := \frac{V_0(a)}{a} \left(\tilde{g}_i + \sum_j W_{ij} \frac{\Delta_j}{a} \right) := \frac{V_0(a)}{a} \left(\tilde{g}_i + \sum_j \left(\tilde{U}_{j,i} + \frac{m\omega^2 a^2}{V_0(a)} \delta_{ij} \right) \frac{\Delta_j}{a} \right) = 0 \quad \forall i$$

Here $\tilde{g}$ and $\tilde{U}$ are dimensionless and the term $m\omega^2 a^2 / [V_0(a) \max(\tilde{U})] =: m\omega^2 / V_{\max}$ chosen of order 0.01-10. Now I use the two free parameters h and the angle φ_d of the dipole moment $\hat{\mathbf{d}}$ to set $G_i = 0$ or close to zero for different values of b . Close to zero means here that $\max(|G_i|)$ compared to $\sqrt{\hbar/(2m\omega)} \max(\tilde{U})/a$ is small. Here $\sqrt{\hbar/(2m\omega)}/a \sim 10^{-3}$ is the displacement from a single motional excitation in units of a .

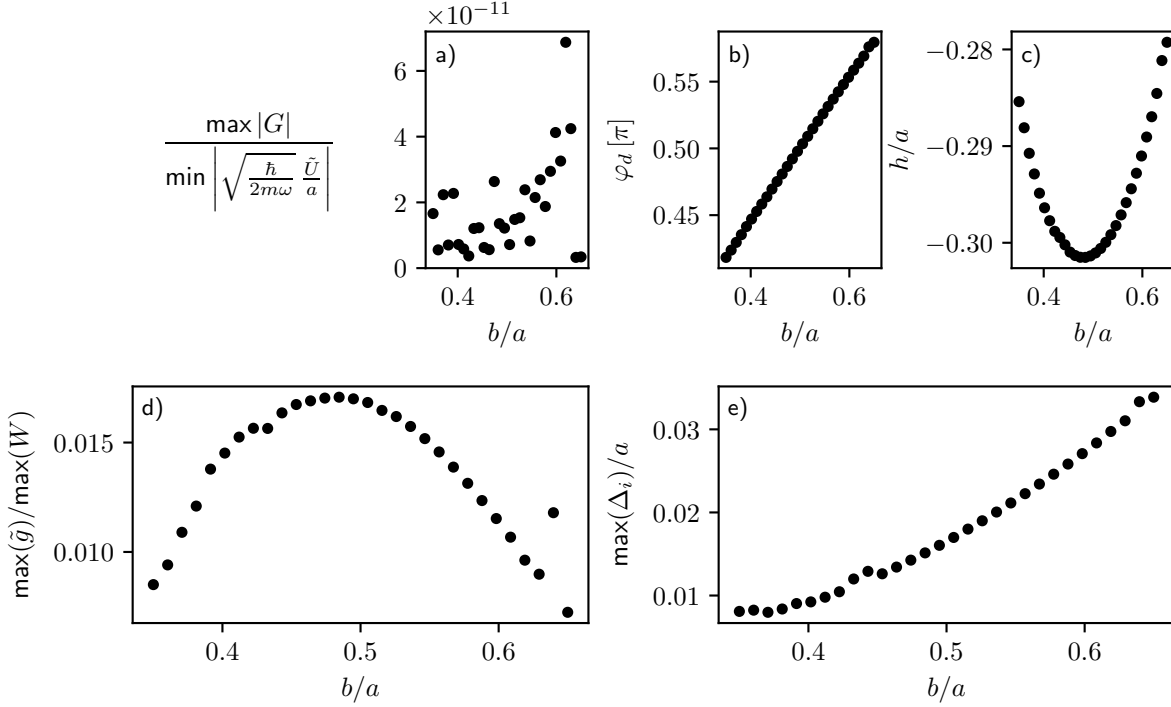


Figure 1: **Parameters for negligible first order.**

Fig. 1 and Fig. 2 tell us that the Δ -displacements are of order 10^{-2} , an order of magnitude bigger than the single motional mode displacement. Nevertheless the shift is small compared to the atomic distance a . Realizing a vanishing first order in this way, the Hamiltonian takes a quadratic form in the creation and annihilation operators and can hence be written as

$$H = \sum_n \hbar\omega a_n^\dagger a_n + \frac{\hbar}{2m\omega} \sum_{n,m} U_{n,m} (a_n a_m + a_n^\dagger a_m^\dagger + a_n a_m^\dagger + a_n^\dagger a_m) \\ = \mathbf{a}^\dagger A \mathbf{a}$$

with $\mathbf{a} = (a_1, \dots, a_N, a_1^\dagger, \dots, a_N^\dagger)$ the vector of operators and a coupling matrix A , which takes the following form

$$A = \frac{\hbar}{2m\omega} \begin{pmatrix} \tilde{U} & U \\ U & \tilde{U} \end{pmatrix} \\ \text{with } \tilde{U} = U + m\omega^2.$$

This allows for a topological characterization of the bosonic system, according to [7]. A first step is to consider the Heisenberg equations of motion for the operators which read

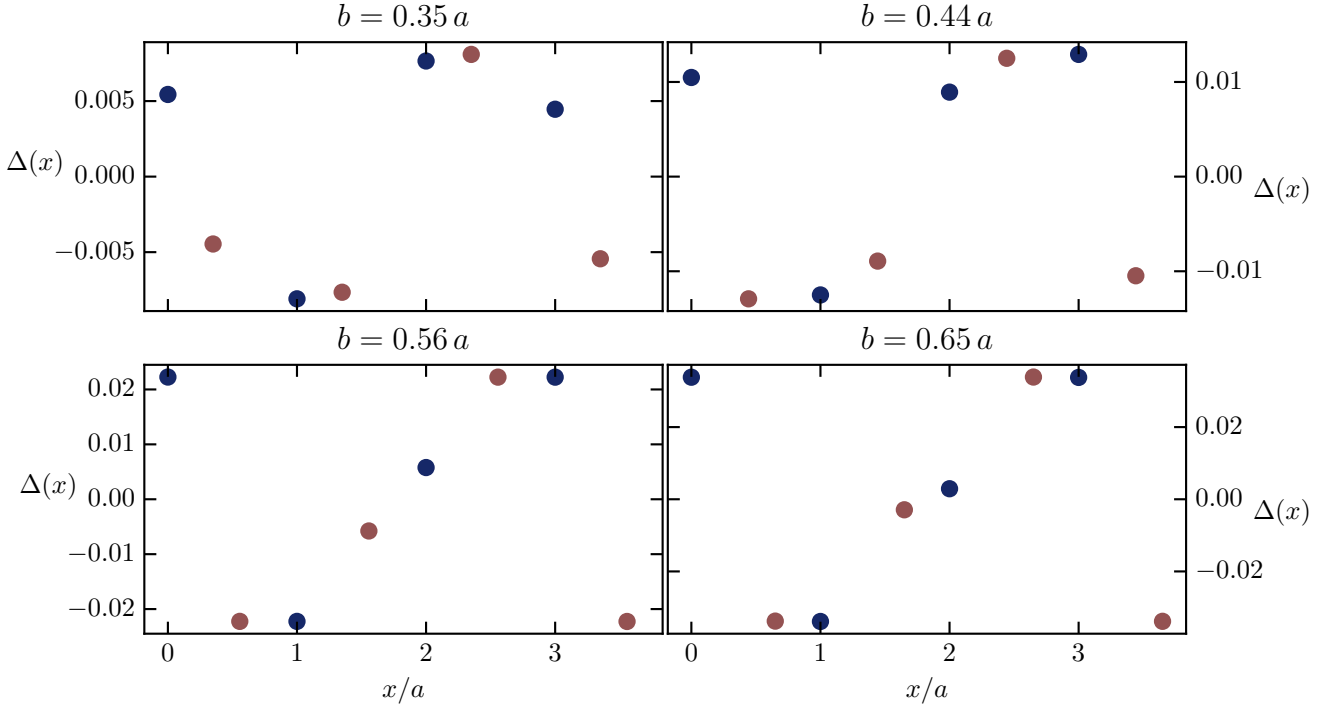


Figure 2: **Displacements.**

$$\begin{aligned}
\frac{d}{dt} a_j &= i ([H, a_j]) = -i \hbar \omega a_j - i \frac{\hbar}{2m\omega} \sum_n (U_{j,n} a_n + U_{n,j} a_n + U_{j,n} a_n^\dagger + U_{n,j} a_n^\dagger) \\
&= -i \hbar \omega a_j - i \frac{\hbar}{m\omega} \sum_n (U_{j,n} a_n + U_{j,n} a_n^\dagger) = -2i (A \vec{a})_j \\
\frac{d}{dt} a_j^\dagger &= i ([H, a_j^\dagger]) = i \hbar \omega a_j^\dagger + i \frac{\hbar}{2m\omega} \sum_n (U_{j,n} a_n + U_{n,j} a_n + U_{j,n} a_n^\dagger + U_{n,j} a_n^\dagger) \\
&= 2i (A \vec{a})_{j+N}
\end{aligned}$$

This lets one write the equations of motion in a different form

$$\begin{aligned}
\Rightarrow \quad \frac{d}{dt} \vec{a} &= -i \tilde{H} \vec{a} \\
\text{where } \tilde{H} &= \frac{\hbar}{m\omega} \begin{pmatrix} \bar{U} & U \\ -U & -\bar{U} \end{pmatrix} =: \begin{pmatrix} h & \Delta \\ -\Delta & -h \end{pmatrix}
\end{aligned}$$

And it is this matrix $\tilde{H}$ that determines the topological properties of the system, since it provides the eigenmodes of the dynamics. $\tilde{H}$ is in the Bogoliubov de Gennes form and a real matrix. Therefore it is time reversal symmetric. Particle-hole symmetry is also fulfilled, as $\tilde{H}^* = -\Sigma_x \tilde{H} \Sigma_x$ with

$$\Sigma_x = \begin{pmatrix} 0 & \mathbb{1} \\ \mathbb{1} & 0 \end{pmatrix}$$

As a consequence it is then also chiral symmetric, as $\tilde{H} = -\Sigma_x \tilde{H} \Sigma_x$. It is hence in the CI symmetry class, which allows for a topological characterization in one dimension. Since H is not hermitian it does not necessarily have only real eigenvalues. For the topological classification to be possible, the Hamiltonian has to be, however, dynamically stable, which means no complex eigenvalues are allowed, as they lead to unbounded dynamics. When looking at the spectrum of the BdG-Hamiltonian over the

parameter that lets the first order be neglected, one observes always large imaginary parts. This is seen in Fig.3. As a consequence the system is not classifiable by the formalism described in [7].

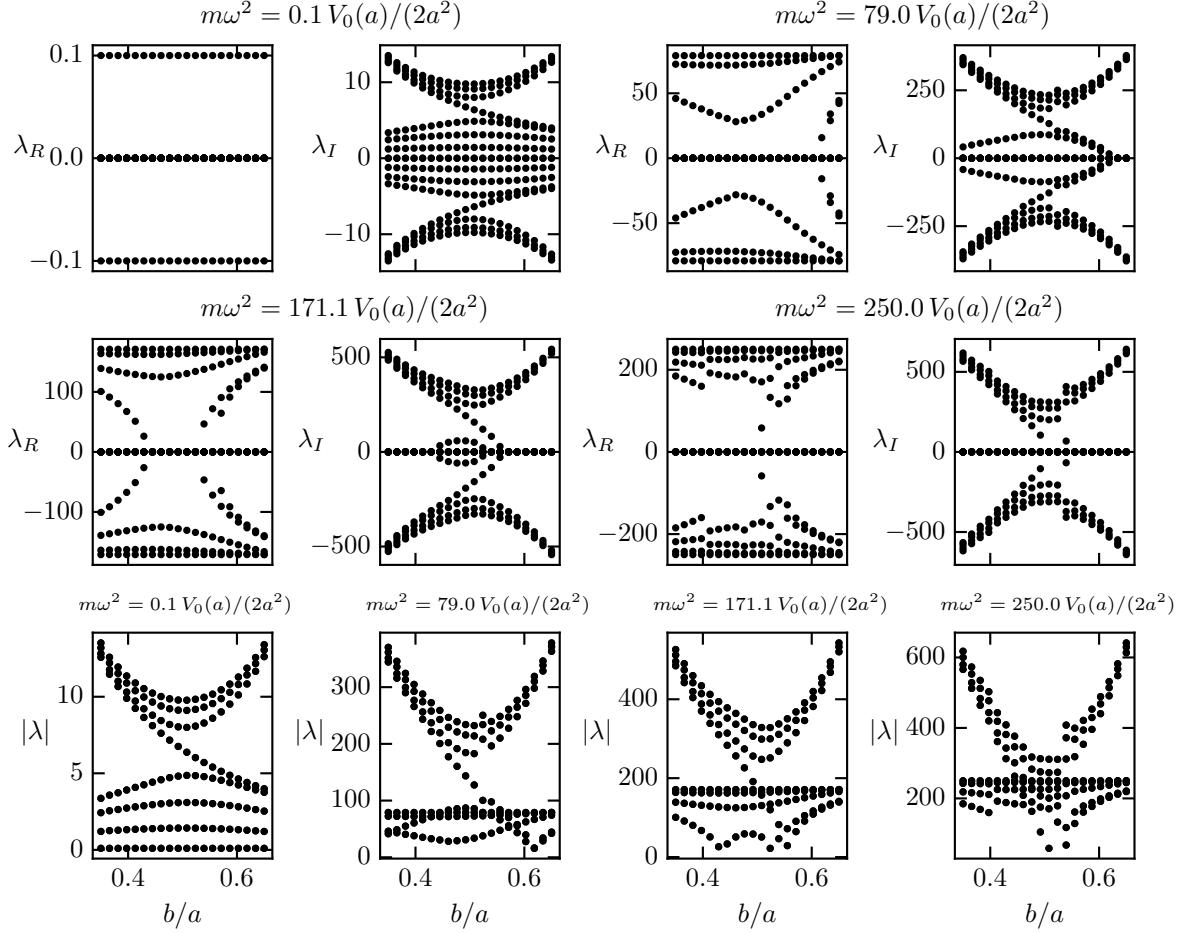


Figure 3: Spectrum of BdG Hamiltonian for different trap frequencies.

Momentum space Hamiltonian

Let $c_{\mathbf{R},\alpha}^A$ and $c_{\mathbf{R},\alpha}^B$ be the phonon annihilation operators of sublattice A and B respectively at the unit cell, located at $\mathbf{R}$, where $\alpha \in \{x, y\}$ labels the the direction of movement. Let further L be the set of all unit cells of the lattice, and $C = \{c_{\mathbf{R},\alpha}^A, c_{\mathbf{R},\alpha}^B\}_{\mathbf{R} \in L, \alpha \in \{x, y\}}$ the set of all annihilation operators. The Hamiltonian in real space can then be written as

$$H = \sum_{\substack{\mathbf{R}, \mathbf{R}' \in L \\ a_{\mathbf{R}} \in C_{\mathbf{R}}, b_{\mathbf{R}'} \in C_{\mathbf{R}'}}} U_{ab}(\mathbf{R} - \mathbf{R}') (a_{\mathbf{R}}^\dagger b_{\mathbf{R}'} + a_{\mathbf{R}} b_{\mathbf{R}'}^\dagger + a_{\mathbf{R}}^\dagger b_{\mathbf{R}'}^\dagger + a_{\mathbf{R}} b_{\mathbf{R}'})$$

For two dimensional periodic boundary conditions one can introduce the Fourier transformed operators, $a \in C$

$$a_{\mathbf{k}} = \frac{1}{\sqrt{L}} \sum_{\mathbf{R} \in N} e^{-i\mathbf{k} \cdot \mathbf{R}} a_{\mathbf{R}},$$

$k \in \bar{L}$ where $\bar{L}$ is the reciprocal lattice of L and N is the extent of the lattice in one direction. Writing the Hamiltonian in terms of these operators yields

$$\begin{aligned}
H &= \frac{1}{L} \sum_{\substack{\mathbf{R}, \mathbf{R}' \in L \\ \mathbf{k}, \mathbf{k}' \in \bar{L} \\ a_{\mathbf{k}} \in C_{\mathbf{k}}, b_{\mathbf{k}'} \in C_{\mathbf{k}'}}} U_{ab}(\mathbf{R} - \mathbf{R}') \left(e^{-i(\mathbf{k}\mathbf{R} - \mathbf{k}'\mathbf{R}')} a_{\mathbf{k}}^\dagger b_{\mathbf{k}'} + e^{i(\mathbf{k}\mathbf{R} - \mathbf{k}'\mathbf{R}')} a_{\mathbf{k}} b_{\mathbf{k}'}^\dagger \right. \\
&\quad \left. + e^{-i(\mathbf{k}\mathbf{R} + \mathbf{k}'\mathbf{R}')} a_{\mathbf{k}}^\dagger b_{\mathbf{k}'}^\dagger + e^{i(\mathbf{k}\mathbf{R} + \mathbf{k}'\mathbf{R}')} a_{\mathbf{k}} b_{\mathbf{k}'} \right) \\
&= \frac{1}{L} \sum_{\substack{\mathbf{R}, \Delta\mathbf{R} = \mathbf{R} - \mathbf{R}' \\ \mathbf{k}, \mathbf{k}' \\ a, b}} U_{ab}(\Delta\mathbf{R}) \left(e^{-i\mathbf{R}(\mathbf{k} - \mathbf{k}') - i\Delta\mathbf{R}\mathbf{k}'} a_{\mathbf{k}}^\dagger b_{\mathbf{k}'} + e^{i\mathbf{R}(\mathbf{k} - \mathbf{k}') + i\Delta\mathbf{R}\mathbf{k}'} a_{\mathbf{k}} b_{\mathbf{k}'}^\dagger \right. \\
&\quad \left. + e^{-i\mathbf{R}(\mathbf{k} + \mathbf{k}') + i\Delta\mathbf{R}\mathbf{k}'} a_{\mathbf{k}}^\dagger b_{\mathbf{k}'}^\dagger + e^{i\mathbf{R}(\mathbf{k} + \mathbf{k}') - i\Delta\mathbf{R}\mathbf{k}'} a_{\mathbf{k}} b_{\mathbf{k}'} \right) \\
&= \sum_{\Delta\mathbf{R}, \mathbf{k}, a, b} U_{ab}(\Delta\mathbf{R}) \left(e^{-i\Delta\mathbf{R}\mathbf{k}} a_{\mathbf{k}}^\dagger b_{\mathbf{k}} + e^{i\Delta\mathbf{R}\mathbf{k}} a_{\mathbf{k}} b_{\mathbf{k}}^\dagger + e^{-i\Delta\mathbf{R}\mathbf{k}} a_{\mathbf{k}}^\dagger b_{-\mathbf{k}}^\dagger + e^{i\Delta\mathbf{R}\mathbf{k}} a_{\mathbf{k}} b_{-\mathbf{k}} \right) \\
&= \sum_{\mathbf{k}, a, b} \left(U_{ab}(\mathbf{k}) a_{\mathbf{k}}^\dagger b_{\mathbf{k}} + U_{ab}^*(\mathbf{k}) a_{\mathbf{k}} b_{\mathbf{k}}^\dagger + U_{ab}(\mathbf{k}) a_{\mathbf{k}}^\dagger b_{-\mathbf{k}}^\dagger + U_{ab}^*(\mathbf{k}) a_{\mathbf{k}} b_{-\mathbf{k}} \right) =: \sum_{\mathbf{k} \geq 0} H_{\mathbf{k}, -\mathbf{k}} \\
&\text{where } U_{ab}(\mathbf{k}) = \sum_{\Delta\mathbf{R}} U_{ab}(\Delta\mathbf{R}) e^{-i\Delta\mathbf{R}\mathbf{k}}
\end{aligned}$$

The block of the Hamiltonian in momentum space $H_{\mathbf{k}, -\mathbf{k}}$ can be written in a more compact form, by introducing the vector of operators $\beta_{\mathbf{k}} = (a_{\mathbf{k}}, b_{\mathbf{k}}, a_{-\mathbf{k}}, b_{-\mathbf{k}}, a_{\mathbf{k}}^\dagger, b_{\mathbf{k}}^\dagger, a_{-\mathbf{k}}^\dagger, b_{-\mathbf{k}}^\dagger)$ and the coupling matrix $B_{\mathbf{k}}$, which takes the following form

$$B_{\mathbf{k}} = \begin{pmatrix} u_{aa} & u_{ab} & 0 & 0 & 0 & 0 & u_{aa} & u_{ab} \\ u_{ab}^* & u_{aa} & 0 & 0 & 0 & 0 & u_{ab}^* & u_{aa} \\ 0 & 0 & u_{aa} & u_{ab}^* & u_{aa} & u_{ab}^* & 0 & 0 \\ 0 & 0 & u_{ab} & u_{aa} & u_{ab} & u_{aa} & 0 & 0 \\ 0 & 0 & u_{aa} & u_{ab}^* & u_{aa} & u_{ab}^* & 0 & 0 \\ 0 & 0 & u_{ab} & u_{aa} & u_{ab} & u_{aa} & 0 & 0 \\ u_{aa} & u_{ab} & 0 & 0 & 0 & 0 & u_{aa} & u_{ab} \\ u_{ab}^* & u_{aa} & 0 & 0 & 0 & 0 & u_{ab}^* & u_{aa} \end{pmatrix}$$

such that $H_{\mathbf{k}, -\mathbf{k}} = \beta_{\mathbf{k}}^\dagger B_{\mathbf{k}} \beta_{\mathbf{k}}$. By reordering $\beta_{\mathbf{k}}$ into $\tilde{\beta}_{\mathbf{k}} = (a_{\mathbf{k}}, b_{\mathbf{k}}, a_{-\mathbf{k}}^\dagger, b_{-\mathbf{k}}^\dagger, a_{-\mathbf{k}}, b_{-\mathbf{k}}, a_{\mathbf{k}}^\dagger, b_{\mathbf{k}}^\dagger)$ the matrix $B_{\mathbf{k}}$ becomes block diagonal with

$$\tilde{B}_{\mathbf{k}} = \begin{pmatrix} u_{aa} & u_{ab} & u_{aa} & u_{ab} & 0 & 0 & 0 & 0 \\ u_{ab}^* & u_{aa} & u_{ab}^* & u_{aa} & 0 & 0 & 0 & 0 \\ u_{aa} & u_{ab} & u_{aa} & u_{ab} & 0 & 0 & 0 & 0 \\ u_{ab}^* & u_{aa} & u_{ab}^* & u_{aa} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & u_{aa} & u_{ab}^* & u_{aa} & u_{ab}^* \\ 0 & 0 & 0 & 0 & u_{ab} & u_{aa} & u_{ab} & u_{aa} \\ 0 & 0 & 0 & 0 & u_{aa} & u_{ab}^* & u_{aa} & u_{ab}^* \\ 0 & 0 & 0 & 0 & u_{ab} & u_{aa} & u_{ab} & u_{aa} \end{pmatrix}$$

Accordingly the Hamiltonian can be diagonalized in terms of momentum, if one introduce the particle-hole space for a specific $\mathbf{k}$ as corresponding to the annihilation operator vector $\mathbf{a}_{\mathbf{k}} = (a_{\mathbf{k}}, b_{\mathbf{k}}, a_{-\mathbf{k}}^\dagger, b_{-\mathbf{k}}^\dagger)$ with $(a_{\mathbf{k}}, b_{\mathbf{k}})$ annihilating particles and $(a_{-\mathbf{k}}^\dagger, b_{-\mathbf{k}}^\dagger)$ annihilating holes.

$$H_{\mathbf{k}, -\mathbf{k}} = \mathbf{a}_{\mathbf{k}}^\dagger A_{\mathbf{k}} \mathbf{a}_{\mathbf{k}} + \mathbf{a}_{-\mathbf{k}}^\dagger A_{-\mathbf{k}} \mathbf{a}_{-\mathbf{k}}$$

$$\text{with } A_{\mathbf{k}} = \bar{A}_{-\mathbf{k}} = \begin{pmatrix} u_{aa} & u_{ab} & u_{aa} & u_{ab} \\ u_{ab}^* & u_{aa} & u_{ab}^* & u_{aa} \\ u_{aa} & u_{ab} & u_{aa} & u_{ab} \\ u_{ab}^* & u_{aa} & u_{ab}^* & u_{aa} \end{pmatrix}$$

This can in principal also be written in a more compact form as

$$H_{\mathbf{k}, -\mathbf{k}} = \begin{pmatrix} a_{\mathbf{k}}^\dagger & b_{\mathbf{k}}^\dagger & a_{-\mathbf{k}} & b_{-\mathbf{k}} \end{pmatrix} \tilde{A}_{\mathbf{k}} \begin{pmatrix} a_{\mathbf{k}} \\ b_{\mathbf{k}} \\ a_{-\mathbf{k}}^\dagger \\ b_{-\mathbf{k}}^\dagger \end{pmatrix}$$

$$\text{with } \tilde{A}_{\mathbf{k}} = \begin{pmatrix} 2\text{Re}(u_{aa}) & 2u_{ab} & 2\text{Re}(u_{aa}) & 2u_{ab} \\ 2u_{ab}^* & 2\text{Re}(u_{aa}) & 2u_{ab}^* & 2\text{Re}(u_{aa}) \\ 2\text{Re}(u_{aa}) & 2u_{ab} & 2\text{Re}(u_{aa}) & 2u_{ab} \\ 2u_{ab}^* & 2\text{Re}(u_{aa}) & 2u_{ab}^* & 2\text{Re}(u_{aa}) \end{pmatrix}$$

The matrix $A_{\mathbf{k}}$ takes the form

$$A_{\mathbf{k}} = \begin{pmatrix} B & C \\ C^* & B^T \end{pmatrix}$$

with $B^\dagger = B$ and $C^T = C$. In general for a Hamiltonian H which is written as

$$H = (\bar{a}^\dagger, \bar{a}) \begin{pmatrix} P & Q \\ R & S \end{pmatrix} \begin{pmatrix} \bar{a} \\ \bar{a}^\dagger \end{pmatrix}$$

where $\bar{a} = (a_{\mathbf{k}}, b_{\mathbf{k}}, a_{-\mathbf{k}}, b_{-\mathbf{k}})$ and P, Q, R, S are matrices, the Heisenberg equations of motion determined by the commutators of H with the operators a_i and $a_i^\dagger$ can be calculated as

$$\begin{aligned} H &= \bar{a}^\dagger P \bar{a} + \bar{a}^\dagger Q \bar{a}^\dagger + \bar{a} R \bar{a} + \bar{a} S \bar{a}^\dagger \\ &= \sum_{ij} a_i^\dagger P_{ij} a_j + a_i^\dagger Q_{ij} a_j^\dagger + a_i R_{ij} a_j + a_i S_{ij} a_j^\dagger \\ [H, a_k] &= \sum_j P_{kj} a_j + Q_{kj} a_j^\dagger + a_j^\dagger Q_{jk} + a_j S_{jk} \\ &= P \bar{a} + S^T \bar{a} + Q \bar{a}^\dagger + Q^T \bar{a}^\dagger \\ [H, a_k^\dagger] &= -\sum_j a_j^\dagger P_{jk} + R_{kj} a_j^\dagger + a_j^\dagger R_{jk} + S_{kj} a_j^\dagger \\ &= -(P^T \bar{a}^\dagger + R \bar{a} + R^T \bar{a} + S \bar{a}^\dagger) \end{aligned}$$

Thus in the case of the Hamiltonian $H_{\mathbf{k}, -\mathbf{k}}$, where $P^T = S$ and $R^T = R$, the Heisenberg equations of motion can be written as

$$\begin{aligned} \frac{d}{dt} \mathbf{a}_j &= -2i \sum_{l=1}^4 B_{jl} \mathbf{a}_l + C_{jl} \mathbf{a}_{4+l} \quad \text{for } j = 1, \dots, 4 \\ \frac{d}{dt} \mathbf{a}_j &= 2i \sum_{l=1}^4 C_{jl}^* \mathbf{a}_l + B_{jl}^* \mathbf{a}_{4+l} \quad \text{for } j = 4, \dots, 8 \\ \Rightarrow \frac{d}{dt} \mathbf{a} &= -2i \begin{pmatrix} B & C \\ -C^* & -B^* \end{pmatrix} \mathbf{a} =: -i \tilde{H}_{\mathbf{k}} \mathbf{a} \end{aligned}$$

Mathematical approach to the BdG Hamiltonian

I want to introduce a more mathematical approach to the Hamiltonian.

$$\begin{aligned}\mathcal{H} &= \frac{1}{2}(\mathbf{a}^\dagger)^t h \mathbf{a} + \frac{1}{2}\mathbf{a}^t h \mathbf{a}^\dagger + \frac{1}{2}(\mathbf{a}^\dagger)^t \Delta \mathbf{a}^\dagger + \frac{1}{2}\mathbf{a}^t \Delta^\dagger \mathbf{a}, \quad h = h^\dagger, \quad \Delta = \Delta^t \\ &= \frac{1}{2}(\mathbf{h} + \bar{\mathbf{h}} + \Delta + \Delta^\dagger)\end{aligned}$$

This can be written in matrix form as

$$\mathcal{H} = \frac{1}{2} \begin{pmatrix} \mathbf{a}^\dagger \\ \mathbf{a} \end{pmatrix}^t \tilde{A} \begin{pmatrix} \mathbf{a} \\ \mathbf{a}^\dagger \end{pmatrix}, \quad \tilde{A} = \begin{pmatrix} h & \Delta \\ \Delta^\dagger & h \end{pmatrix}$$

In the topological framework we want to consider single-particle Hamiltonians, defined on a single particle space $\mathfrak{h} = \mathbb{C}^N \otimes \ell^2(\mathbb{Z}^d)$ where $\mathbb{C}^N$ treats the internal degrees of freedom of each unit cell and $\ell^2(\mathbb{Z}^d)$ represents the lattice structure. One can then define

$$\mathbf{h} : \mathfrak{h} \rightarrow \mathfrak{h}.$$

But the Δ -terms are two-particle operators. The solution to this problem is to interpret the creation part of the stacked operator vector as operators in a hole-space, i.e. hole-annihilation operators and then treating holes and particles as independent degrees of freedom. I show here how this reinterpretation can be carried out. Take for example the last term of the Hamiltonian. It takes two particles and annihilates them, it can be viewed as bilinear form on the two particle space, reducing the particle space by two particles

$$\Delta^\dagger : \mathfrak{h} \times \mathfrak{h} \rightarrow \mathbb{C} : |\psi\rangle |\phi\rangle \mapsto \Delta^\dagger(|\psi\rangle |\phi\rangle)$$

But for plugging in a certain vector $|\psi\rangle$ and leaving the other spot open, we can define a map $\tilde{\Delta}^\dagger(|\psi\rangle) : \mathfrak{h} \rightarrow \mathcal{L}(\mathfrak{h}, \mathbb{C}) \rightarrow \mathfrak{h}^*$, where $\mathcal{L}(\mathfrak{h}, \mathbb{C})$ is the space of linear maps from $\mathfrak{h}$ to $\mathbb{C}$. But each such map can be represented in the dual space by a vector $\mathfrak{L}(\mathfrak{h}, \mathbb{C}) \rightarrow \mathfrak{h}^* : \Delta^\dagger(|\psi\rangle, \cdot) \mapsto |\tilde{\Delta}^\dagger(\psi)\rangle \in \mathfrak{h}^*$ such that $\Delta^\dagger(|\psi\rangle |\phi\rangle) = \langle \tilde{\Delta}^\dagger(\psi) | \phi \rangle$. But the linearity of $\Delta^\dagger$ results in

$$|\tilde{\Delta}^\dagger(\lambda\psi)\rangle = \bar{\lambda} |\tilde{\Delta}^\dagger(\psi)\rangle$$

This makes $\tilde{\Delta}^\dagger$ an anti-linear operation. One can resolve this issue by introduce a different space $\bar{\mathfrak{h}}$ which possesses a scalar multiplication of the form

$$\cdot : \mathbb{C} \times \bar{\mathfrak{h}} \rightarrow \bar{\mathfrak{h}} : (\lambda, |\varphi\rangle) \mapsto \bar{\lambda} |\varphi\rangle$$

By this way one can define a linear operator over single particle Hamiltonians

$$\Delta_s^\dagger : \mathfrak{h} \rightarrow \bar{\mathfrak{h}} : |\psi\rangle \mapsto |\tilde{\Delta}^\dagger(\psi)\rangle$$

In a similar way one can define

$$\begin{aligned}\Delta : \mathbb{C} \rightarrow \mathfrak{h} \times \mathfrak{h} : 1 &\mapsto \Delta |\psi\rangle |\phi\rangle \\ \leadsto \Delta_s : \bar{\mathfrak{h}} \rightarrow \mathfrak{h} : |\psi\rangle &\mapsto \langle \psi | \Delta(1) = \Delta |\phi\rangle\end{aligned}$$

where Δ_s can be divided into a simple linear map $M : \bar{\mathfrak{h}} \rightarrow \mathfrak{h}$ that maps each basis vector $|b\rangle_{\bar{\mathfrak{h}}}$ in $\bar{\mathfrak{h}}$ to the appropriate basis vector $|b\rangle_{\mathfrak{h}}$ in $\mathfrak{h}$, thus $\bar{\lambda} |b\rangle_{\bar{\mathfrak{h}}} \mapsto \lambda |b\rangle_{\mathfrak{h}}$, followed by adjunction and multiplication with $\Delta(1)$.

$$\begin{aligned}\Delta_s : \bar{\mathfrak{h}} \rightarrow \mathfrak{h} : |\psi\rangle &\mapsto (M(|\psi\rangle))^t \Delta(1) = \Delta |\phi\rangle \\ \leadsto \Delta : \mathbb{C} \rightarrow \mathfrak{h} \times \mathfrak{h} : 1 &\mapsto M(|\psi\rangle_{\bar{\mathfrak{h}}}) \Delta_s(|\psi\rangle_{\bar{\mathfrak{h}}})\end{aligned}$$

At this state one identify the space $\mathfrak{h}$ with the single particle space and $\bar{\mathfrak{h}}$ with the single hole-space, where particle and holes are interpreted as independent degrees of freedom. The total space is than

represented by the Nambu-spinor $(|\psi\rangle_{\mathfrak{h}}, |\varphi\rangle_{\bar{\mathfrak{h}}})$ with particle degrees of freedom $|\psi\rangle$ and hole degrees of freedom $|\phi\rangle$. This lets the effect of the Hamiltonian on such a Nambu-state be written as

$$\mathcal{H} \begin{pmatrix} |\psi\rangle \\ |\phi\rangle \end{pmatrix} = \begin{pmatrix} \mathbf{h} & \Delta \\ \Delta^\dagger & \bar{\mathbf{h}} \end{pmatrix} \begin{pmatrix} |\psi\rangle \\ |\phi\rangle \end{pmatrix}$$

with $\bar{\mathbf{h}} : \bar{\mathfrak{h}} \rightarrow \bar{\mathfrak{h}}$

Each single particle operator can be separated into operators that act on the internal system of a unit cell and one, that mediates between unit cells u^x . Considering periodic boundaries the operators only depend on the distance x between the cells and one can write for example

$$\mathbf{h} = \sum_x h_x u^x$$

where $h_x : \mathbb{C} \rightarrow \mathbb{C}$ treats the internal degrees of freedom and $u^x : \ell^2(\mathbb{Z}^d) \rightarrow \ell^2(\mathbb{Z}^d)$ is just the shift operator. Fourier transforming a state $\psi(y) = |\psi\rangle_{\text{int}} |\psi\rangle_{\text{Lat}}(y) \in \mathfrak{h}$, $\phi(y) = |\phi\rangle_{\text{int}} |\phi\rangle_{\text{Lat}}(y) \in \bar{\mathfrak{h}}$, where y is the position of a unit cell, is defined as

$$\begin{aligned} \mathcal{F}(\psi)(k) &= \sum_y e^{-iky} \psi(y) \\ \mathcal{F}(\phi)(k) &= \sum_y e^{iky} \phi(y) \end{aligned}$$

By this one can determine the Fourier transform of the operators by their action on the states.

$$\begin{aligned} \mathcal{F}(\mathbf{h}\psi)(k) &= \sum_{y,x} e^{-iky} h_x u^x \psi(y) \\ &= \sum_{y,x} e^{-iky} h_x \psi(y-x) \\ &= \sum_{z,x} e^{-ikz} e^{-ikx} h_x \psi(z) \\ &= \sum_x e^{-ikx} h_x \mathcal{F}(\psi)(k) \\ &=: \mathbf{h}(k) \mathcal{F}(\psi)(k) =: \mathbf{h}(k) \psi(k) \end{aligned}$$

As $\bar{\mathbf{h}} = \sum_x \bar{h}_x u^x$, where the bar means complex conjugation, we can write

$$\begin{aligned} \mathcal{F}(\bar{\mathbf{h}}\phi)(k) &= \sum_{y,x} e^{iky} \bar{h}_x u^x \phi(y) \\ &= \sum_x e^{ikx} \bar{h}_x \mathcal{F}(\phi)(k) = \overline{\sum_x e^{-ikx} h_x \mathcal{F}(\phi)(k)} \\ &= \bar{\mathbf{h}}(k) \phi(k) \end{aligned}$$

For $\Delta_s = \sum_x \Delta_x u^x$ and $\Delta_s^\dagger = \sum_x \Delta_x^\dagger u^x$ we find the Fourier transform as

$$\begin{aligned} \mathcal{F}(\Delta_s \phi)(k) &= \sum_{y,x} e^{-iky} \Delta_x u^x \phi(y) \\ &= \sum_x e^{-ikx} \Delta_x \mathcal{F}(\phi)(-k) \\ &= \Delta_s(k) \phi(-k) \\ \mathcal{F}(\Delta_s^\dagger \psi)(k) &= \sum_{y,x} e^{iky} \Delta_x^\dagger u^x \psi(y) \\ &= \sum_x e^{ikx} \Delta_x^\dagger \mathcal{F}(\psi)(-k) = \overline{\sum_x e^{-ikx} \Delta_x \mathcal{F}(\psi)(-k)} \\ &= \Delta_s^\dagger(k) \psi(-k) \end{aligned}$$

If we write the space over all k in the form $\Psi = \oplus_k (\psi(k), \phi(-k))$ one can write the Hamiltonian a block diagonal $\mathcal{H} = \oplus_k A_k$ with

$$A_k = \begin{pmatrix} \mathbf{h}(k) & \Delta_s(k) \\ \Delta_s^\dagger(-k) & \bar{\mathbf{h}}(-k) \end{pmatrix}$$

The dynamical matrix that has to be analyzed with respect to topological properties is still

$$H_k = \begin{pmatrix} \mathbf{h}(k) & \Delta_s(k) \\ -\Delta_s^\dagger(-k) & -\mathbf{h}(-k) \end{pmatrix}$$

If the matrix H_k has chiral symmetry, which is true if H has chiral symmetry, this means that there is a unitary matrix U such that

$$UH_kU^\dagger = \begin{pmatrix} 0 & S \\ T & 0 \end{pmatrix} =: H'$$

The J matrix transforms under U to J' , which is still hermitian and squares to the identity. Setting

$$J' = \begin{pmatrix} A & B \\ B^\dagger & C \end{pmatrix}$$

one can write the explicitly

$$\begin{aligned} H'^\dagger J' &= J' H' \\ \Leftrightarrow \begin{pmatrix} T^\dagger B^\dagger & T^\dagger C \\ S^\dagger A & S^\dagger B \end{pmatrix} &= \begin{pmatrix} BT & AS \\ CT & B^\dagger S \end{pmatrix} \\ \Leftrightarrow S &= A^{-1} T^\dagger C \\ T &= B^{-1} T^\dagger B^\dagger \\ S &= (B^\dagger)^{-1} S^\dagger B \end{aligned}$$

The equation $S = A^{-1} T^\dagger C$ gives the insight that calculating the winding with S or T is equivalent. This becomes clear when one remembers the definition of the winding number for a hermitian block-off-diagonal matrix with blocks h and $h^\dagger$, i.e.

$$w = \frac{1}{2\pi i} \int_{\text{BZ}} dk \text{Tr} [h^{-1}(k) \partial_k h(k)].$$

Investigating the kernel of the integral when replacing h by S , one can put the winding of the two blocks in relation.

$$\begin{aligned} \text{Tr} [S^{-1} \partial_k S] &= \text{Tr} [C^{-1} (T^\dagger)^{-1} A \partial_k (A^{-1} T^\dagger C)] \\ &= \text{Tr} [(T^\dagger)^{-1} \partial_k T^\dagger] \end{aligned}$$

this is the same relation as between the hermitian blocks ($w(h) = -w(h^\dagger)$). Going back to the Rydberg atom system, the computed H_k has the following form, which is independent of the geometry of the system.

$$\begin{aligned} H_k &= \begin{pmatrix} R_k & P_k \\ -P_k & -R_k \end{pmatrix} \\ R_k &= \begin{pmatrix} w(\mathbf{k}) & u(\mathbf{k}) \\ \bar{u}(\mathbf{k}) & w(\mathbf{k}) \end{pmatrix} \quad P_k = \begin{pmatrix} v(\mathbf{k}) & u(\mathbf{k}) \\ \bar{u}(\mathbf{k}) & v(\mathbf{k}) \end{pmatrix} \end{aligned}$$

Exploiting the symmetric structure of the matrix one can write

$$\begin{aligned} H_k &= \tau_z \otimes R_k + i\tau_y \otimes P_k \\ R_k &= w(\mathbf{k}) \sigma_0 + \text{Re}[u(\mathbf{k})] \sigma_x - \text{Im}[u(\mathbf{k})] \sigma_y \\ P_k &= v(\mathbf{k}) \sigma_0 + \text{Re}[u(\mathbf{k})] \sigma_x - \text{Im}[u(\mathbf{k})] \sigma_y \\ \Rightarrow H_k &= w(\mathbf{k}) \tau_z \otimes \sigma_0 + v(\mathbf{k}) i\tau_y \otimes \sigma_0 + \text{Re}[u(\mathbf{k})] (\tau_z + i\tau_y) \otimes \sigma_x - \text{Im}[u(\mathbf{k})] (\tau_z + i\tau_y) \otimes \sigma_y \end{aligned}$$

where $\tau_\alpha, \sigma_\alpha$ $\alpha \in \{0, x, y, z\}$ are the usual Pauli matrices. Since every 4×4 matrix X can be written in the form

$$X = \sum_{\alpha, \beta \in \{0, x, y, z\}} x_{\alpha, \beta} \tau_\alpha \otimes \sigma_\beta$$

one can conclude that the only matrix commuting with H_k is the identity, i.e. $[X, H_k] = 0 \Rightarrow X = \mathbb{1}$. As no two different pauli matrices commute, the occurrence of τ_z and τ_y implies $x_{\alpha,\beta} = 0$ for $\alpha \neq 0$, similarly the occurrence of σ_x and σ_y implies $x_{\alpha,\beta} = 0$ for $\beta \neq 0$. Thus H_k is irreducible as it has no unitary symmetries. This allows to perform a topological classification. Exploiting the chiral symmetry, one can change to a basis where the chiral transformation $\Sigma_x^k = (\mathbb{1}, -\mathbb{1})$. In this basis H_k takes the form

$$H'_k = \begin{pmatrix} 0 & S_k \\ T_k & 0 \end{pmatrix}$$

$$S_k = \begin{pmatrix} w(\mathbf{k}) - v(\mathbf{k}) & 0 \\ 0 & w(\mathbf{k}) - v(\mathbf{k}) \end{pmatrix} \quad T_k = \begin{pmatrix} w(\mathbf{k}) + v(\mathbf{k}) & 2u(\mathbf{k}) \\ 2\bar{u}(\mathbf{k}) & w(\mathbf{k}) + v(\mathbf{k}) \end{pmatrix}$$

As both matrices S_k and T_k are hermitian the Hamiltonian has vanishing winding number, indicating a topological trivial phase. This is confirmed when looking at the eigenstates inside the gap of Fig. 3, when $m\omega^2 = 10 V_{\max}$. I want to mention that if there are no pairing terms, i.e. $\Delta = 0$, the Hamiltonian is block diagonal and therefore not irreducible. In this case the introduction of holes is redundant and the winding number can be calculated from a single block, which can lead to non-vanishing winding numbers. It is thereby shown that the pairing terms prohibit the occurrence of topologically non-trivial phases in 1D for motional degrees of freedom stemming from a second order Taylor expansion of an interaction potential.

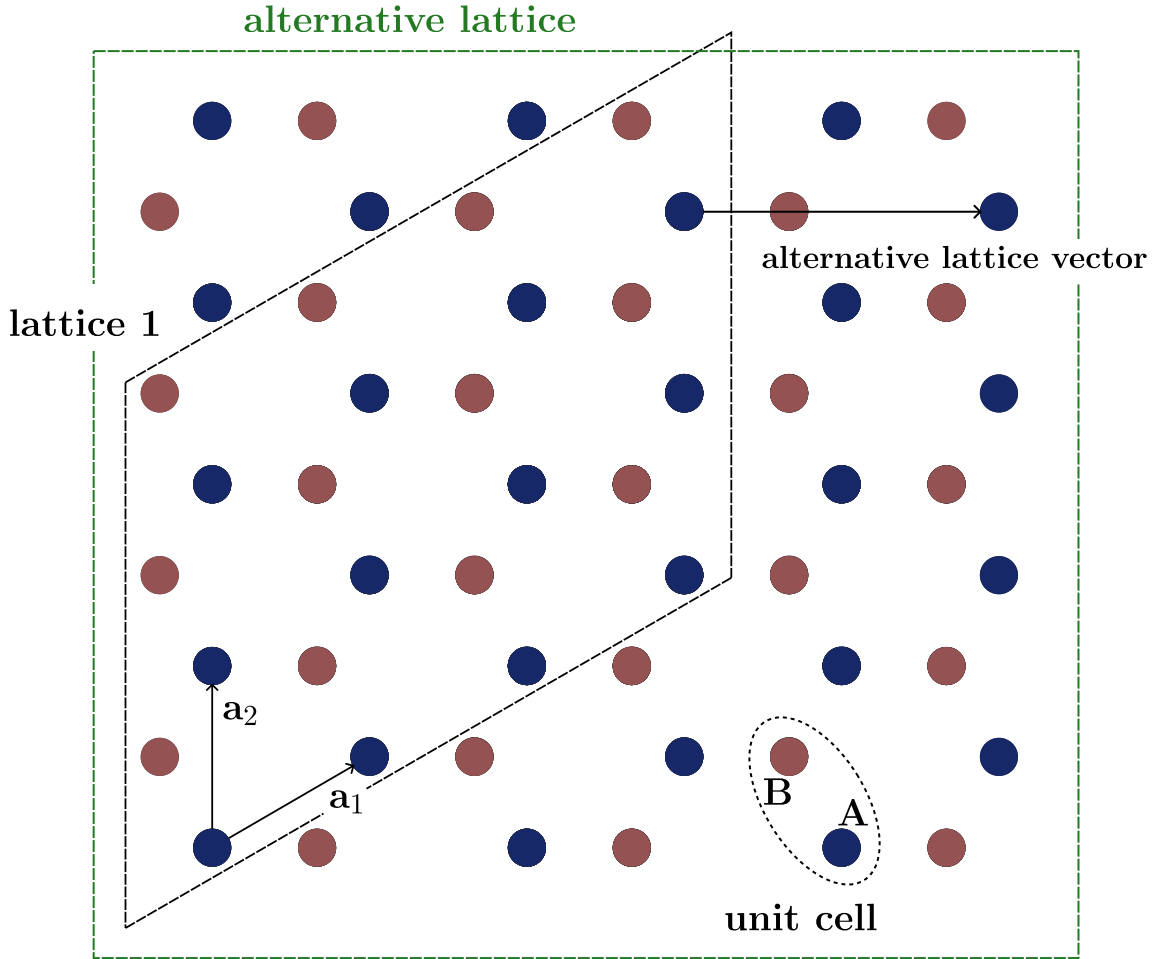


Figure 4: **Lattice definition** The main configuration of the grid, lattice 1, is the parallelogram spanned by the lattice vectors $\mathbf{a}_1$ and $\mathbf{a}_2$. This is visualized by the black dashed frame. There is also the possibility of defining a square lattice which is highlighted by the green dashed frame.

For the case where only one direction of the lattice obeys periodic boundary conditions, while the other direction has open boundaries, the Fourier transform is only performed in the periodic direction.

For this purpose I introduce the lattice vectors $\mathbf{a}_1$ and $\mathbf{a}_2$, which span the lattice. The lattice can be split into subspaces $L_n = \{j \cdot \mathbf{a}_n\}_{j \in M_n}$ with $n \in \{1, 2\}$ and M_n is chosen in such a way, that L_n spans from one edge of the lattice to the other. Hence the position of any unit cell $\mathbf{R}$ can be expressed as the linear combination of two vectors $\mathbf{r}_1 \in L_1$ and $\mathbf{r}_2 \in L_2$, i.e. $\mathbf{R} = \mathbf{r}_1 + \mathbf{r}_2$. I can now also introduce the reciprocal lattice vectors $\mathbf{k}_1$ and $\mathbf{k}_2$ with the property $\mathbf{a}_i \cdot \mathbf{k}_j = 2\pi\delta_{ij}$. I perform the Fourier transformation as an example for the lattice 1 from Fig. 4.

Here the lattice direction represented by $\mathbf{a}_2$ shall be periodic. I introduce the one dimensionally Fourier transformed operators

$$a_{\mathbf{r}_1, \mathbf{k}} = \frac{1}{\sqrt{N}} \sum_{\mathbf{r}_2 \in L_2} e^{-i\mathbf{k}\mathbf{r}_2} a_{\mathbf{r}_1 + \mathbf{r}_2}, \quad \mathbf{k} \in \left\{ \frac{1}{N}, \dots, 1 \right\} \cdot \mathbf{k}_2$$

The Hamiltonian can now be transformed via

$$\begin{aligned} H &= \frac{1}{N} \sum_{\substack{\mathbf{r}_1, \mathbf{r}'_1 \in L_1 \\ \mathbf{r}_2, \mathbf{r}'_2 \in L_2}} \sum_{\substack{a, b \in C \\ \mathbf{k}, \mathbf{k}' \in \tilde{L}}} H^{a,b}(\mathbf{R} - \mathbf{R}') e^{-i\mathbf{k}\mathbf{r}_2} e^{i\mathbf{k}'\mathbf{r}'_2} a_{\mathbf{r}_1, \mathbf{k}}^\dagger b_{\mathbf{r}'_1, \mathbf{k}'} \\ &= \frac{1}{N} \sum_{\substack{\mathbf{r}_1, \mathbf{r}'_1 \\ \mathbf{r}_2, \Delta\mathbf{r}_2 = \mathbf{r}_2 - \mathbf{r}'_2}} \sum_{\substack{a, b \\ \mathbf{k}, \mathbf{k}'}} H^{a,b}(\Delta\mathbf{R}) e^{-i\mathbf{R}(\mathbf{k} - \mathbf{k}')} e^{-i\Delta\mathbf{R}\mathbf{k}'} a_{\mathbf{r}_1, \mathbf{k}}^\dagger b_{\mathbf{r}'_1, \mathbf{k}'} \\ &= \sum_{\substack{\mathbf{r}_1, \mathbf{r}'_1 \\ \Delta\mathbf{r}_2, \mathbf{k} \\ a, b}} H^{a,b}(\Delta\mathbf{R}) e^{-i\Delta\mathbf{r}_2\mathbf{k}} a_{\mathbf{r}_1, \mathbf{k}}^\dagger b_{\mathbf{r}'_1, \mathbf{k}} =: \sum_{\substack{\mathbf{r}_1, \mathbf{r}'_1 \\ \mathbf{k}, a, b}} H^{a,b}(\mathbf{k}, \Delta\mathbf{R}) a_{\mathbf{r}_1, \mathbf{k}}^\dagger b_{\mathbf{r}'_1, \mathbf{k}} \\ \text{where } H^{a,b}(\mathbf{k}, \Delta\mathbf{R}) &= \sum_{\Delta\mathbf{r}_2} H^{a,b}(\Delta\mathbf{R}) e^{-i\Delta\mathbf{r}_2\mathbf{k}} = \sum_{\Delta\mathbf{r}_2} H^{a,b}(\Delta\mathbf{R}) e^{-i\Delta\mathbf{R}\mathbf{k}} \end{aligned}$$

The last equality holds, when $\mathbf{k}$ is in the direction of one of the reciprocal lattice vectors. However, as we perform only a one dimensional Fourier transform, I think a restriction to a purely one dimensional treatment is necessary. Thus $\mathbf{k}_n$ should only be chosen with respect to the condition $\mathbf{k}_n \mathbf{a}_n = 2\pi$, which gives the natural choice $\mathbf{k}_n \parallel \mathbf{a}_n$. For the case of the alternative square lattice I think, we should consider the number of rows twice as big as the number of columns and choose an alternative lattice vector into the x-direction, as depicted in Fig. 4.

What the one dimensional Fourier transform does, is projecting the system onto the one-dimensional chain not of atoms, but of interacting rows or columns of atoms. In this sense the division into sub-directions or sub-bases does not have to be restricted to the lattice vectors. The only thing needed is a labeling of the different rows or columns with the complementary direction having a periodic structure. This allows the division of the alternative square lattice as described above.

Aligned Dipole-Dipole Interactions

In this section the derivative of the dipole interaction will be performed in the case where all dipoles point into the same direction. The subscripts labeling the different positions are dropped in order to make the notation plainer.

$$V_{dd}(\mathbf{r}) = V_0 \left(\frac{1}{r^3} - 3 \frac{(\mathbf{r} \cdot \hat{\mathbf{d}})^2}{r^5} \right), \quad (0.1)$$

with $\mathbf{r} = \mathbf{R}_i - \mathbf{R}_j$ beeing the distance between any to sites i and j ($i \neq j$). The first derivative with respect to component α of $\mathbf{r}$, $\alpha \in \{x, y, z\}$ reads

$$\frac{1}{V_0} \frac{\partial V_{dd}}{\partial r_\alpha} = -\frac{3r_\alpha}{r^5} + \frac{15r_\alpha(\mathbf{r} \cdot \hat{\mathbf{d}})^2}{r^7} - \frac{6(\mathbf{r} \cdot \hat{\mathbf{d}})d_\alpha}{r^5},$$

the second derivative reads consequently

$$\begin{aligned}
\frac{1}{V_0} \frac{\partial^2 V_{dd}}{\partial r_\alpha \partial r_\beta} &= -3 \frac{\delta_{\alpha\beta}}{r^5} + 15 \frac{r_\alpha r_\beta}{r^7} + 15 \frac{\delta_{\alpha\beta} (\mathbf{r} \cdot \hat{\mathbf{d}})^2}{r^7} - 15 \cdot 7 \frac{r_\alpha r_\beta (\mathbf{r} \cdot \hat{\mathbf{d}})^2}{r^9} \\
&\quad + 30 \frac{d_\alpha r_\beta + d_\beta r_\alpha}{r^7} (\mathbf{r} \cdot \hat{\mathbf{d}}) - 6 \frac{d_\alpha d_\beta}{r^5} \\
&= \delta_{\alpha\beta} \left(-3 \frac{1}{r^5} + 15 \frac{(\mathbf{r} \cdot \hat{\mathbf{d}})^2}{r^7} \right) - 6 \frac{d_\alpha d_\beta}{r^5} + 30 \frac{d_\alpha r_\beta + d_\beta r_\alpha}{r^7} (\mathbf{r} \cdot \hat{\mathbf{d}}) \\
&\quad + 15 r_\alpha r_\beta \left(\frac{1}{r^7} - 7 \frac{(\mathbf{r} \cdot \hat{\mathbf{d}})^2}{r^9} \right).
\end{aligned}$$

General Dipole-Dipole Interactions

In the case where not all dipole moments are aligned, the properties of the interaction get extended to

$$V_{dd}(\mathbf{r}) = V_0 \left(\frac{\hat{\mathbf{d}}_1 \cdot \hat{\mathbf{d}}_2}{r^3} - 3 \frac{(\mathbf{r} \cdot \hat{\mathbf{d}}_1)(\mathbf{r} \cdot \hat{\mathbf{d}}_2)}{r^5} \right).$$

The first derivative reads

$$\frac{1}{V_0} \frac{\partial V_{dd}}{\partial r_\alpha} = -\frac{3(\hat{\mathbf{d}}_1 \cdot \hat{\mathbf{d}}_2) r_\alpha}{r^5} + \frac{15 r_\alpha (\mathbf{r} \cdot \hat{\mathbf{d}}_1)(\mathbf{r} \cdot \hat{\mathbf{d}}_2)}{r^7} - \frac{3((\mathbf{r} \cdot \hat{\mathbf{d}}_1) d_{2,\alpha} + (\mathbf{r} \cdot \hat{\mathbf{d}}_2) d_{1,\alpha})}{r^5},$$

the second derivative reads consequently

$$\begin{aligned}
\frac{1}{V_0} \frac{\partial^2 V_{dd}}{\partial r_\alpha \partial r_\beta} &= -3 \frac{\delta_{\alpha\beta} (\hat{\mathbf{d}}_1 \cdot \hat{\mathbf{d}}_2)}{r^5} + 15 \frac{r_\alpha r_\beta (\hat{\mathbf{d}}_1 \cdot \hat{\mathbf{d}}_2)}{r^7} + 15 \frac{\delta_{\alpha\beta} (\mathbf{r} \cdot \hat{\mathbf{d}}_1)(\mathbf{r} \cdot \hat{\mathbf{d}}_2)}{r^7} - 15 \cdot 7 \frac{r_\alpha r_\beta (\mathbf{r} \cdot \hat{\mathbf{d}}_1)(\mathbf{r} \cdot \hat{\mathbf{d}}_2)}{r^9} \\
&\quad + 15 \left[\frac{d_{1,\alpha} r_\beta + d_{1,\beta} r_\alpha}{r^7} (\mathbf{r} \cdot \hat{\mathbf{d}}_2) + \frac{d_{2,\alpha} r_\beta + d_{2,\beta} r_\alpha}{r^7} (\mathbf{r} \cdot \hat{\mathbf{d}}_1) \right] - 3 \frac{d_{1,\alpha} d_{2,\beta} + d_{2,\alpha} d_{1,\beta}}{r^5} \\
&= \delta_{\alpha\beta} \left(-3 \frac{\hat{\mathbf{d}}_1 \cdot \hat{\mathbf{d}}_2}{r^5} + 15 \frac{(\mathbf{r} \cdot \hat{\mathbf{d}}_1)(\mathbf{r} \cdot \hat{\mathbf{d}}_2)}{r^7} \right) - 3 \frac{d_{1,\alpha} d_{2,\beta} + d_{2,\alpha} d_{1,\beta}}{r^5} \\
&\quad + 15 \left[\frac{d_{1,\alpha} r_\beta + d_{1,\beta} r_\alpha}{r^7} (\mathbf{r} \cdot \hat{\mathbf{d}}_2) + \frac{d_{2,\alpha} r_\beta + d_{2,\beta} r_\alpha}{r^7} (\mathbf{r} \cdot \hat{\mathbf{d}}_1) \right] \\
&\quad + 15 r_\alpha r_\beta \left(\frac{\hat{\mathbf{d}}_1 \cdot \hat{\mathbf{d}}_2}{r^7} - 7 \frac{(\mathbf{r} \cdot \hat{\mathbf{d}}_1)(\mathbf{r} \cdot \hat{\mathbf{d}}_2)}{r^9} \right).
\end{aligned}$$

For the the Van der Waals interaction the potential has to be squared. Therefore the second derivative reads

$$\begin{aligned}
\frac{\partial V_{dd}^2}{\partial r_\alpha} &= 2 V_{dd} \frac{\partial V_{dd}}{\partial r_\alpha} \\
\frac{\partial^2 V_{dd}^2}{\partial r_\alpha \partial r_\beta} &= 2 \frac{\partial V_{dd}}{\partial r_\alpha} \frac{\partial V_{dd}}{\partial r_\beta} + 2 V_{dd} \frac{\partial^2 V_{dd}}{\partial r_\alpha \partial r_\beta}
\end{aligned}$$

Winding Number

A chiral Hamiltonian is defined through the property, that it anticommutes with a unitary hermitian operator Γ

$$\Gamma H \Gamma = -H \quad \text{with} \quad \Gamma^2 = 1. \quad (0.2)$$

From this property follows that the Hamiltonian can be written in block off-diagonal form

$$H = \begin{pmatrix} 0 & h \\ h^\dagger & 0 \end{pmatrix}, \quad (0.3)$$

where h is, in general, a non-Hermitian matrix. This structure also applies for the matrix in momentum space and allows the definition of a winding number, which characterizes the topological properties of the system. In the basis of Eq. (0.3) the chiral operator Γ is diagonal and can be written as

$$\Gamma = \begin{pmatrix} \mathbb{1} & 0 \\ 0 & -\mathbb{1} \end{pmatrix}. \quad (0.4)$$

If the Hamiltonian is 2×2 dimensional, the winding number can be defined through the decomposition of H in terms of the Pauli matrices

$$H = \mathbf{n} \cdot \boldsymbol{\sigma}, \quad (0.5)$$

where $\mathbf{n} = (n_x, n_y, 0)$ is a vector in the xy -plane. The winding number is then given by the winding of the vector $\mathbf{n}$ around the origin when traversing the Brillouin zone. Identifying the xy -plane of $\mathbf{n}$ with the complex plane, which defines a path $\mathbf{n}(k) \rightarrow \gamma(k)$, the winding number can be calculated through

$$w = \frac{1}{2\pi i} \oint_{\gamma_k} \frac{1}{z} dz = \frac{1}{2\pi i} \int_{\text{BZ}} dk \frac{1}{\gamma(k)} \frac{d\gamma(k)}{dk}.$$

In terms of the Hamiltonian above this reads

$$w = \frac{1}{2\pi i} \int_{\text{BZ}} dk \frac{h'(k)}{h(k)}.$$

Generalizing the winding number to higher dimensional Hamiltonians is possible through the use of the block off-diagonal form. The winding number then reads [8, 9]

$$w = \frac{1}{2\pi i} \int_{\text{BZ}} dk \text{Tr} [h^{-1}(k) \partial_k h(k)].$$

Diagonalizing h as $h = S \Lambda S^{-1}$ with the diagonal matrix $\Lambda = \text{diag}(\lambda_1, \lambda_2, \dots)$ and the invertible matrix S , the trace can be rewritten as

$$\begin{aligned} \text{Tr} [h^{-1} \partial_k h] &= \text{Tr} [S \Lambda^{-1} S^{-1} \partial_k (S \Lambda S^{-1})] \\ &= \text{Tr} [\Lambda^{-1} \partial_k \Lambda + S^{-1} \partial_k S + (\partial_k S^{-1}) S] \\ \text{Tr} [S^{-1} \partial_k S + S \partial_k S^{-1}] &= \text{Tr} [\partial_k (S S^{-1})] = 0, \\ \Rightarrow \text{Tr} [h^{-1} \partial_k h] &= \text{Tr} [\Lambda^{-1} \partial_k \Lambda] = \sum_i \frac{\partial_k \lambda_i}{\lambda_i} = \partial_k \left(\sum_i \ln(\lambda_i) \right) = \partial_k \ln(\det[h]). \end{aligned}$$

Another useful definition of the winding number is through the so-called Q -matrix, which is defined through the projectors onto the eigenstates of the Hamiltonian. As the Hamiltonian is chiral, the eigenvalues come in pairs $\pm E$. We introduce a labeling of the eigenvalues, such that $E_j = -E_{-j}$ with $j = 1, 2, \dots, N$ and N being half the dimension of the Hamiltonian. The corresponding eigenstates are labeled accordingly ($|\psi_j\rangle$) and posses the property $\Gamma |\psi_j\rangle = |\psi_{-j}\rangle$. The Q -matrix is defined as

$$Q = \sum_{j=1}^N (|\psi_j\rangle \langle \psi_j| - |\psi_{-j}\rangle \langle \psi_{-j}|).$$

As a consequence of the chiral symmetry, the Q -matrix anticommutes with Γ and can therefore be written in block off-diagonal form in the basis, where Γ is diagonal, which is also the case in which H is block off-diagonal.

$$\begin{aligned} \Gamma Q \Gamma &= \sum_{j=1}^N (|\psi_{-j}\rangle \langle \psi_{-j}| - |\psi_j\rangle \langle \psi_j|) = -Q \\ \Rightarrow Q &= \begin{pmatrix} 0 & q \\ q^\dagger & 0 \end{pmatrix} \end{aligned}$$

The fact that the winding number of the q -matrix is the same as that of h can be shown with the help of the following two properties.

$$\begin{aligned} Q^2 = 1 & \Rightarrow qq^\dagger = \mathbb{1}, \quad q^\dagger q = \mathbb{1} \\ QHQ = H & \Rightarrow qh^\dagger q = h^\dagger \end{aligned}$$

From the second property follows that $q^\dagger h = h^\dagger q$. Thus $h^\dagger q$ is a hermitian matrix and therefore has no winding number, since the eigenvalues are real. Hence

$$\begin{aligned} 0 &= \text{Tr} [q^\dagger (h^\dagger)^{-1} \partial_k (h^\dagger q)] \\ &= \text{Tr} [q q^\dagger (h^\dagger)^{-1} \partial_k h^\dagger + h^\dagger (h^\dagger)^{-1} q^\dagger \partial_k q] \\ \Rightarrow \text{Tr} [q^{-1} \partial_k q] &= -\text{Tr} [(h^\dagger)^{-1} \partial_k h^\dagger] \\ &= \text{Tr} [h^{-1} \partial_k h] \end{aligned}$$

There is another useful property, for which we consider the basis in which $\Gamma = \sigma_z \otimes \mathbb{1}_N$. This is the same diagonal basis as previously used. In this basis we can represent all eigenvalues of H as a stack of N -component vectors. The eigenvector for negative energies is written as

$$|\psi_i^-\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} |h_{1i}\rangle \\ |h_{2i}\rangle \end{pmatrix} \quad i \in \{1, \dots, N\}$$

Because of the chirality of the Hamiltonian the eigenvector with positive energy is already determined by this choice and reads

$$|\psi_i^+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} |h_{1i}\rangle \\ -|h_{2i}\rangle \end{pmatrix}$$

The matrix Q is represented in the following form

$$\begin{aligned} Q &= \sum_i \begin{pmatrix} 0 & |h_{1i}\rangle \langle h_{2i}| \\ |h_{2i}\rangle \langle h_{1i}| & 0 \end{pmatrix} \\ \Rightarrow q &= \sum_i |h_{1i}\rangle \langle h_{2i}| \end{aligned}$$

If we now compute the kernel of the winding number with respect to q we get

$$\begin{aligned} \text{Tr}[q^\dagger \partial_k q] &= \text{Tr} \left[\sum_{i,j} |h_{2i}\rangle \langle h_{1i}| \partial_k (|h_{1j}\rangle \langle h_{2j}|) \right] \\ &= \sum_{i,j} \text{Tr} [|h_{2i}\rangle \langle h_{1i}| (\partial_k |h_{1j}\rangle) \langle h_{2j}|] + \sum_{i,j} \text{Tr} [|h_{2i}\rangle \langle h_{1i}| |h_{1j}\rangle (\partial_k \langle h_{2j}|)] \\ &= \sum_i \langle h_{1i}| (\partial_k |h_{1i}\rangle) + \sum_i (\partial_k \langle h_{2i}|) |h_{2i}\rangle \end{aligned}$$

When computing the winding number the second term contributes with a minus sign compared to its complex conjugated self, i.e.

$$\int_{\text{BZ}} dk (\partial_k \langle h_{2i}|) |h_{2i}\rangle = - \int_{\text{BZ}} dk \langle h_{2i}| (\partial_k |h_{2i}\rangle)$$

Thus the winding number reads

$$\begin{aligned} w &= \frac{1}{2\pi i} \int_{\text{BZ}} dk \text{Tr}[q^\dagger \partial_k q] \\ &= \frac{1}{2\pi i} \sum_i \int_{\text{BZ}} dk (\langle h_{1i}| \partial_k |h_{1i}\rangle - \langle h_{2i}| \partial_k |h_{2i}\rangle) \end{aligned}$$

This can be expressed with respect to the full eigenvectors of the Hamiltonian via

$$\begin{aligned} w &= \frac{1}{\pi i} \sum_i \int_{\text{BZ}} dk \langle \psi_i^- | \partial_k | \psi_i^- \rangle \\ &= \frac{1}{\pi i} \sum_i \int_{\text{BZ}} dk \langle \psi_i^+ | \partial_k | \psi_i^- \rangle \\ &= \frac{1}{2\pi i} \sum_i \int_{\text{BZ}} dk (\langle h_{1i}| \partial_k |h_{1i}\rangle - \langle h_{2i}| \partial_k |h_{2i}\rangle) \end{aligned}$$

The quantity $\sum_i \langle \Gamma \psi_i^- | \partial_k | \psi_i^- \rangle$ can be related to a polarisation [10].

Time-reversal symmetry

Time reversal symmetry is represented by an anti-unitary operator T . The effect of time-reversal on an operator is the following.

$$\hat{O} \rightarrow T \hat{O} T^{-1},$$

with $T^{-1} = T^\dagger$, as the norm of vectors has to be conserved. T can be decomposed into two operators $T = \tau K$, where K is the complex conjugation operator acting on everything to the right. T can satisfy one of conditions

$$T^2 = \pm 1$$

For spin-less particles in real space usually $\tau = \text{id}$ and $T^2 = 1$, whereas for particles with spin $\tau = -i\sigma_y$ can be chosen in real space, which fulfills $T^2 = -1$.

Take a look at the momentum space Hamiltonian without spin degrees of freedom

$$H = \sum_{\mathbf{k}} H(\mathbf{k}) a_{\mathbf{k}}^\dagger a_{\mathbf{k}}$$

$$\text{with } a_{\mathbf{k}} = \frac{1}{\sqrt{N^d}} \sum_{\mathbf{R}} e^{-i\mathbf{k}\mathbf{R}} a_{\mathbf{R}}$$

Time-reversal transforms the operators according to

$$T a_{\mathbf{R}} T^{-1} = a_{\mathbf{R}}$$

$$\Rightarrow T a_{\mathbf{k}} T^{-1} = a_{-\mathbf{k}}$$

Symmetry regarding time-reversal ($H = T H T^{-1}$) has thus the form

$$H = T H T^{-1}$$

$$= \sum_{\mathbf{k}} H^*(\mathbf{k}) a_{-\mathbf{k}}^\dagger a_{-\mathbf{k}}$$

$$= \sum_{\mathbf{k}} H^*(-\mathbf{k}) a_{\mathbf{k}}^\dagger a_{\mathbf{k}}$$

$$\iff H(\mathbf{k}) = H^*(-\mathbf{k})$$

Short Answer

The statement is in the book, but as an asserted result with references rather than a proof—and the step from that result to “states decay exponentially away from the boundary” is left to the reader. Both gaps are easy to close, and the full proof of the key estimate is in one of the book’s own references.

0.1 What the Book Actually Says

1. **The claim, unproven, in §2.2.1 (p. 27):** After introducing the half-space Hamiltonian $\hat{H}$ on $\mathbb{C}^N \otimes \ell^2(\mathbb{Z}^{d-1} \times \mathbb{N})$, the authors write that when $\text{Ch}_d \neq 0$, the spectrum of $\hat{H}$ extends into the bulk gap, and that

“the electron states corresponding to the spectrum inside the bulk insulating gap are exponentially localized near the boundary, hence the terminology boundary states and boundary spectrum.”

This is a physics statement with a pointer to §2.4.3 for an example—no argument attached.

2. **The quantitative result, §2.4.3, Proposition 2.4.10 (p. 52):** This is the mathematical content you want. With $\hat{H} = \Pi_d H \Pi_d^* + \tilde{H}$ as in (2.55)–(2.56) (Dirichlet restriction plus a boundary Hamiltonian whose hoppings are cut off at distance R from the surface), and assuming the Bulk Gap Hypothesis (BGH) for the bulk component H :

For any smooth φ with support in the bulk insulating gap,

$$\sup_{\omega} |\langle x, n | \varphi(\hat{H}_{\omega}) | y, m \rangle| \leq A_M (1 + |x - y|)^{-M} e^{-\beta(n+m)}, \quad n, m \in \mathbb{N}, \quad x, y \in \mathbb{Z}^{d-1},$$

with M any integer and $A_M, \beta > 0$.

Here n, m are the coordinates normal to the boundary. Note the asymmetry: exponential decay perpendicular to the surface, only polynomial (but arbitrarily fast) decay parallel to it—which is exactly right, since the boundary states must stay extended along the edge.

The book gives no proof, only the remark that “these estimates are by now standard with proofs based on the functional calculus introduced by Dynkin [57], often also referred to as the Helffer–Sjöstrand formula [89]”, citing Elbau–Graf [58], Schulz–Baldes [194], and Prodan [166].

3. **The bulk ingredient, Proposition 2.4.2 (p. 47):** Under BGH, the Fermi projection satisfies:

$$\sup_{\omega} |\langle x | \chi(H_{\omega} \leq \mu) | y \rangle| \leq \gamma e^{-\beta|x-y|},$$

“by a well-known Combes–Thomas estimate”. This is the bulk analogue; the same Combes–Thomas input is what drives Proposition 2.4.10.

—

0.2 The Missing Step: From Prop. 2.4.10 to “States Decay Exponentially”

This one line is not in the book. Let Δ' be a closed interval strictly inside the gap Δ , let $\hat{P} = \chi_{\Delta'}(\hat{H}_{\omega})$, and pick $\varphi \in C^{\infty}$ with $\varphi \equiv 1$ on Δ' and $\text{supp } \varphi \subset \Delta$. Then $\varphi(\hat{H})\hat{P} = \hat{P}$, so any $\psi \in \text{Ran } \hat{P}$ satisfies $\psi = \varphi(\hat{H})\psi$ and

$$|\psi(x, n)| = |\langle x, n | \varphi(\hat{H}) | \psi \rangle| \leq \|\psi\| \cdot \left(\sum_{y, m} |\langle x, n | \varphi(\hat{H}) | y, m \rangle|^2 \right)^{1/2}.$$

Insert Proposition 2.4.10 with $2M > d - 1$: the m -sum of $e^{-2\beta m}$ converges, the y -sum of $(1 + |x - y|)^{-2M}$ converges, and both are independent of x and n . Hence

$$|\psi(x, n)| \leq C e^{-\beta n} \|\psi\|, \quad \text{uniformly in } \omega \text{ and in } x.$$

So every state in the spectral subspace of $\hat{H}$ associated with energies in the gap decays exponentially in the direction normal to the boundary, with a rate independent of the disorder configuration.

Taking $x = y$ and $n = m$ in Proposition 2.4.10 gives the same conclusion for the local density of states (LDOS) in the gap: it is bounded by $A e^{-2\beta n}$, which is the version that survives the fact that for a homogeneous edge, the boundary spectrum is typically continuous—there are then no normalizable eigenstates at all, and “exponentially localized at the boundary” must be phrased as a statement about spectral projections / the LDOS rather than about eigenvectors. That is presumably why the book states it in kernel form.

Two Caveats Implicit in the Hypotheses:

- β degrades as $\text{supp } \varphi$ approaches the gap edges (the estimate requires a positive distance δ to $\sigma(H)$), so nothing is claimed for states at the band edges.
- Localization is strictly transverse.

—

0.3 Why Proposition 2.4.10 Is True—and Where the Proof Is

The proof is in reference [166]:

E. Prodan, *The edge spectrum of Chern insulators with rough boundaries*, J. Math. Phys. **50**, 083517 (2009) ([arXiv:0809.2569](https://arxiv.org/abs/0809.2569)), Proposition 2 and Appendix 6.2.

It is for $d = 2$ with an arbitrarily rough boundary, but the argument is dimension-independent and is precisely the source of the book's statement. The mechanism, in the same notation:

1. **Combes–Thomas for the bulk resolvent (his Prop. 1, App. 6.1):** Conjugating with the boost $U_q |n\rangle = e^{-q \cdot n} |n\rangle$ gives $H_q = H_0 + W_q$ with $\|W_q\| = \zeta(q) \rightarrow 0$ as $q \rightarrow 0$, hence

$$|\langle n, \alpha | (H_0 - z)^{-1} | n', \beta \rangle| \leq \frac{e^{-q|n-n'|}}{\text{dist}(z, \sigma(H_0)) - \zeta(q)}$$

for q small enough that the denominator is positive. This is the book's [53]/Prop. 2.4.2 input, made explicit for exactly the tight-binding class of (2.12).

2. **The boundary is a finite-range perturbation:** Writing $H = H_0 - \Delta V$, where ΔV collects the hoppings cut by the boundary (and, in the book's setting, also $\tilde{H}$ from (2.56)), ΔV is supported in a strip of finite width around the surface. Then the twice-iterated resolvent identity

$$R(z) = R_0(z) + R_0(z) \Delta V R_0(z) + R_0(z) \Delta V R(z) \Delta V R_0(z)$$

is used, and since the support of φ lies in the gap of H_0 , the purely- R_0 terms are analytic there and drop out of the functional calculus. What survives has one bulk resolvent running from x to the strip and another from the strip to y —and by step 1, each contributes $e^{-q \cdot (\text{distance to boundary})}$, giving $e^{-q(n_1+n'_1)}$. The identity expresses the simple principle that exponential decay of the bulk resolvent makes $R(z) - R_0(z)$ exponentially localized near the edge, because ΔV is strictly localized there.

3. **Handling the boundary spectrum singularity:** The technical subtlety is that the middle factor $R(\lambda \pm i\varepsilon)$ is evaluated on the boundary spectrum, so it blows up as $\varepsilon \rightarrow 0$. Prodan handles this by sandwiching it between the normalized vectors $\Delta V |m, \alpha\rangle / \|\cdot\|$ and reading the $\varepsilon \rightarrow 0$ limit as an integral against a spectral probability measure, so only $\sup_\lambda$ of the (bounded) remaining factors is needed (his Eqs. 70–79).
4. **The parallel direction:** The parallel direction is treated separately (App. 6.2(ii)) with an almost-analytic extension / Helffer–Sjöstrand representation:

$$\varphi(H) = \frac{1}{2\pi} \int \bar{\partial} f_N(H - z)^{-1} d^2 z.$$

Because $\zeta(q) \sim q$ for small q , one may take $q \approx |\text{Im } z|/M$, and integrating $v^{N-1} e^{-v|n-n'|/M}$ yields decay $(1 + |n - n'|)^{-N}$ for every N . Multiplying the two estimates gives exactly the mixed form of the book's Prop. 2.4.10.

Other Treatments of the Same Estimate:

- P. Elbau, G.-M. Graf, *Equality of bulk and edge Hall conductance revisited*, Commun. Math. Phys. **229**, 415–432 (2002) (the book's [58]);
- H. Schulz-Baldes, *Persistence of spin edge currents in disordered quantum spin Hall systems*, Commun. Math. Phys. **324**, 589–600 (2013) ([194]);
- For the underlying estimate itself: J.-M. Combes, L. Thomas, Commun. Math. Phys. **34**, 251–270 (1973).
- M. Aizenman, S. Warzel, *Random Operators* (AMS 2015), Ch. 10, has a clean textbook Combes–Thomas proof.

0.4 Two Other Places the Book Makes the Point Without the General Estimate

1. **The explicit periodic model, §2.2.4 (pp. 33–35):** For the Dirac-type model restricted to $x_d \geq 0$, the authors state that

“for $\hat{E}_k$ inside the bulk insulating gap, the solutions to the Schrödinger equation must be sought in the form $\psi_k(x) = \xi_k \otimes (\lambda_k)^x$, $|\lambda_k| < 1$,”

and then derive $\lambda_k = -(m + \sum_j \cos k_j)$ together with $(i\gamma_d + \gamma_0)\xi_k = 0$; the edge band exists precisely where the constraint $|\lambda_k| < 1$ holds, cf. (2.28).

This is the transfer-matrix picture: an energy in the gap admits no real transverse momentum, only complex ones, so the only admissible solutions are evanescent, with localization length $1/|\log |\lambda_k||$. It is an argument, but only for this model and only for periodic, Dirichlet-truncated systems. (Same style of argument in the chiral model, §2.3.3, p. 43 ff.; and §1.2, p. 4, has the degenerate case $m = 0$ where the zero modes are supported exactly on the boundary site.)

2. **The C^* -algebraic shadow of the statement, §4.3.1 (pp. 100–101):** In the exact sequence $0 \rightarrow \mathcal{E}_d \rightarrow \hat{\mathcal{A}}_d \rightarrow \mathcal{A}_d \rightarrow 0$, the evaluation ev is a C^* -homomorphism and therefore commutes with continuous functional calculus (this is used explicitly in the proof of Prop. 4.3.1).

Hence for φ supported in the gap, $\langle (\cdot) \varphi(\hat{h}) \rangle = \varphi(\langle (\cdot) \hat{h} \rangle) = \varphi(h) = 0$, i.e., $\varphi(\hat{h}) \in \mathcal{E}_d \cong \mathcal{A}_{d-1} \otimes \mathcal{K}$ by Prop. 3.2.4(i)—“functions of $\hat{H}$ supported in the bulk gap are boundary observables”, compact in the transverse direction. The authors invoke exactly this when they note that $\tilde{u}_\Delta - 1_N$ lies in the smooth boundary algebra “according to Proposition 2.4.10”. This is the qualitative, structural version; Prop. 2.4.10 is what upgrades compactness to an explicit exponential rate.

Summary

The result you want is **Proposition 2.4.10**, the deduction to states is the Cauchy–Schwarz line shown in Section 2, and the proof of 2.4.10 itself you will find in Prodan (2009), App. 6.2—not in the book.

1D analysis of topological states

In order to have a chiral Hamiltonian I will choose the orientation of the dipoles in a way, such that the chirality remains. This will be done for each lattice configuration separately, i.e. the dipoles depend on the lattice geometry.

In order to make the achievement of chirality more feasible I only consider nearest neighbour interactions. When talking of chirality the chirality of the matrix U , defined above, is in mind. Recall that

$$U_{n,m} = -\frac{\partial^2 V_{n,m}(\mathbf{r}_{nm})}{\partial \mathbf{r}_{nm}^2} \quad \text{if } n \neq m$$

$$U_{n,n} = \sum_{j; j \neq n} \frac{\partial^2 V_{n,j}(\mathbf{r}_{nj})}{\partial \mathbf{r}_{nj}^2}$$

Chirality implies that all couplings between sites on the same sublattice vanish. The nearest neighbour restriction already assures that couplings between different sites of the same sublattice vanish. Left is thus only the onsite potential due to the Taylor expansion of the dipole–dipole potential. The aim is now to choose the orientation of the dipoles in such a way that the sum in $U_{n,n}$ is equal to zero.

Be aware that so far I didn’t write explicitly the dependence of U from the direction of atom movement x, y . Hence $U_{n,m}, U_{n,n}$ are themselves two times two matrixes $(U_{n,m}^{\alpha,\beta})_{\alpha,\beta \in \{x,y\}}$ and the Hamiltonian expressed explicitly with respect to the x, y -phonon modes reads

$$H = \sum_{n,\alpha} \omega a_{n,\alpha}^\dagger a_{n,\alpha} + \frac{2}{m\omega} \sum_{n,m,\alpha,\beta} U_{n,m}^{\alpha,\beta} a_{n,\alpha}^\dagger a_{m,\beta}$$

The terms with $a_{n,\alpha}^\dagger a_{n,\alpha}$ can thus be eliminated by redefining the trap frequency.

$$\omega_\alpha = \omega + \frac{2}{m\omega} U_{n,n}^{\alpha,\alpha}$$

The terms coupling the two directions of phonon modes on the same side can not be internalized by another quantity. Here the degrees of freedom related to the dipole orientation are exploited to tune the $U_{n,n}^{x,y}$ values to zero, as mentioned above. Because of the nearest neighbour nature of the consideration only two distances play a role, the intracell distance $\mathbf{r}_I$ and the intercell distance $\mathbf{r}_O$.

$$U_{n,n}^{x,y} = \frac{\partial^2 V(\mathbf{r}_I)}{\partial r_{I,x} \partial r_{I,y}} + \frac{\partial^2 V(\mathbf{r}_O)}{\partial r_{O,x} \partial r_{O,y}}$$

The dipoles are then chosen in such a way that this sum vanishes. For open boundary conditions the values of $U_{n,n}^{\alpha,\beta}$ at the borders are different from the bulk values. In the case of $\alpha = \beta$ the difference can be compensated by changed trap frequencies at the boundary. For the xy coupling this is not possible and the interaction due to both neighboring atoms has to made zero individually such that x - and y -modes are completely decoupled from each other.

The topologically non-trivial phase actually breaks down, when those border-terms are not eliminated. This is confirmed by simulations. It might be, at first, counter intuitive that reducing the effect of an open border is decremental to the observation of edge states. It makes, however perfect sense, when remembering that those boundary terms in our case destroy the chiral symmetry. It seems to be necessary for the bulk-boundary condition to hold that not only the momentum space Hamiltonian with periodic boundary conditions has to satisfy chiral symmetry but also the real space Hamiltonian with open boundary conditions. Otherwise one could choose the Hamiltonian describing the border in a completely arbitrary manner and still observe edge states, what would be quite a strong statement.

What one observes is that, if the same direction edge coupling is kept in, no separated energies in the bulk gaps arise. At the other hand if only the boundary terms, coupling the two phonon directions, remain and $U_{n,n}^{\alpha,\alpha}$ is completely absorbed into the trap frequencies separated energies inside the bulk energy gaps can be observed. These separated energies, however, are not located at zero and converge for the atom number going to infinity to a non-zero value, what makes them non-topological.

Aligned dipoles in plane

I consider a geometry with two sublattices repeating itself in the x -axis and separated by the distance h from each other into the y -axis. Different configurations of this geometry are realized by moving the sublattices against each other in the x -direction. I consider an even amount of lattice, with N atoms in each sublattice. The n th atom of each sublattice A and B takes the position.

$$\begin{aligned} \mathbf{r}_n^A &= n \cdot a \mathbf{e}_x \\ \mathbf{r}_n^B &= n \cdot a \mathbf{e}_x + b \mathbf{e}_x + h \mathbf{e}_y, \end{aligned}$$

where a is the lattice constant, and the parameters b and h can be chosen to realize different lattice configurations with the restriction that b and h have to be chosen in a way, such that the nearest neighbor restriction makes sense. For any b -value it is possible to find h and the dipole moment $\mathbf{d}$ in a way that fulfils the above conditions.

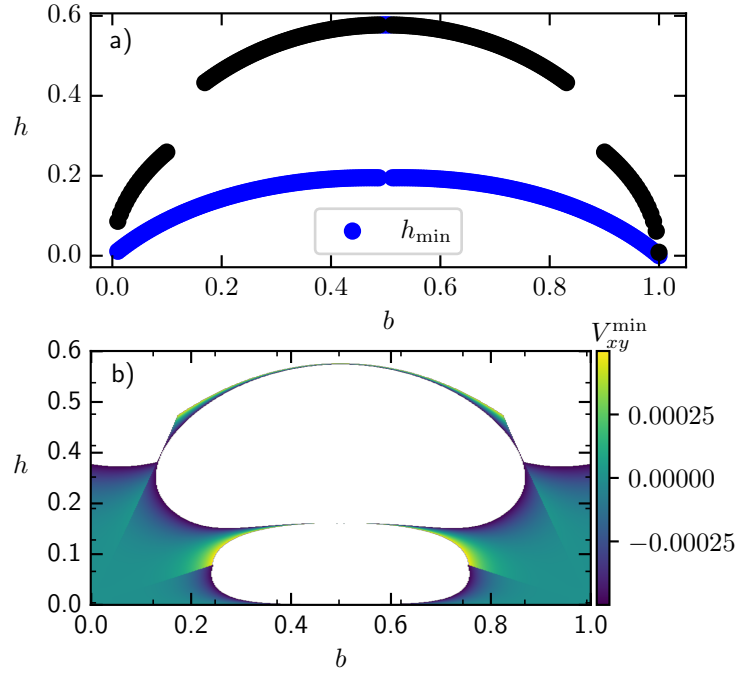


Figure 5: **Optimal h for chiral symmetry.** In a) The h values are depicted which decouple x - and y -phonons. Those values were determined numerically by computing roots of $V_{xy}(h)$. A grid of h values as initial guesses for the root algorithm was taken to have a better chance of finding all roots. In b) $V_{xy}(b, h)$ is drawn for reference. Larger values of the function have been omitted to make the pattern of small coupling values more visible.

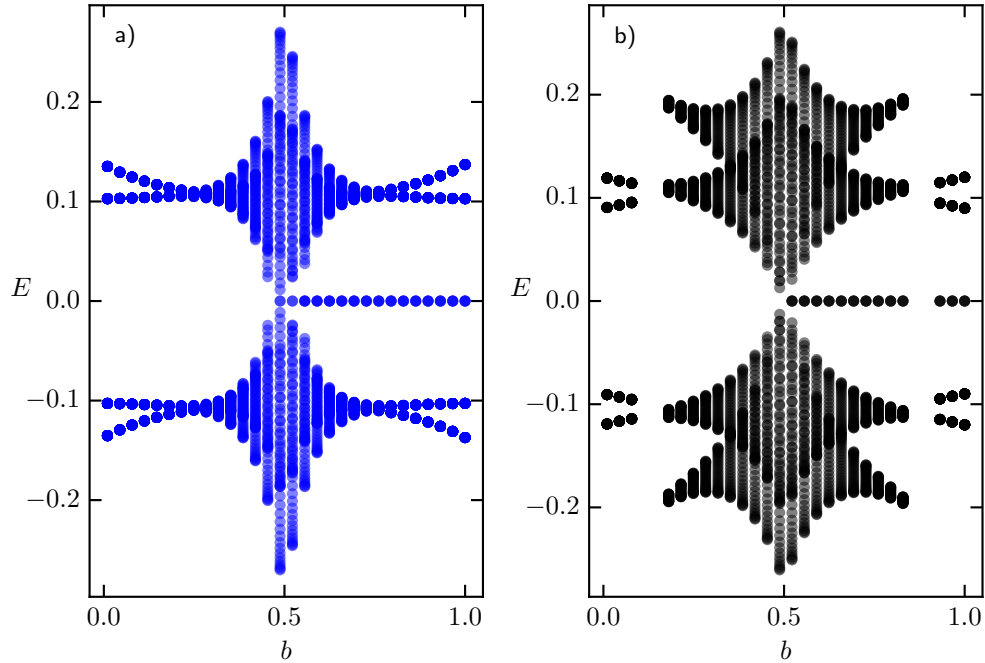


Figure 6: **Energy spectrum for the chiral Hamiltonian.** The diagram was created with the parameter $a = 1$. Blue and black colors depict the Energy spectrum for the h values picked from the blue and black curves in Fig. 5 respectively.

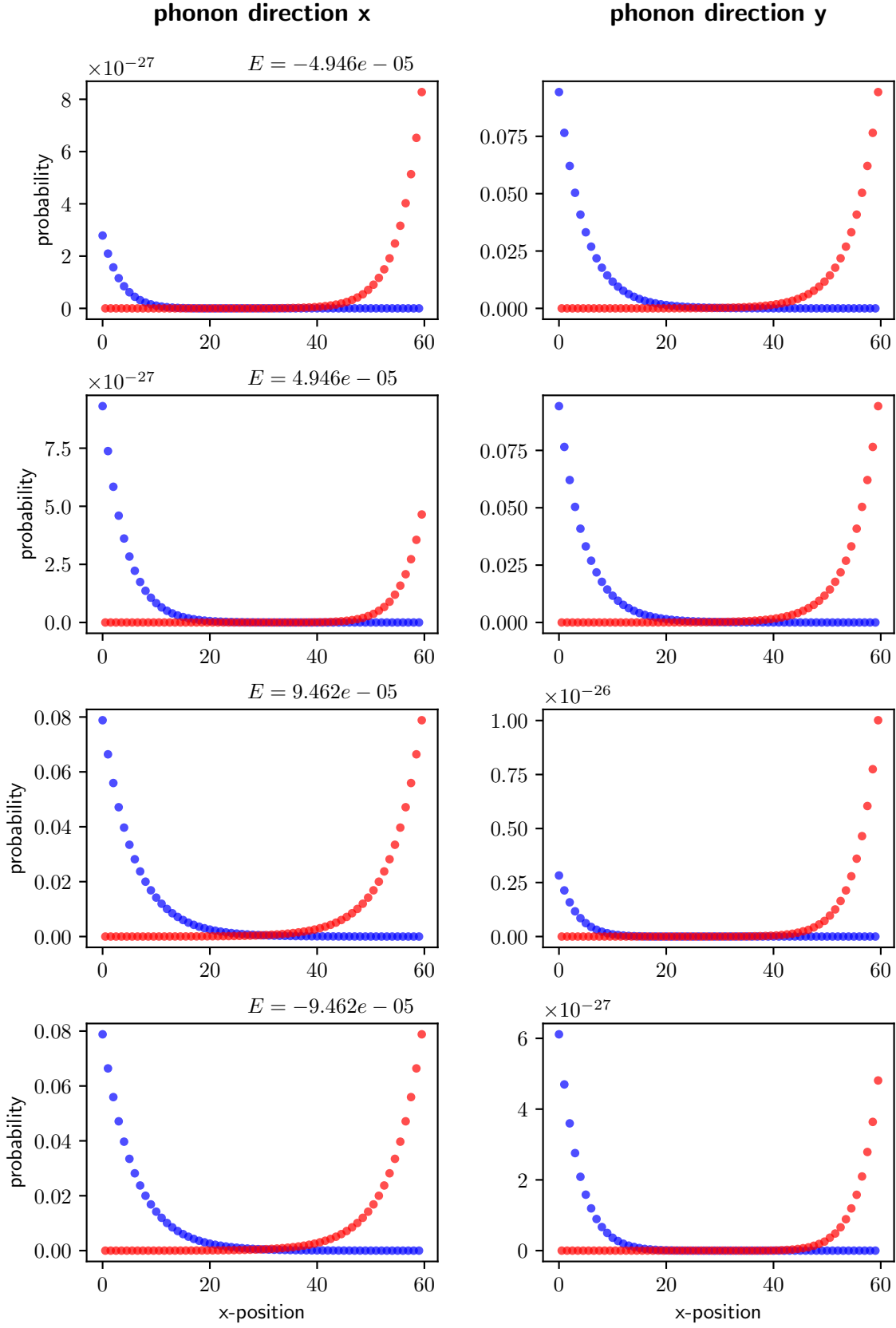


Figure 7: **Edge states.** The figure shows the decay of edge states with the four energies with the smallest absolute value. Each row depicts one energy state. Drawn in blue are the excitations of sublattice A, while the excitation of sublattice B is shown in red.

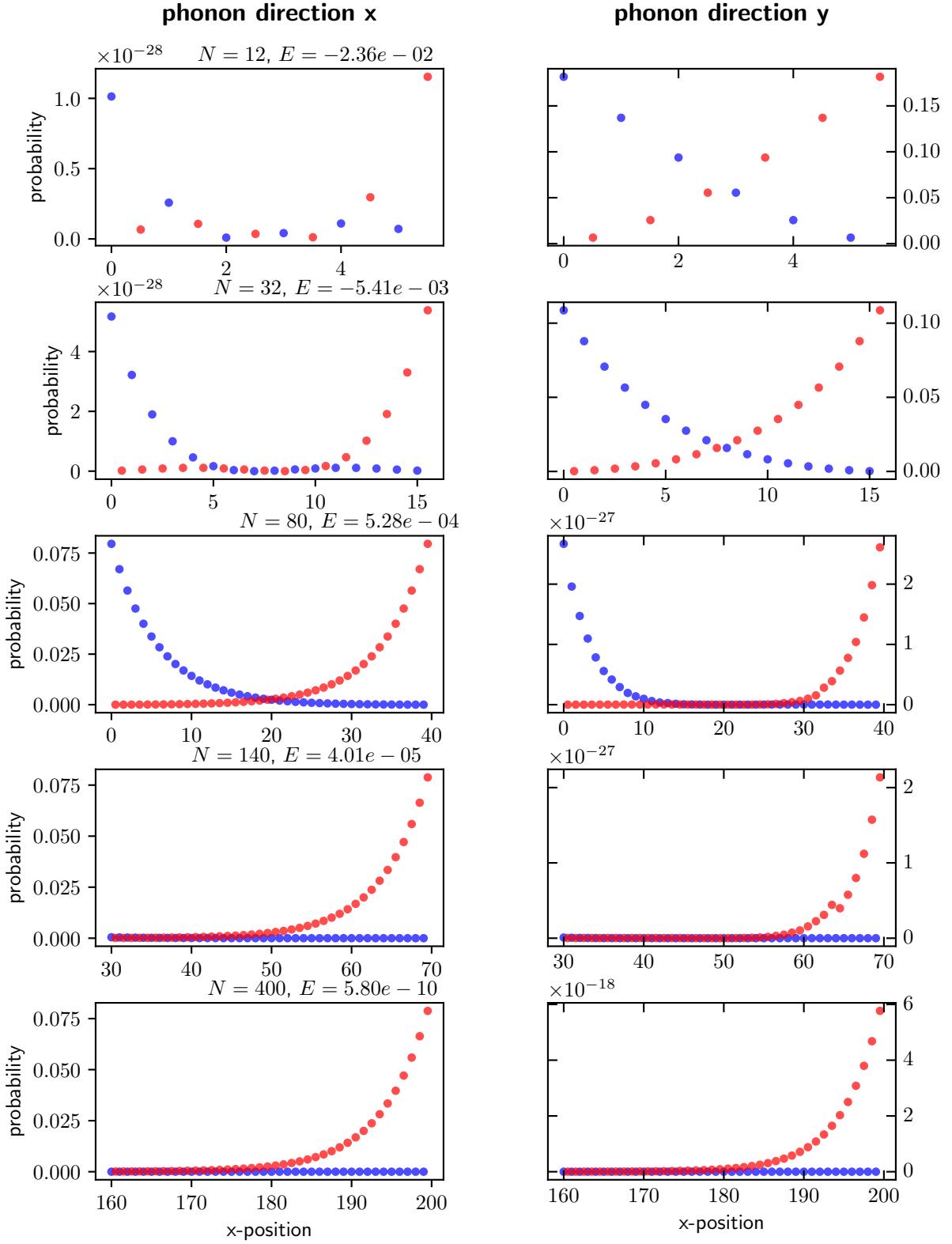
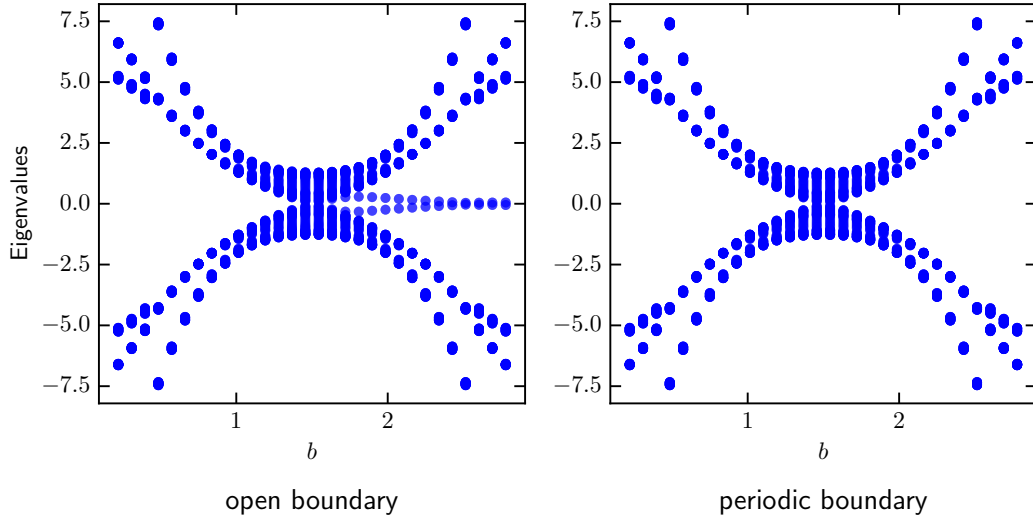


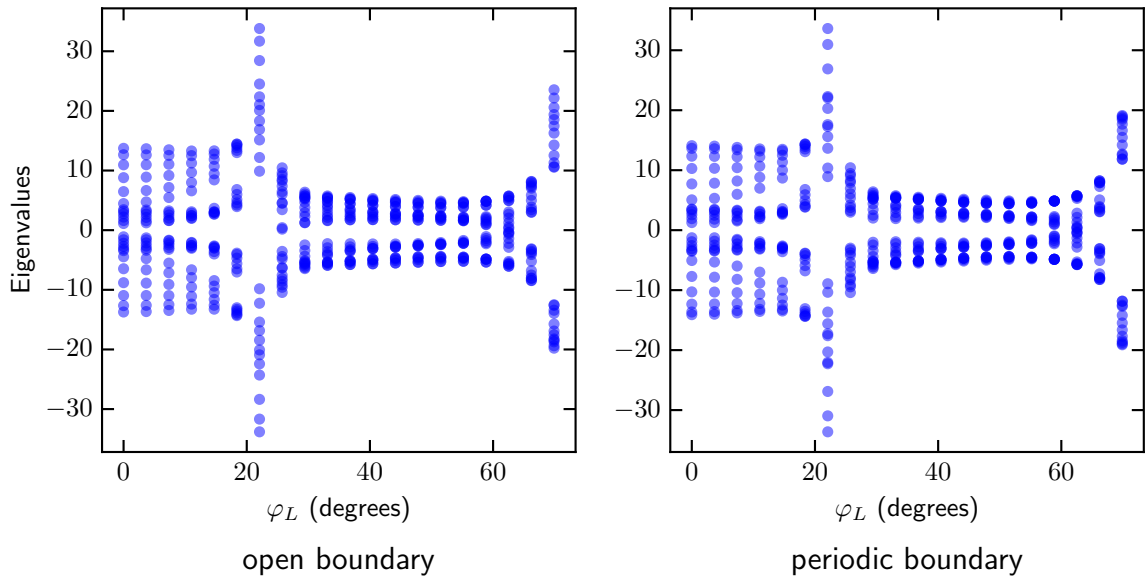
Figure 8: Decay of energy for topological states with atom number.. In each row the eigenstate for the energy with smallest absolute value is plotted. Blue and red show again the excitations of sublattice A and B respectively. When moving down in the rows of the figure the atom number of the system is increased. It shows that with increasing atom number the energy of the topological decays. What also can be observed. That the decay-length of the excitations when moving away from the edges does not change with particle number.

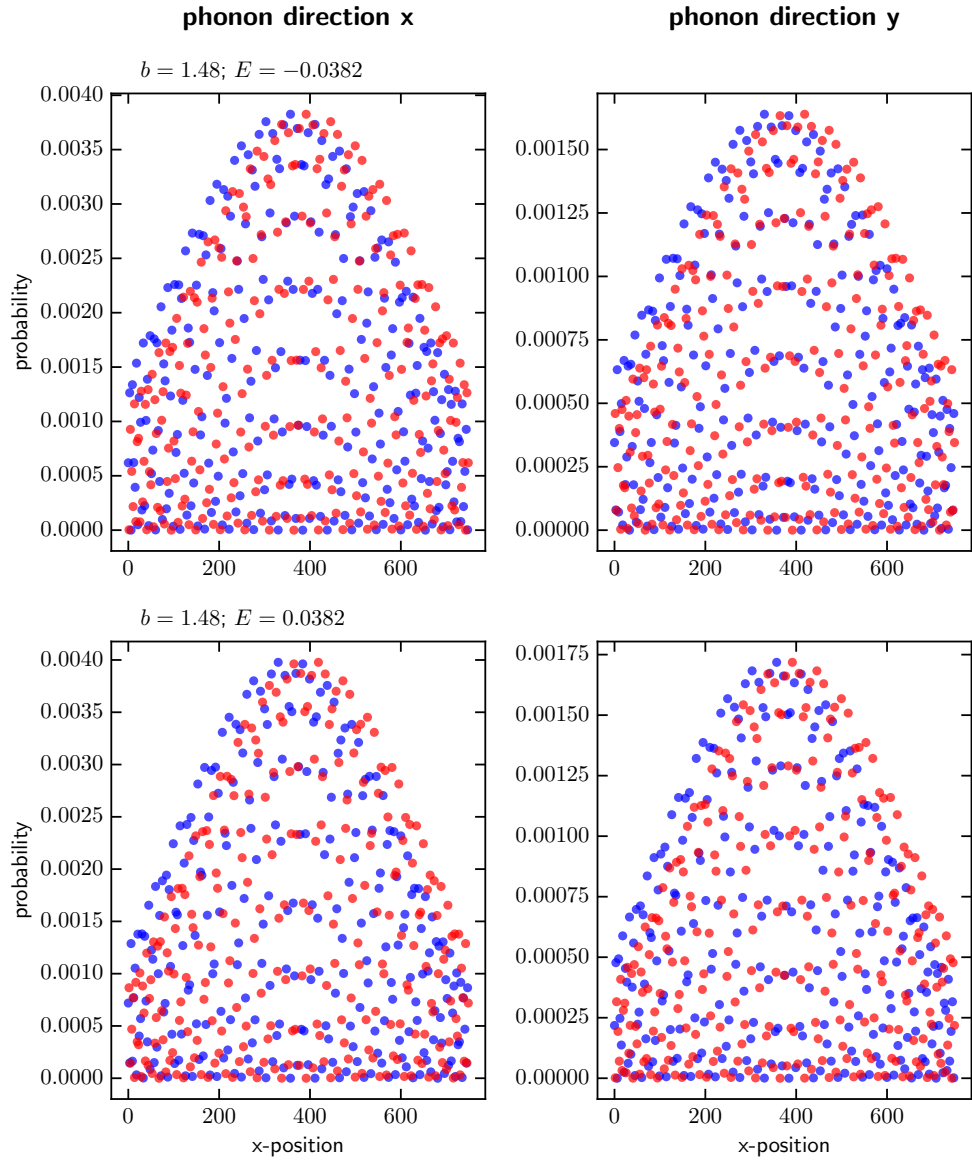


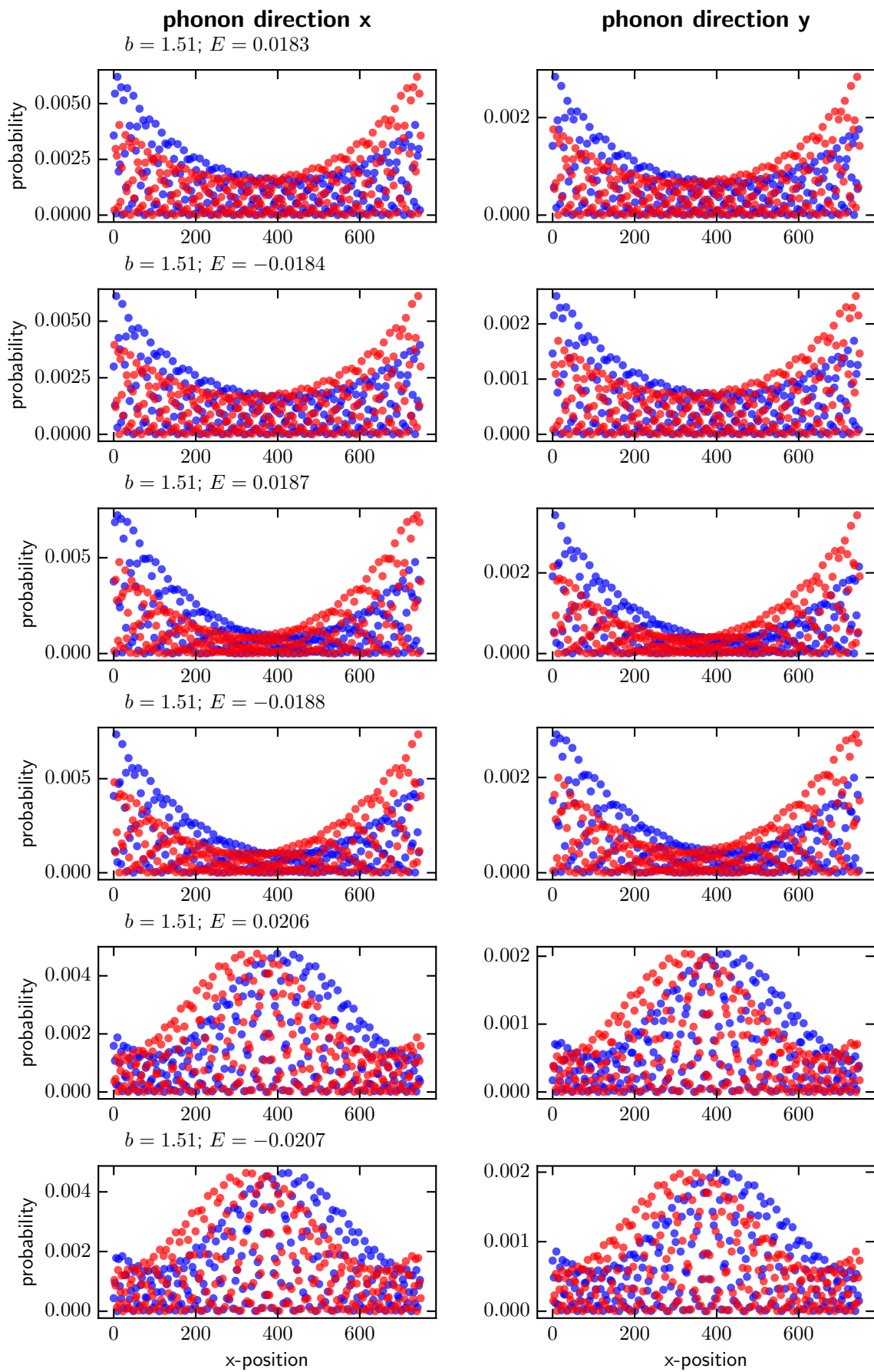
Out of plane dipoles

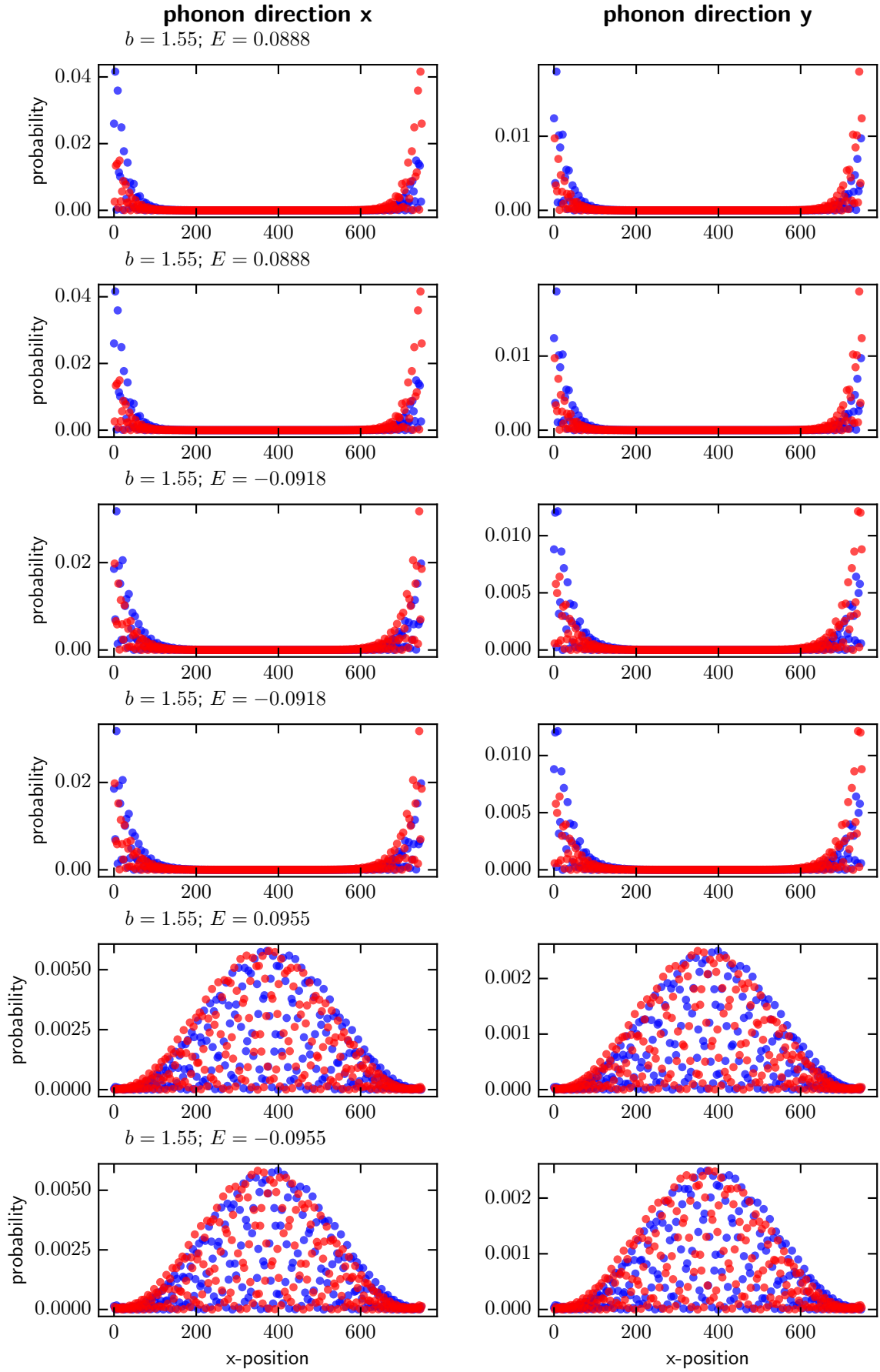
First I will allow the dipoles to point out of the xy -plane in which the one dimensional chain will reside. I restrict the dipoles to the following form in order to have easier calculations.

$$\mathbf{d}_1 = \begin{pmatrix} d_{1,x} \\ 0 \\ \sqrt{-d_{1,x}^2} \end{pmatrix}, \quad \mathbf{d}_2 = \begin{pmatrix} 0 \\ d_{2,y} \\ \sqrt{-d_{2,y}^2} \end{pmatrix}$$









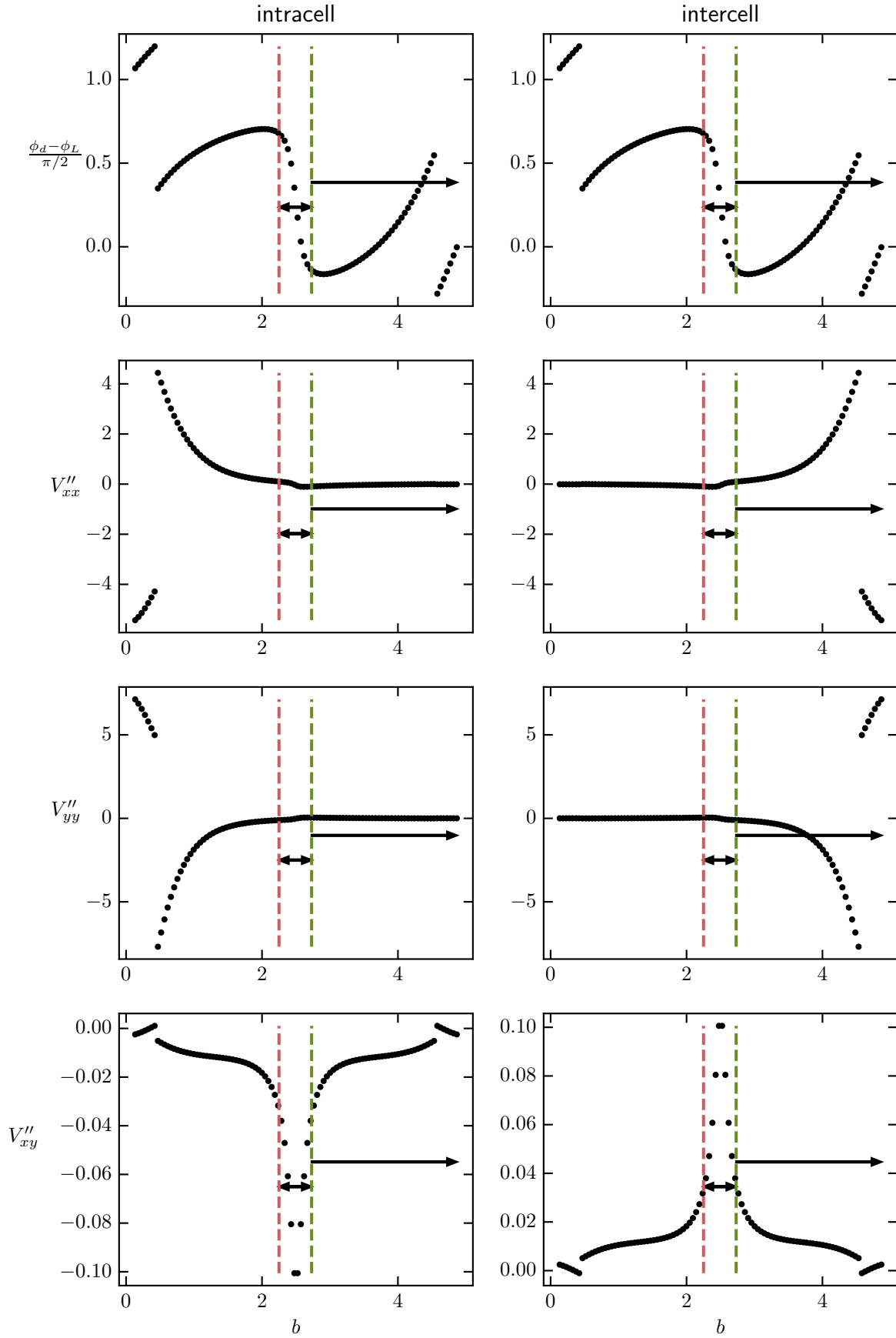
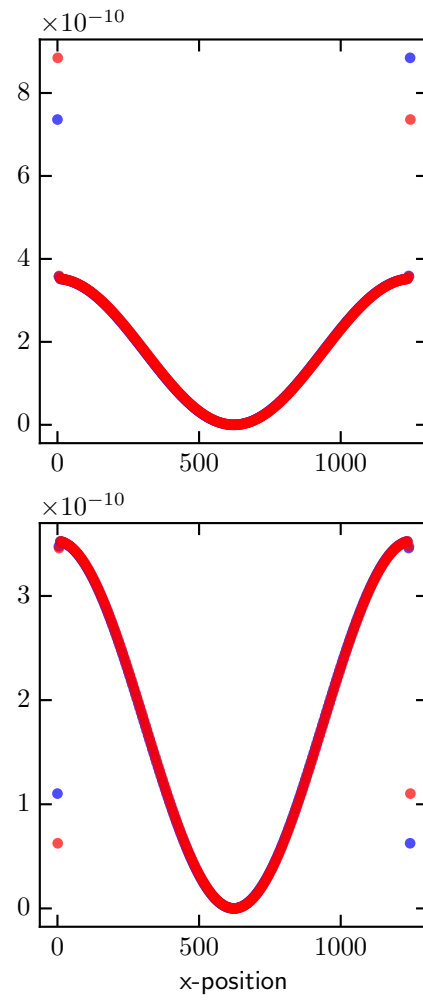
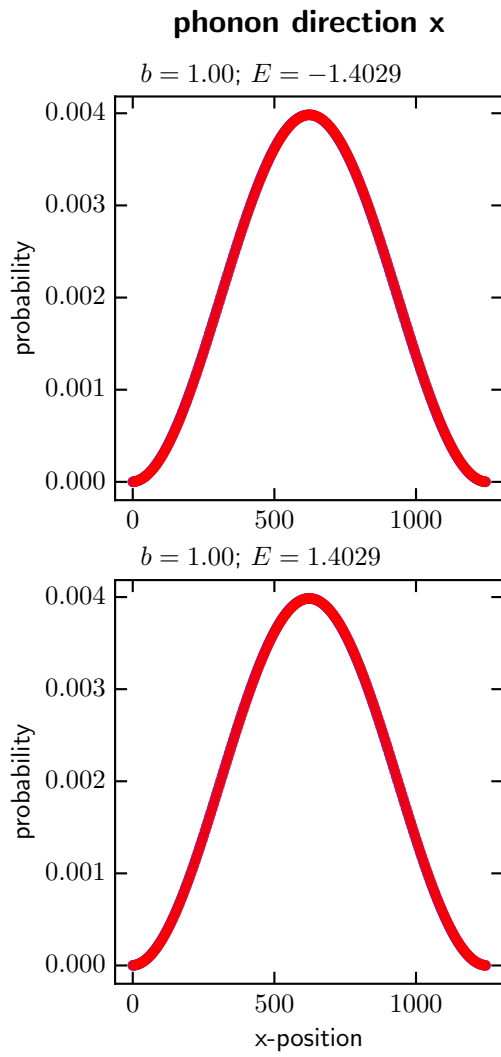
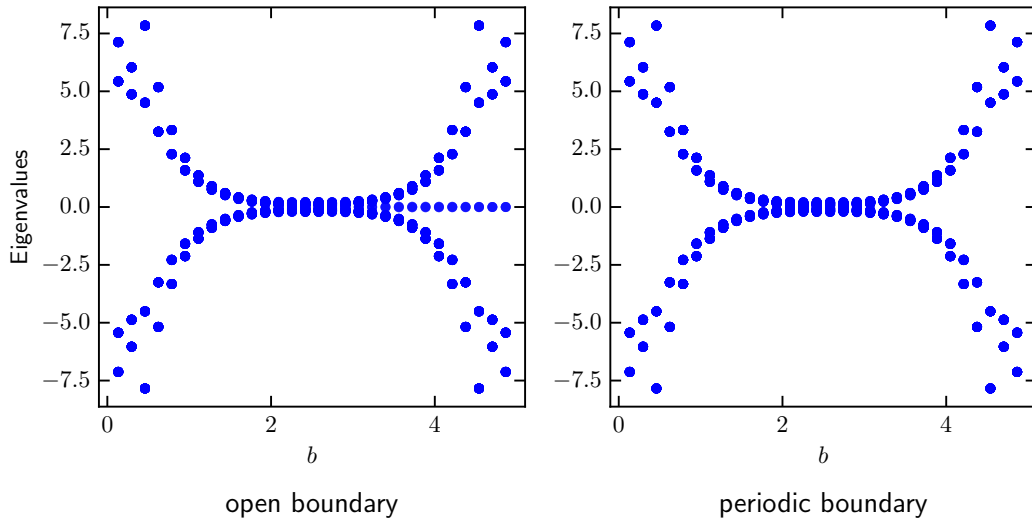
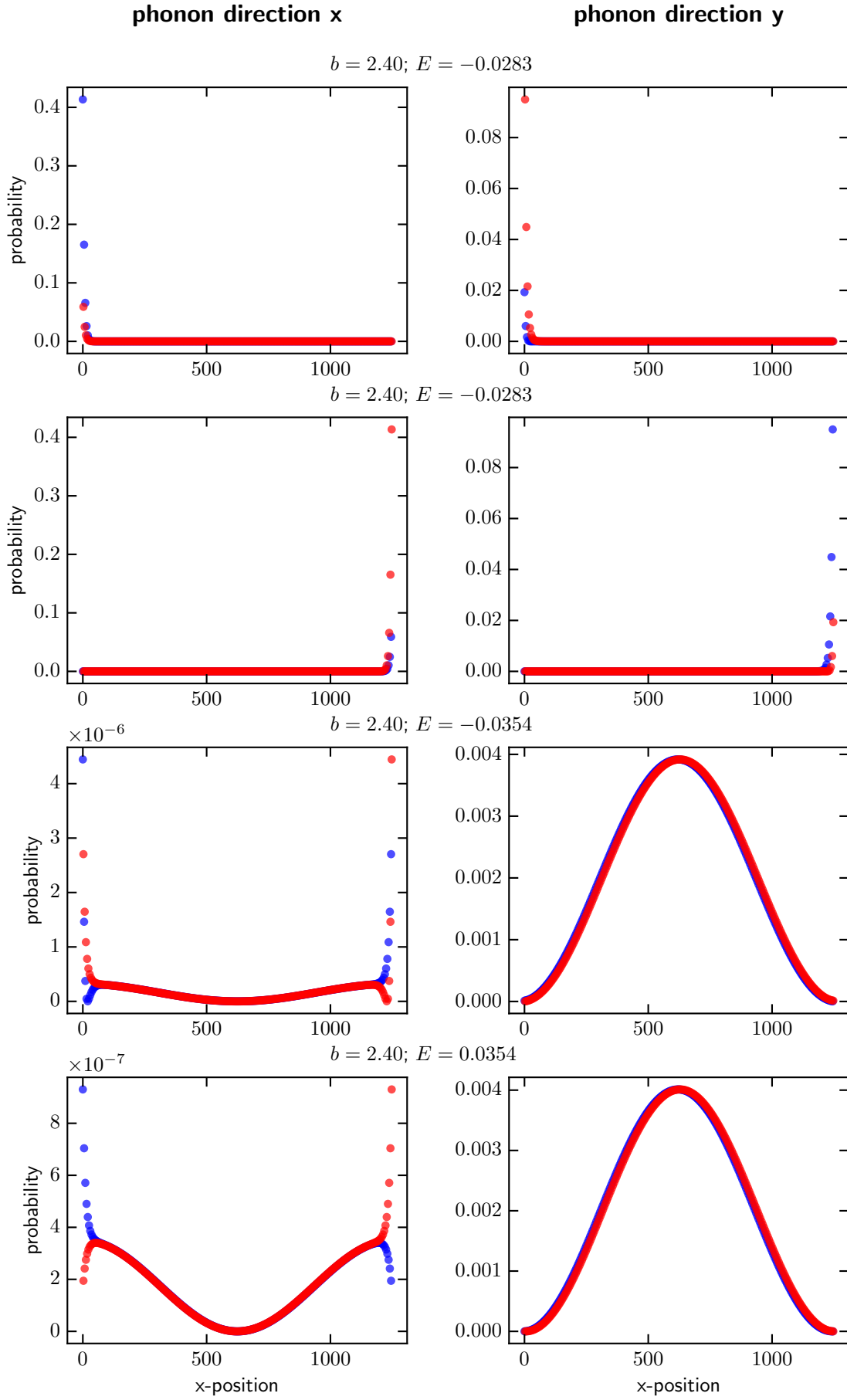
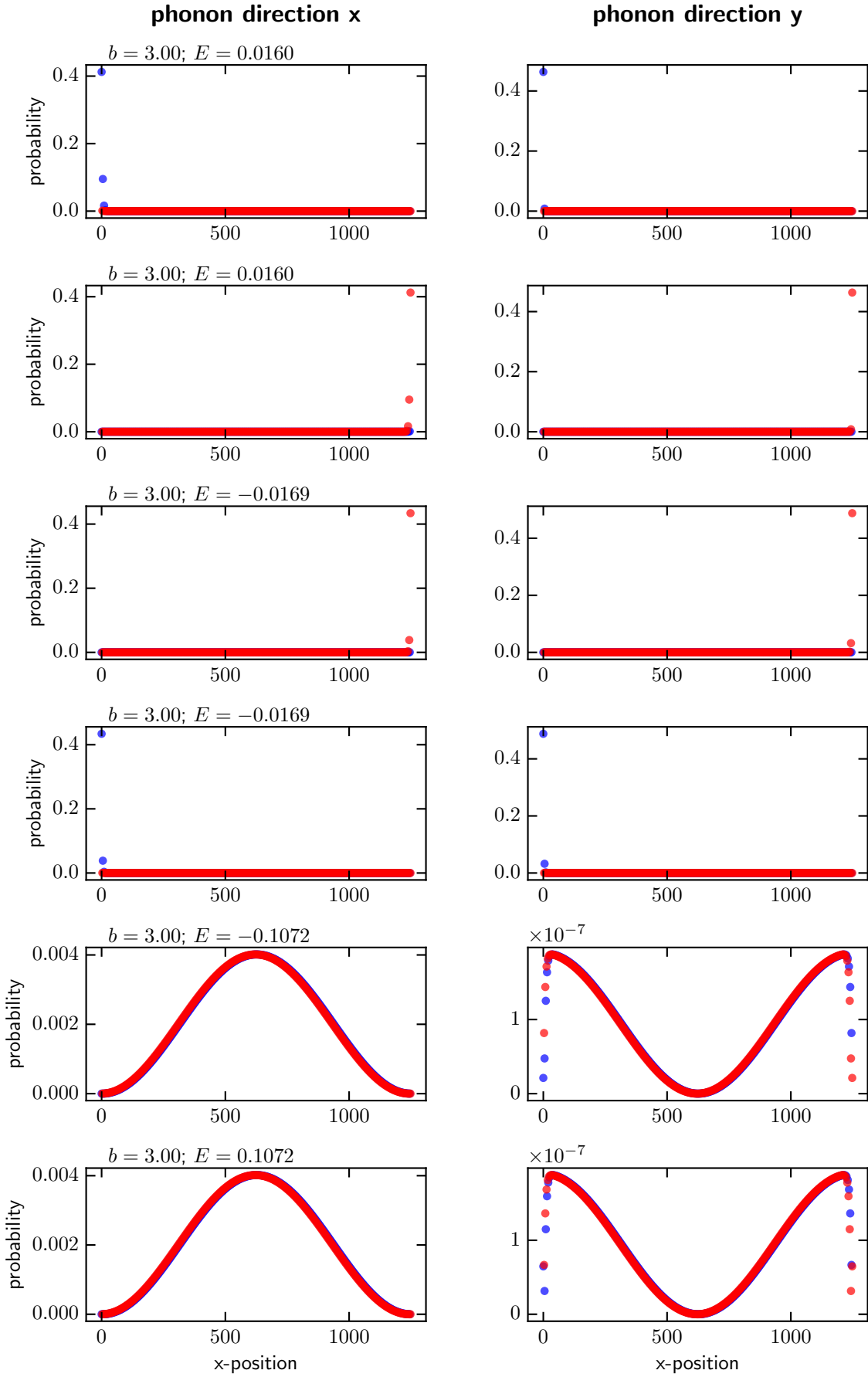


Figure 9: The diagram was created with the parameters $a = 5$, $h = 1.1$. Between the red and the olive dashed line, i.e. between b -values of 2.25-2.73 the winding number is 1, whereas right to the olive dashed line the winding number is 2.







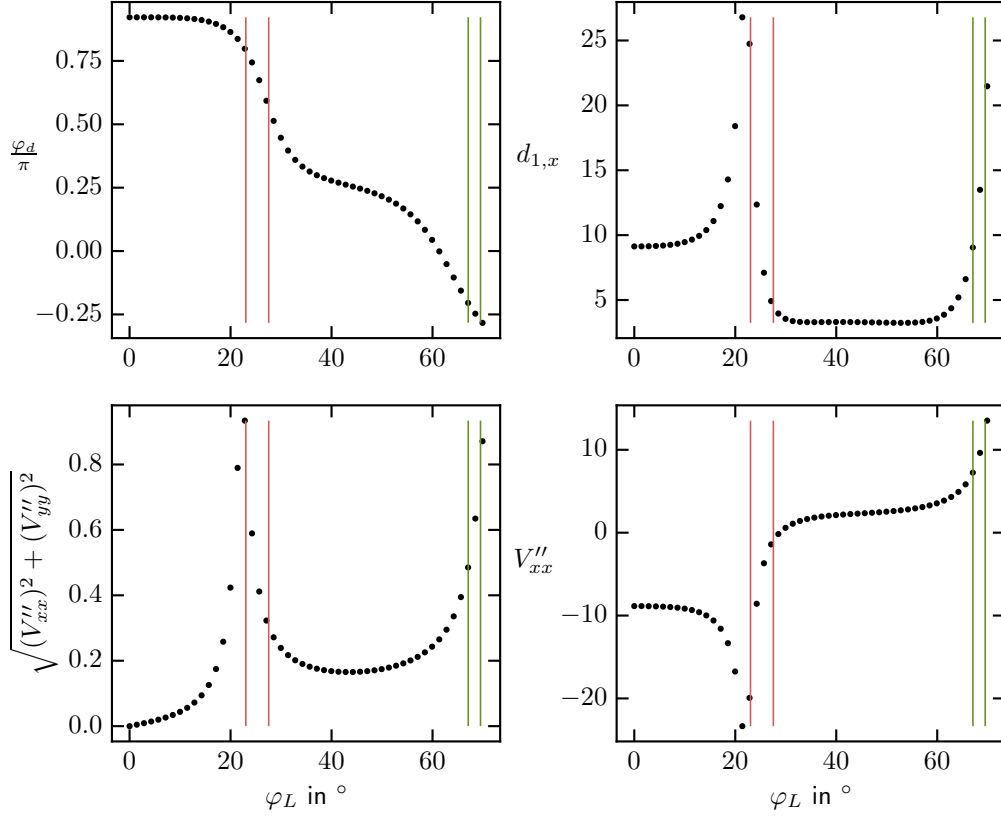


Figure 10: The occurrence of topological states is signaled by the vertical lines. Between the two red as well as between the two green lines topological states can be found.

As chain I consider a zig-zag chain with sublattice vector $\mathbf{a}_S$ and lattice vector $\mathbf{a}_L$.

$$\mathbf{a}_S = \begin{pmatrix} c \\ \epsilon \end{pmatrix}, \quad \mathbf{a}_L = \begin{pmatrix} c+1 \\ 0 \end{pmatrix}$$

For nearest neighbour interaction the condition that lets the momentum space Hamiltonian be chiral under periodic boundary conditions reads

$$0 \stackrel{!}{=} V''_{xy}(\mathbf{a}_S) + V''_{xy}(\mathbf{a}_S - \mathbf{a}_L)$$

$$V''_{xy}(\mathbf{r}_0) := \frac{\partial^2 V}{\partial r_x \partial r_y}(\mathbf{r}_0)$$

With the particular dipole configuration, introduced above, the derived interaction takes the form

$$V''_{xy}(\mathbf{r}) = d_{1,x} d_{2,y} \left(\frac{12}{r^5} - 105 \frac{r_x^2 r_y^2}{r^9} \right) + 15 \sqrt{1 - d_{1,x}^2} \sqrt{1 - d_{2,y}^2} \frac{r_x r_y}{r^7}$$

$$V''_{\alpha\alpha}(\mathbf{r}) = d_{1,x} d_{2,y} \left(45 \frac{r_x r_y}{r^7} - 105 r_\alpha^2 \frac{r_x r_y}{r^9} \right) + \sqrt{1 - d_{1,x}^2} \sqrt{1 - d_{2,y}^2} \left(-\frac{3}{r^5} + 15 \frac{r_\alpha^2}{r^7} \right)$$

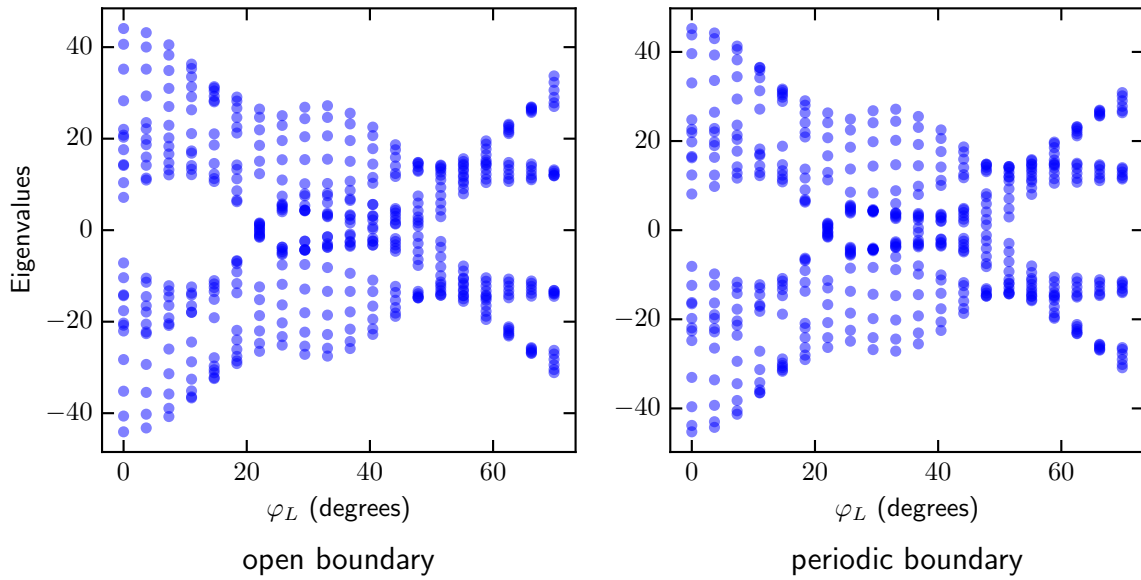
In plane dipole orientation

As a next step I restrict the dipoles to the xy -plane separate the magnitude of the dipole projected onto the plane from the occurrence of topological states. In this scenario I investigate the properties of the dipole restriction of the following form.

$$\mathbf{d}_1 = \begin{pmatrix} d_{1,x} \\ d_{1,y} \\ 0 \end{pmatrix}, \quad \mathbf{d}_2 = \begin{pmatrix} d_2 \\ 0 \\ 0 \end{pmatrix}$$

For this configuration the interaction takes to From

$$\begin{aligned}
 V''_{xy}(\mathbf{r}) &= d_{1,x}d_2 \left(45 \frac{r_x r_y}{r^7} - 105 \frac{r_x^3 r_y}{r^9} \right) + d_{1,y}d_2 \left(\frac{12}{r^5} - 105 \frac{r_x^2 r_y^2}{r^9} \right) \\
 V''_{\alpha\alpha}(\mathbf{r}) &= d_{1,x}d_2 \left[-\frac{3 + \delta_{\alpha,x}}{r^5} + \frac{15}{r^7} (r_x^2 + 2r_\alpha r_x + r_\alpha^2 + 2\delta_{\alpha,x} r_x^2) - 105 \frac{r_x^2 r_\alpha^2}{r^9} \right] \\
 &\quad + d_{1,y}d_2 \left[\frac{15}{r^7} (r_x r_y + 2\delta_{\alpha,x} r_x r_y) - 105 \frac{r_\alpha^2 r_x r_y}{r^9} \right]
 \end{aligned}$$



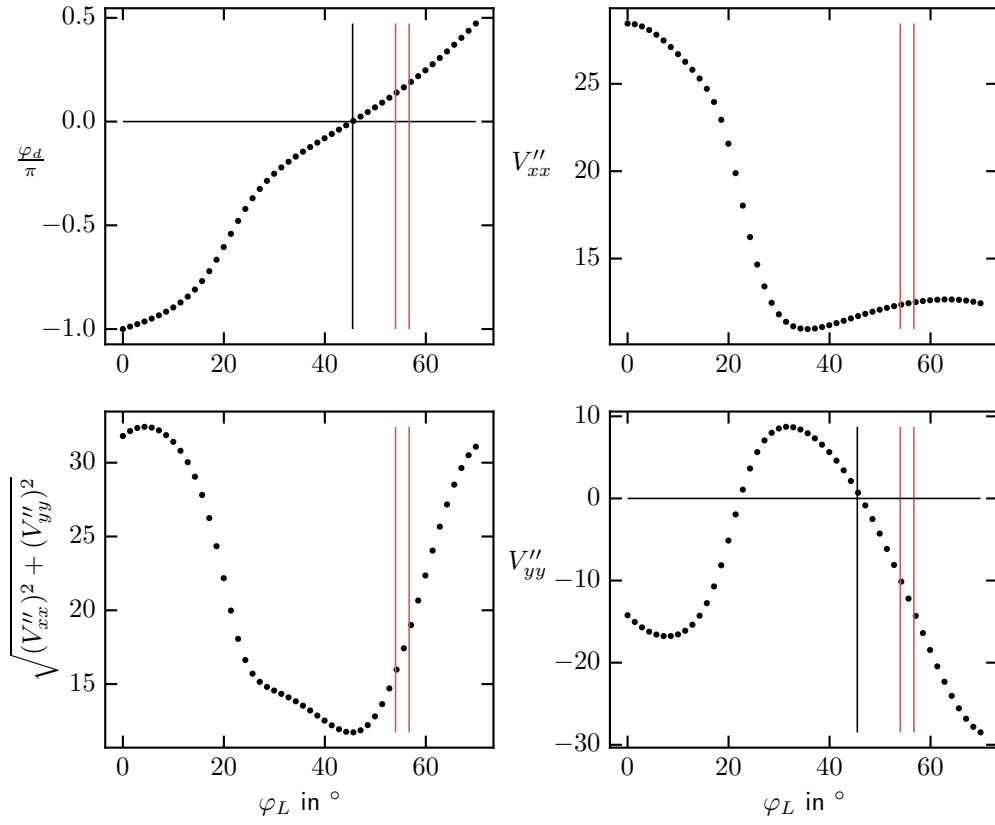


Figure 11: Topological states can again be found in the angle range between the two red vertical lines.

1D hexagonal chain

In this chapter I consider a chain with equal spacing r between the atoms where the angle φ is changed in a manner, such that the first layer of a 2D hexagonal lattice is reached. Because of the one dimensionality of the chain the unit cell contains four atoms (A, B, C, D).

$$\begin{aligned} \mathbf{a}_L &= \begin{pmatrix} 2r(1 + \cos \varphi_L) \\ 0 \end{pmatrix} & \mathbf{a}_B &= \begin{pmatrix} r \cos \varphi_L \\ r \sin \varphi_L \end{pmatrix} \\ \mathbf{a}_C &= \begin{pmatrix} r(1 + \cos \varphi_L) \\ r \sin \varphi_L \end{pmatrix} & \mathbf{a}_D &= \begin{pmatrix} r(1 + 2 \cos \varphi_L) \\ 0 \end{pmatrix} \end{aligned}$$

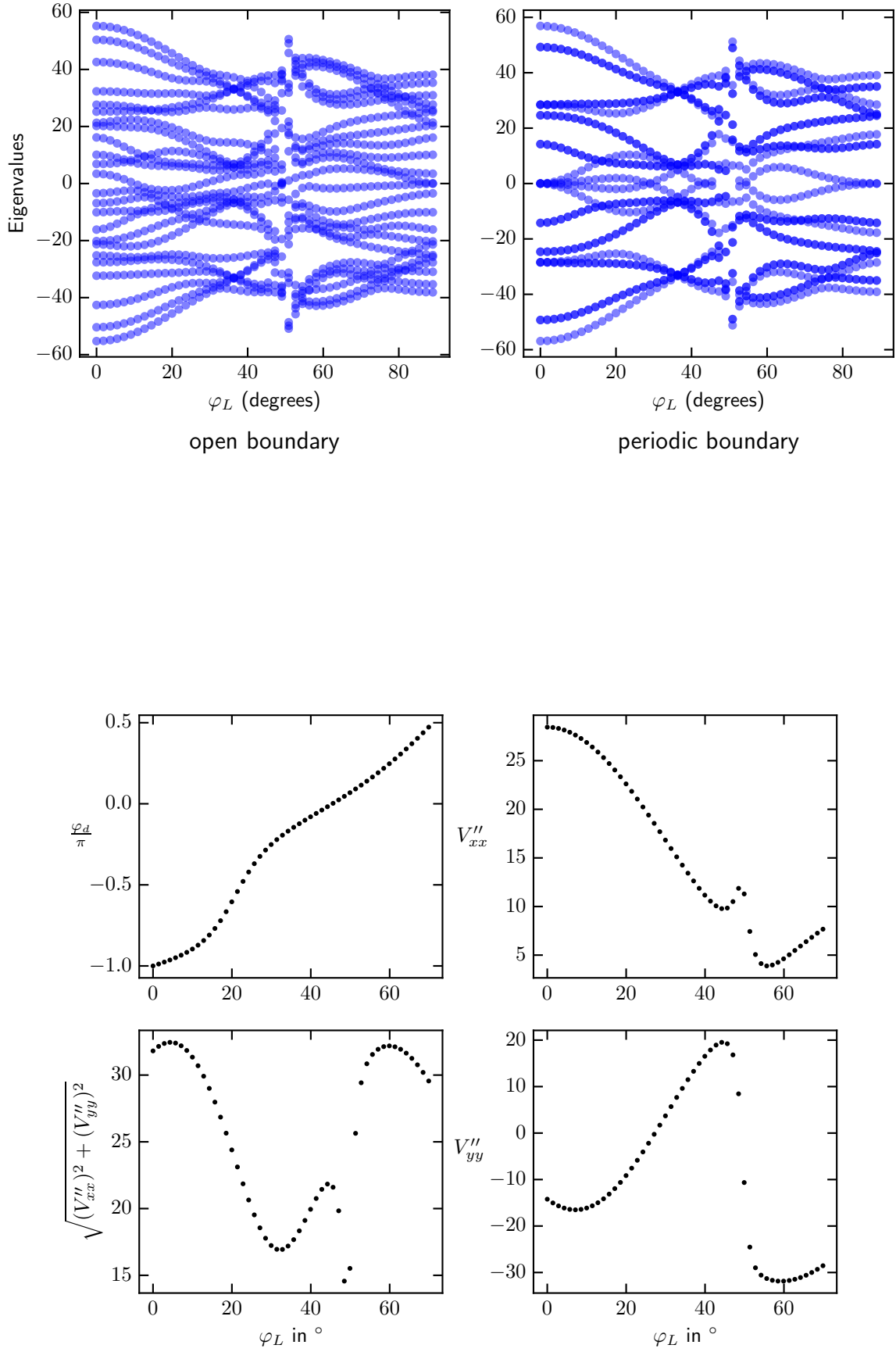
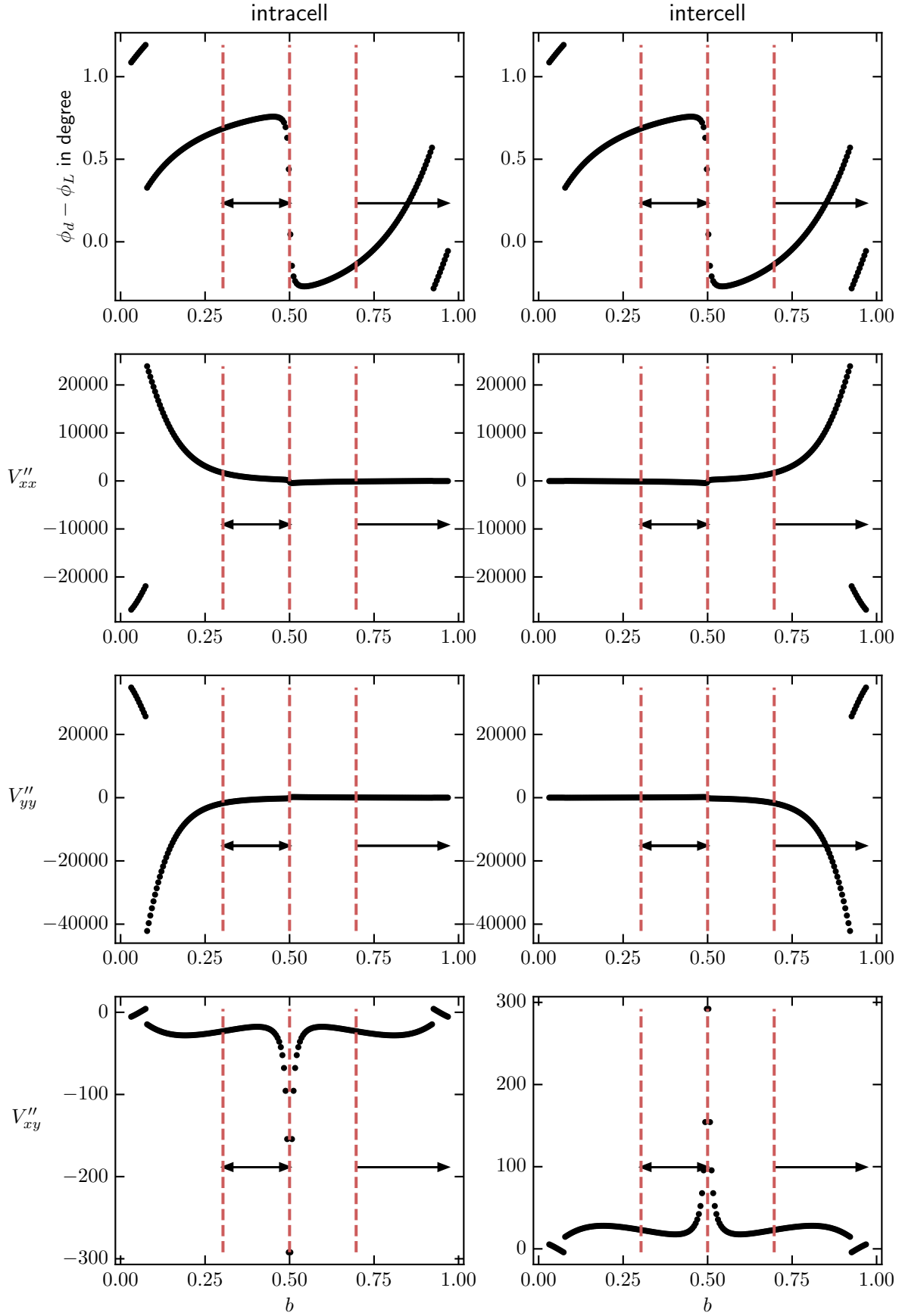
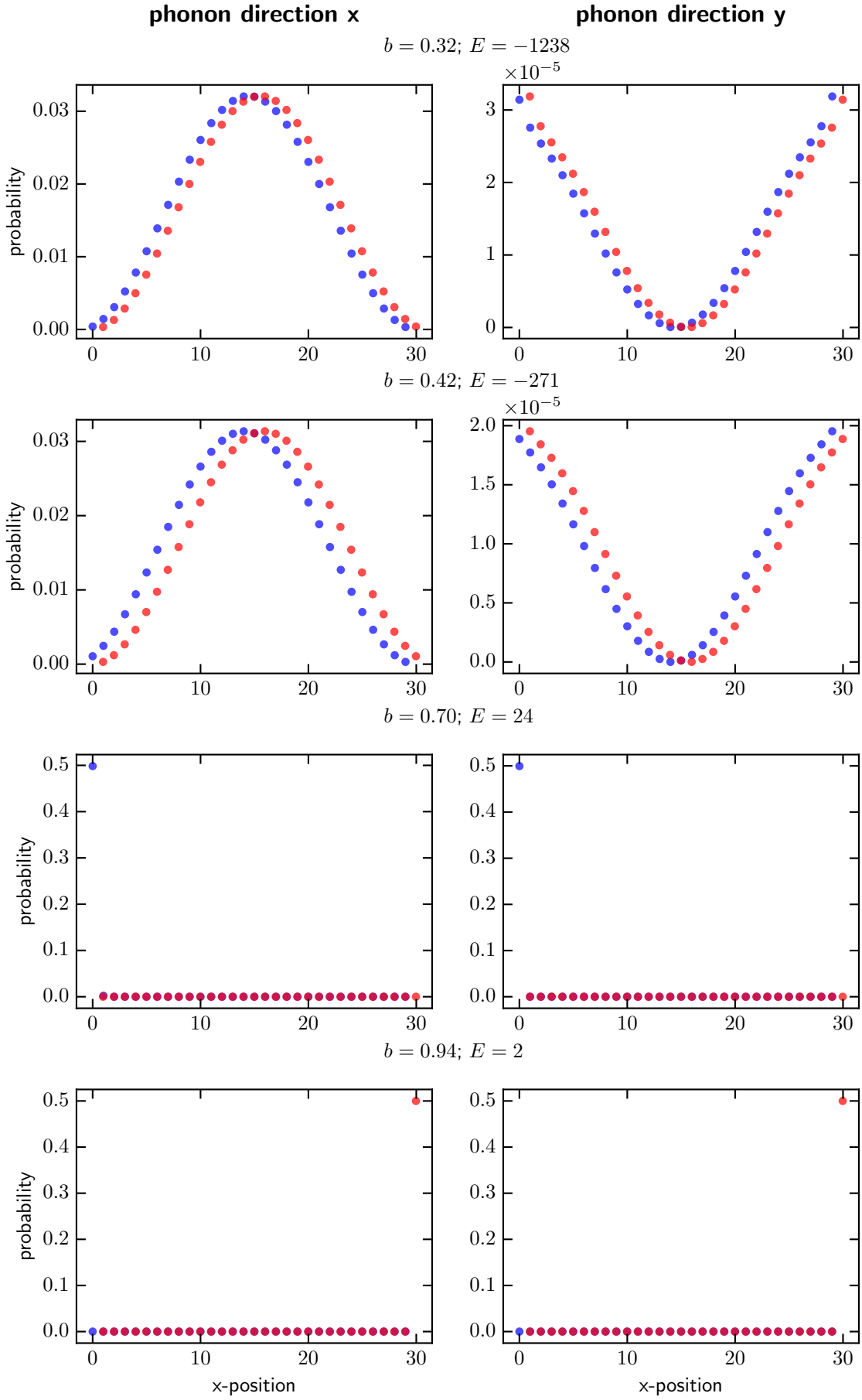
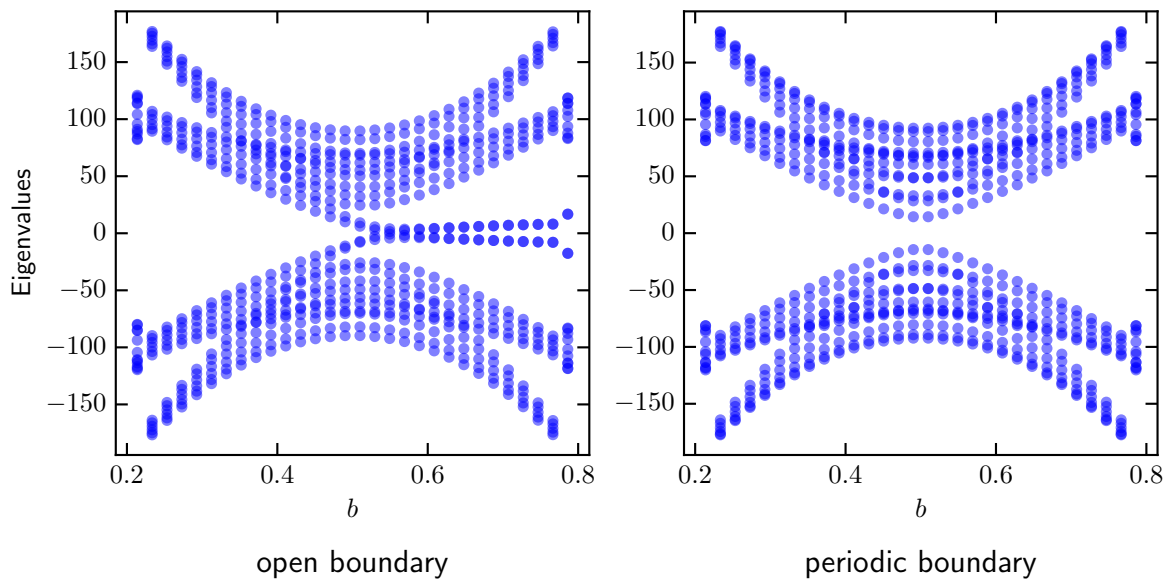
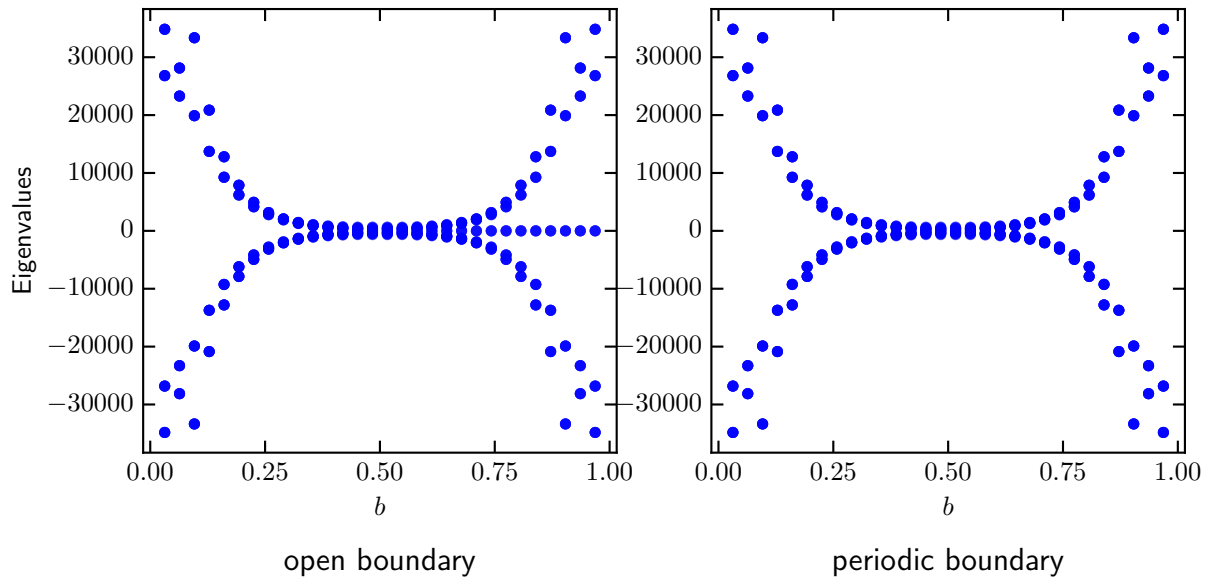


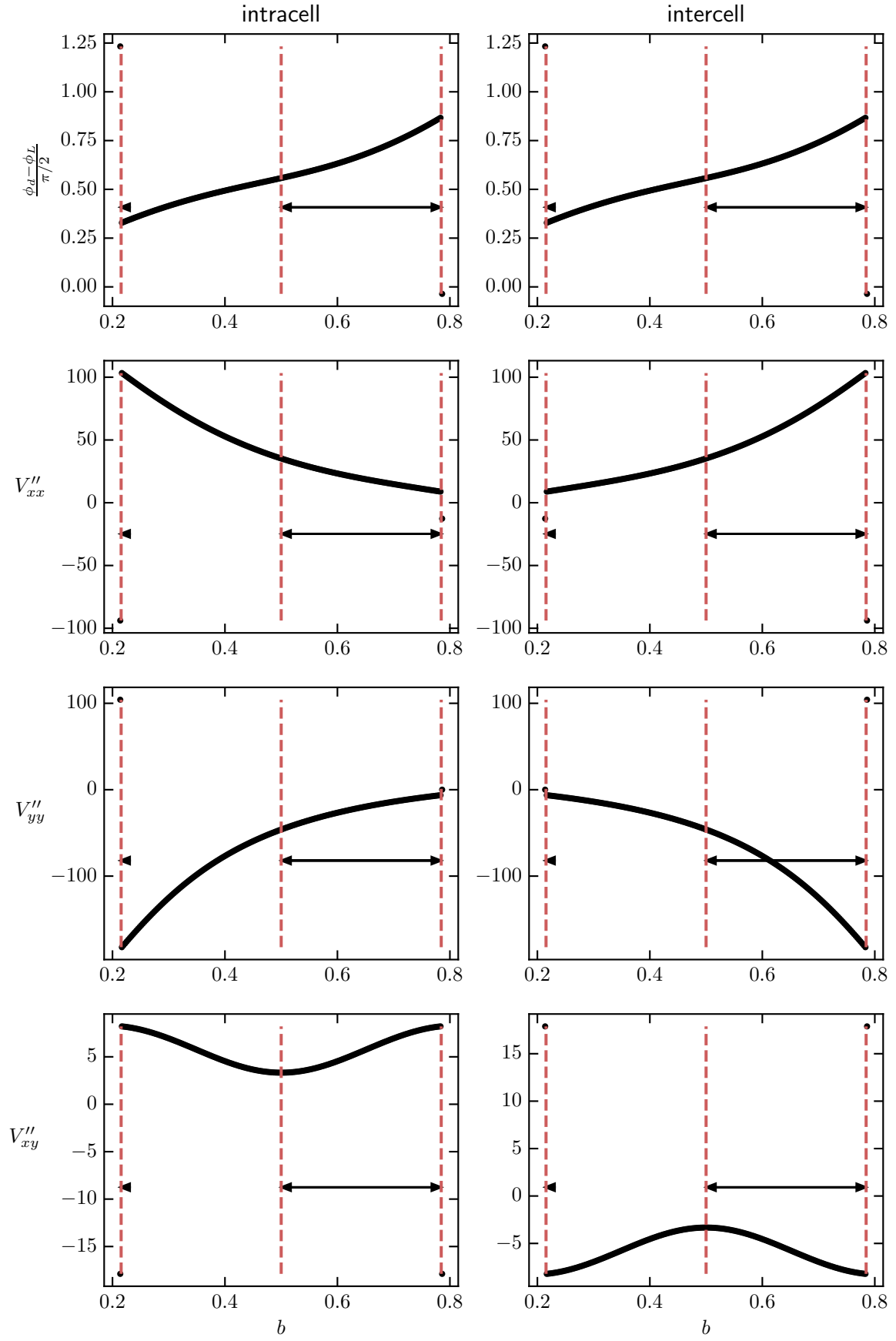
Figure 12: In this configuration with equal spacing between the atoms no topological states occur.

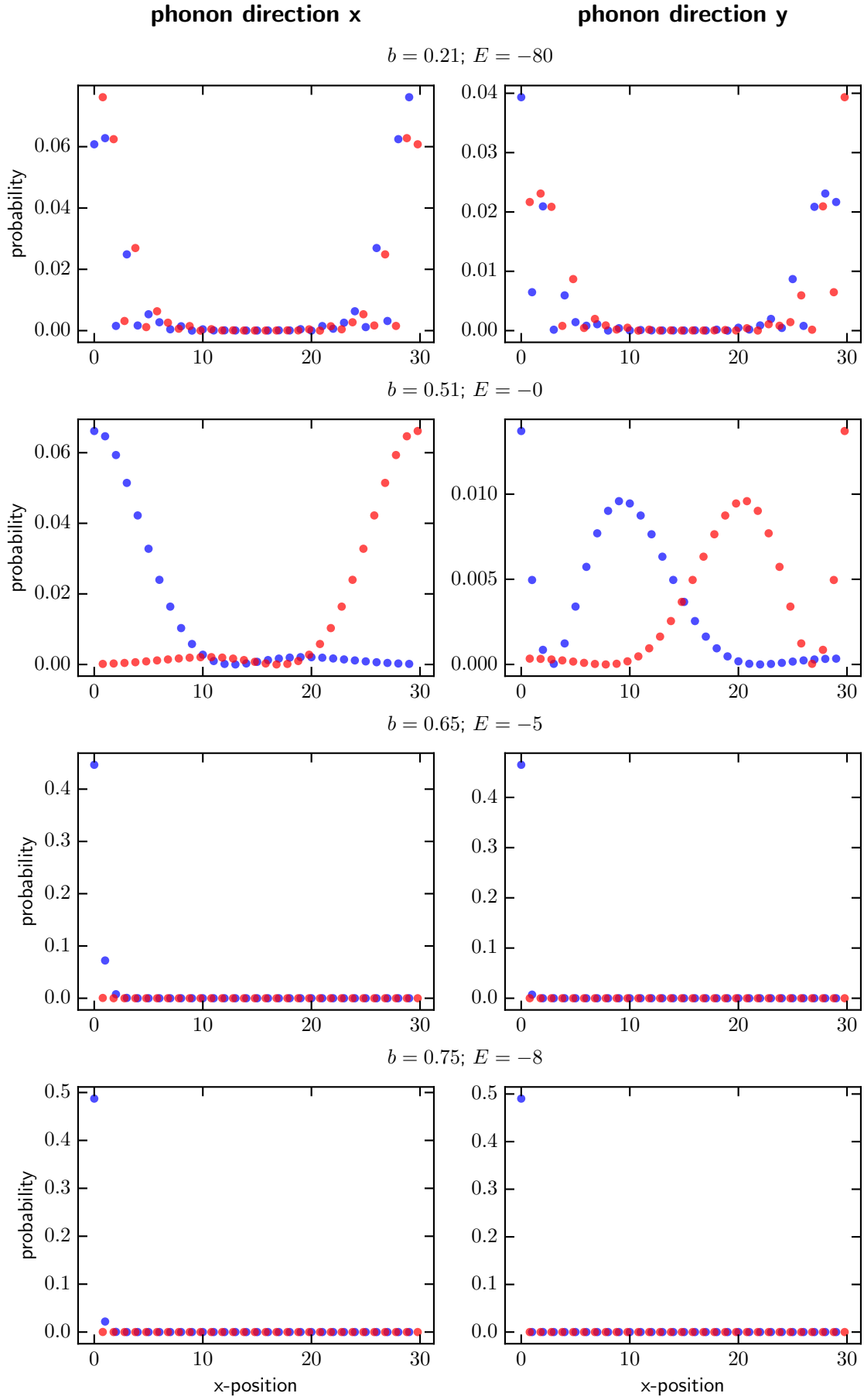
0.5 Aligned dipoles in plane











0.6 1D chain

When allowing displacements in the 2-dimensional plane one can exploit the independence of the individual contributions to H_k , which implies that X has to anti-commute with every contribution individually. Due to the results of Appendix ?? the only possible chiral symmetry in the case where $H_k^{A,A}$ is not identical zero is represented by the unitary

$$S = \mathbb{1} \otimes \sigma_y$$

It requires that the identity contribution in the second slot of the tensor product has to vanish. This can only be true if the xx and yy interaction have opposite signs.

Recall the dependence of the interaction of the direction of $\mathbf{r}$ and $\mathbf{d}$ in form of

$$\begin{aligned} c_{\alpha\beta}(\hat{\mathbf{r}}) &= \delta_{\alpha\beta} (-3 + 15 (\hat{\mathbf{r}} \cdot \hat{\mathbf{d}})^2) - 6 \hat{d}_\alpha \hat{d}_\beta + 30 \frac{\hat{d}_\alpha r_\beta + \hat{d}_\beta r_\alpha}{r} (\hat{\mathbf{r}} \cdot \hat{\mathbf{d}}) + 15 \frac{r_\alpha r_\beta}{r^2} (1 - 7 (\hat{\mathbf{r}} \cdot \hat{\mathbf{d}})^2) \\ &= \delta_{\alpha\beta} (-3 + 15 (\hat{\mathbf{r}} \cdot \hat{\mathbf{d}})^2) + A_{\alpha\beta} \end{aligned}$$

In order for the interaction between atoms of relative distance $\mathbf{r}$ to vanish the following condition has

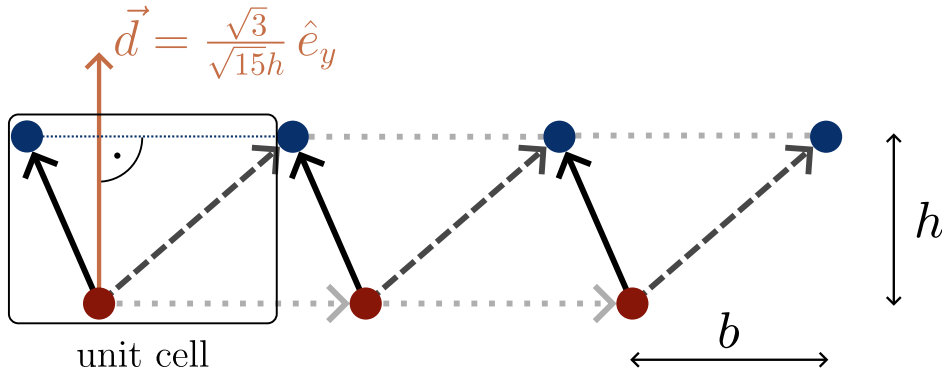


Figure 13: **Zik-zak chain and dipole orientation.** Interactions up to next nearest neighbors are presented. The dipole is orientated in such a way that it is perpendicular to the connection vectors of nearest neighbor distances. Thus the scalar product with the dipole moment is the same for both the solid and the dashed vector.

to hold

$$\begin{aligned} c_{xx}(\hat{\mathbf{r}}) &\stackrel{!}{=} -c_{yy}(\hat{\mathbf{r}}) \\ &= -6 + 30 (\hat{\mathbf{r}} \cdot \hat{\mathbf{d}})^2 + A_{xx} + A_{yy} \\ A_{xx} + A_{yy} &= -6 (\hat{d}_x^2 + \hat{d}_y^2) + 60 \frac{\hat{d}_x r_x + \hat{d}_y r_y}{r} (\hat{\mathbf{r}} \cdot \hat{\mathbf{d}}) + 15 \frac{r_x^2 + r_y^2}{r^2} (1 - 7 (\hat{\mathbf{r}} \cdot \hat{\mathbf{d}})^2) \\ &= 9 \|\hat{\mathbf{d}}_{\parallel}\|^2 - 3 \cdot 15 (\hat{\mathbf{r}} \cdot \hat{\mathbf{d}})^2 \\ \Rightarrow (\hat{\mathbf{r}} \cdot \hat{\mathbf{d}}_{\parallel})^2 &= \frac{3}{15} \end{aligned} \tag{0.6}$$

This condition can only be satisfied when all vectors lie on a straight line. Picturing the zik-zak chain implementation of the system as depicted in Fig. 13 one realizes that this is not possible if one encounters both interactions between and within the sublattices. However, the restriction to only nearest neighbor interactions lets the coupling only take place between sublattices and one can choose the dipole moment in y direction to satisfy Eq. (0.6) as in Fig. 13. In this case the model only contains four different coupling parameters, V_x , V_{xy} for interactions within the unit cell and W_x , W_{xy} for interactions between

unit cells. One can write down the contributions to the Hamiltonian as

$$H_k^{A,A} = \begin{pmatrix} V_x + W_x & V_{xy} + W_{xy} \\ V_{xy} + W_{xy} & -V_x - W_x \end{pmatrix} = (V_x + W_x) \sigma_z + (V_{xy} + W_{xy}) \sigma_x$$

$$H_k^{A,B} = \begin{pmatrix} V_x + W_x e^{-ika} & V_{xy} + W_{xy} e^{-ika} \\ V_{xy} + W_{xy} e^{-ika} & -V_x - W_x e^{-ika} \end{pmatrix} = (V_x + W_x e^{-ika}) \sigma_z + (V_{xy} + W_{xy} e^{-ika}) \sigma_x$$

Transforming to the basis where $S = \sigma_z \otimes \mathbb{1}$ the Hamiltonian takes the form

$$H_k = \begin{pmatrix} 0 & h \\ h^\dagger & 0 \end{pmatrix}$$

with $h = \begin{pmatrix} i(V_{xy} + W_{xy}) - (V_x + W_x) & i(V_{xy} + W_{xy} e^{-ika}) - (V_x + W_x e^{-ika}) \\ i(V_{xy} + W_{xy} e^{ika}) - (V_x + W_x e^{ika}) & i(V_{xy} + W_{xy}) - (V_x + W_x) \end{pmatrix}$

This can be written in shorter and clearer form as

$$h = \begin{pmatrix} a & b + c e^{-ika} \\ b + c e^{ika} & a \end{pmatrix} \quad a, b, c \in \mathbb{C}$$

From this the eigenvalues $\lambda_\pm$ can be calculated as

$$\lambda_\pm = a \pm \sqrt{b^2 + c^2 + 2bc \cos(ka)}$$

Since a is constant the only winding can come from the square root expression. As the discriminant is, however, of the form $r + s t$ with $r, s \in \mathbb{C}$ and $t \in [0, 1]$ the square root cannot describe a closed curve in the complex plain. Hence this implementation of chiral symmetry does not exhibit topologically non-trivial phases.

In order to have a qualitatively different disorder, one has to make the $H_k^{A,A}$ term vanish. Since one cannot choose the dipole in such a way that the xx , yy and xy interaction of one distance vanish at the same time, one has to assume nearest neighbor interactions only. In this scenario one is still left with the onsite terms of U . Exploiting the high control of dipole traps in cold atom experiments one can tune the harmonic potential of each trap individually [11, 12]. Thus by fine tuning the trap frequencies at each site one can in principle cancel the onsite terms corresponding to xx and yy interactions. The interaction between different direction, however, is still present after this step and has to be tuned to zero via the geometry of the lattice and the dipole orientation. This decouples the two direction completely from each other leading to a mere coexistence of two models. It has to be noted that the open system with boundaries has to satisfy the same symmetries as the infinite system in order for the bulk-boundary correspondence to hold. Hence it is not possible to tune the sum in $U_{ii}^{xy} = \sum_r \partial^2 V_{dd} / \partial r_x \partial r_y$ to zero for every site of the open system without forcing every summand to zero. After decoupling into one model for each direction of displacement, it is, however, possible to observe both models in different topological phases at the same time, as indicated by Fig. 14.

Note: The value of the derivative of the dipole-dipole interaction on the unit circle is only dependent on two properties. On the one hand the dipole-dipole interaction is only dependent on the angle between dipole and distance vector. The derivative on the other hand is additionally dependent on the direction of derivation, thus by the choice of the coordinate frame. If the frame is rotated by the angle φ around the z -axis, one can write the old variables as functions of the new ones.

$$r_x = \cos \varphi r'_x + \sin \varphi r'_y$$

$$r_y = -\sin \varphi r'_x + \cos \varphi r'_y$$

From this expression one can write the derivations into the new x - and y -directions.

$$\begin{aligned}\frac{\partial V_{dd}}{\partial r'_x} &= \frac{\partial V_{dd}}{\partial r_x} \cos \varphi - \frac{\partial V_{dd}}{\partial r_y} \sin \varphi \\ \frac{\partial^2 V_{dd}}{(\partial r'_x)^2} &= \cos^2 \varphi \frac{\partial^2 V_{dd}}{\partial r_x^2} - 2 \cos \varphi \sin \varphi \frac{\partial^2 V_{dd}}{\partial r_x \partial r_y} + \sin^2 \varphi \frac{\partial^2 V_{dd}}{\partial r_y^2} \\ \frac{\partial^2 V_{dd}}{(\partial r'_y)^2} &= \sin^2 \varphi \frac{\partial^2 V_{dd}}{\partial r_x^2} + 2 \cos \varphi \sin \varphi \frac{\partial^2 V_{dd}}{\partial r_x \partial r_y} + \cos^2 \varphi \frac{\partial^2 V_{dd}}{\partial r_y^2} \\ \frac{\partial^2 V_{dd}}{\partial r'_x \partial r'_y} &= \cos \varphi \sin \varphi \frac{\partial^2 V_{dd}}{\partial r_x^2} - (\cos^2 \varphi - \sin^2 \varphi) \frac{\partial^2 V_{dd}}{\partial r_x \partial r_y} - \cos \varphi \sin \varphi \frac{\partial^2 V_{dd}}{\partial r_y^2}\end{aligned}$$

All three derivatives in the new frame can only be zero if they were already zero in the old reference frame. As it is not possible to find a dipole orientation that makes this happen for the direction $\hat{e}_x$. It is not possible to find another orientation of the distance vector such that all second derivatives would vanish.

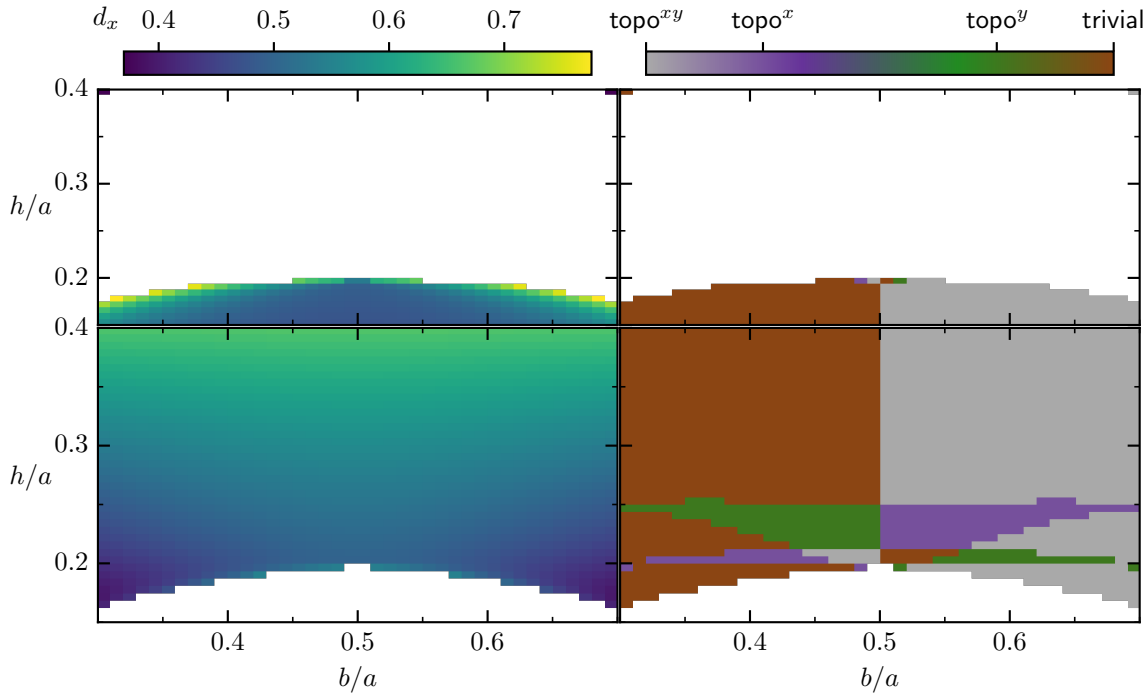


Figure 14: **Topological phase diagram.** There are up to four possible dipole orientations that decouple x and y -directions. Two of them are shown on the left panels by depicting the x -component of the dipole. White space indicates here that there were no appropriate dipoles found. On the right side of the panel the topological phases are depicted. It is shown, whether a topological phase arises for the x , y or both directions, or if the system exhibits a topologically trivial phase.

A different approach to decoupling the two directions of displacement x and y , is by choosing different trap frequencies along each direction (see Section ??). If one can still ensure $|\omega_x - \omega_y| \gg |U| \ll \omega_x, \omega_y$, the two direction of displacement are decoupled as well, since transfer between the x and y -mode are suppressed. In this case one still has free choice over the dipole orientation. Focussing in the following only on one direction, one receives a full one-dimensional model with one mode per site. Eliminating the onsite potential with the help of the trap frequencies, this system can be mapped to an extended SSH-model. The only requirement left for this mapping is the vanishing of interactions within the same sublattice. This can be eliminated by choosing an appropriate dipole orientation, i.e. $d_x = \pm 1/\sqrt{3}$.

Spin-phonon coupling

Transport

I want to allow for the propagation of excitations through the chain. The first step towards this is constructing a Hamiltonian which describes the dynamics of a system of Rydberg atoms, where one atom is excited, hence restricting ourselves to the single excitation subspace. The general coupling between the atoms arises from the interaction between the dipole moments of the atoms,

$$V = \sum_{\mathbf{r}} V(\mathbf{r})$$

$$\text{with } V(\mathbf{r}) = V_0 \left(\frac{\hat{\mathbf{d}}_1 \cdot \hat{\mathbf{d}}_2}{r^3} - 3 \frac{(\mathbf{r} \cdot \hat{\mathbf{d}}_1)(\mathbf{r} \cdot \hat{\mathbf{d}}_2)}{r^5} \right).$$

Each individual atom is characterized by its position in the lattice and its atomic energy level which is captured by the non-interacting Hamiltonian H_0 . The total Hamiltonian thus reads

$$H = H_0 + V$$

We will investigate the system in a regime, where the coupling due to the dipole forces is much smaller than the energy gap Δ , which separates the energy of the atoms from their next higher and lower states, i.e. $\|V\| \ll \Delta$. In order to gain an approximate effective Hamiltonian, which acts in a closed way on the one excitation subspace, I will perform a Schrieffer-Wolff transform up to second order [bravyi_schrieffer-wolff_2011]. Let P_0 be the projector on the one excitation subspace. Then an exact alternative Hamiltonian H' can be constructed by the transformation

$$H' = e^S H e^{-S}$$

for which holds, that it is block-diagonal with respect to the subspaces defined by the projections P_0 and $1 - P_0$. Thus the alternative Hamiltonian H' is closed with respect to the one excitation subspace and one can define the exact effective Hamiltonian as

$$H_{\text{eff}} = P_0 e^S H e^{-S} P_0$$

Let ε be of the order of the coupling $\varepsilon = \|V\|$. The transformation generating operator S can be expanded in orders of V .

$$S = \sum_{n=1}^{\infty} \varepsilon^n S_n$$

I will consider the effective Hamiltonian up to second order in the coupling. This lets H_{eff} read

$$H_{\text{eff}} = H_0 P_0 + P_0 V P_0 + \frac{1}{2} P_0 [S_1, V_{\text{od}}] P_0,$$

with $V_{\text{od}} = P_0 V (1 - P_0) + (1 - P_0) V P_0$ being the off-diagonal part of the interaction. The first term of the generator S has the form

$$S_1 = \mathcal{L}(V_{\text{od}})$$

$$\text{with } \mathcal{L}(X) = \sum_{\substack{i,j \in E \\ E_i \neq E_j}} \frac{\langle i | V | j \rangle}{E_i - E_j} |i\rangle \langle j|$$

The sum in the superoperator $\mathcal{L}$ goes over all energy eigenstates, where the i, j combination of states with equal energy are omitted. It can be written via a sum over all energies in the single excitations subspace E_1 and all the other eigenenergies $\tilde{E}$.

$$\mathcal{L}(X) = \sum_{\substack{i \in E_1 \\ j \in \tilde{E}}} \frac{\langle i|V|j\rangle}{E_i - E_j} |i\rangle \langle j| - \text{h.c.}$$

From this the commutator term in the effective Hamiltonian can be calculated.

$$P_0 [S_1, V_{\text{od}}] P_0 = 2 \sum_{\substack{i \in E_1 \\ j \in \tilde{E}}} \frac{|\langle i|V|j\rangle|^2}{E_i - E_j} |i\rangle \langle i| =: H_{\text{eff},2}$$

Thus finally the effective Hamiltonian reads

$$H_{\text{eff}} = H_0 P_0 + P_0 V P_0 + \sum_{\substack{i \in E_1 \\ j \in \tilde{E}}} \frac{|\langle i|V|j\rangle|^2}{E_i - E_j} |i\rangle \langle i|$$

For the effective Hamiltonian the contribution $H_0 P_0$ is just a constant term and can be neglected. Further because of the large relative size of Δ over the interaction strength, I will only include energies into the above sum which differ from the single excitation subspace by Δ . Further the zero excitations subspace does not couple to the one with one excitation in the considered order of expansion. Thus only eigenstates in the double excitation subspace have to be taken into account for the sum.

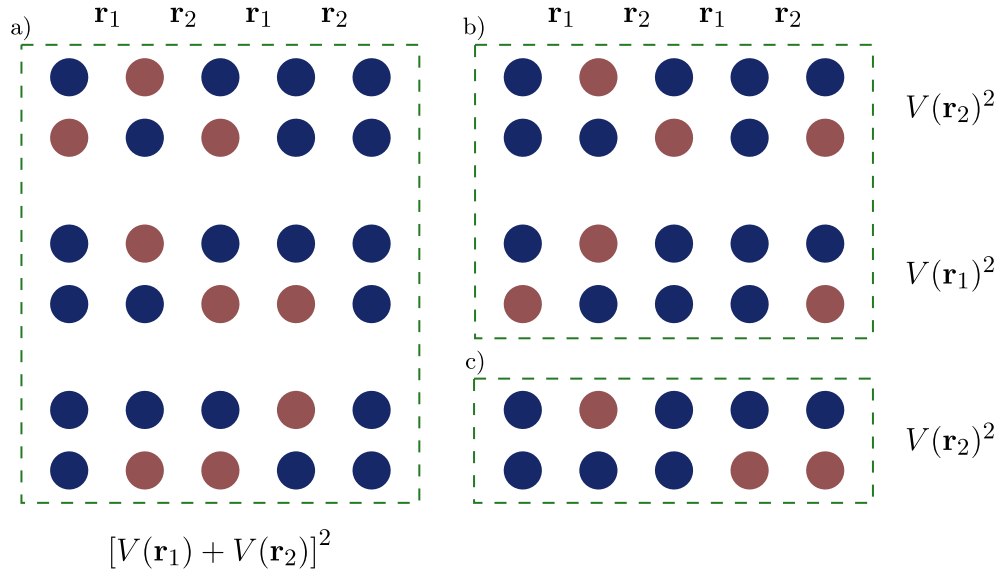


Figure 15: **Coupling schemes.** There are different ways double excited states couple to the single excited ones. The top row always depicts a state of the single, while the bottom row depicts a state of the double excited subspace. Generally configurations of two different subspaces are coupled to each other, when for two neighboring atoms one configuration has the inverted excitation scheme with respect to the other. I assign them to three categories. The first category is depicted in a). It couples two neighboring atoms twice. In the picture all possible cases are shown. For b) only two atoms couple two each other. Here the coupling takes place at a pair of $|e\rangle$ and $|g\rangle$ in both configurations. If the excitation in the single excitation subspace is at position $2 < i < N - 1$, there are $N - 4$ different states in the double excited subspace that contribute a term $V(r_1)^2$ and the same number for $V(r_2)^2$. Category c) couples two unexcited sites in the single excitations subspace to two excited sites in the double excited subspace. For this case there are $N/2 - 2$ states that contribute $V(r_1)^2$ and $N/2 - 2$ states that contribute $V(r_2)^2$, if the single excited state is excited at atom $2 < i < N - 1$.

In the following the effective Hamiltonian will be calculated for the specific configuration we are looking at. We consider a one-dimensional staggered chain of N atoms, where the atoms are separated

by alternating distances $\mathbf{r}_1$ and $\mathbf{r}_2$. For a single atom there are only two states available, excited $|e\rangle$ and unexcited $|g\rangle$. The states $|i\rangle$, which are summed over, represent the states where only atom i is excited and the others are in the unexcited state. Going to a regime where only nearest neighbor interactions contribute we can write using the coupling schemes of Fig. 22

$$H_{\text{eff},2} = \sum_{i=1}^N \tilde{V}_i |i\rangle \langle i|$$

$$\tilde{V}_i = -\frac{1}{\Delta} \begin{cases} 3 [V(\mathbf{r}_1) + V(\mathbf{r}_2)]^2 + (N-4 + \frac{N}{2} - 2) V^2(\mathbf{r}_1) + (N-4 + \frac{N}{2} - 2) V^2(\mathbf{r}_2) & \text{if } 1 \neq i \neq N \\ [V(\mathbf{r}_1) + V(\mathbf{r}_2)]^2 + (N-3 + \frac{N}{2} - 1) V^2(\mathbf{r}_1) + (\frac{N}{2} - 1) V^2(\mathbf{r}_2) & \text{if } i = 1 \\ 2 [V(\mathbf{r}_1) + V(\mathbf{r}_2)]^2 + (N-3 + \frac{N}{2} - 2) V^2(\mathbf{r}_1) + (N-4 + \frac{N}{2} - 1) V^2(\mathbf{r}_2) & \text{if } i = 2 \\ 2 [V(\mathbf{r}_1) + V(\mathbf{r}_2)]^2 + (N-4 + \frac{N}{2} - 1) V^2(\mathbf{r}_1) + (N-3 + \frac{N}{2} - 2) V^2(\mathbf{r}_2) & \text{if } i = N-1 \\ [V(\mathbf{r}_1) + V(\mathbf{r}_2)]^2 + (\frac{N}{2} - 1) V^2(\mathbf{r}_1) + (N-3 + \frac{N}{2} - 1) V^2(\mathbf{r}_2) & \text{if } i = N \end{cases}$$

The part involving interactions within the same subspace can be written as

$$P_0 V P_0 = \sum_{\substack{i=1 \\ j=i+1}}^{N-1} V(\mathbf{r}_{i,j}) \sigma_i^+ \sigma_j^- + \text{h.c.},$$

where $\mathbf{r}_{i,j}$ is the distance between the i th and j th site. I now also want to write $H_{\text{eff},2}$ in terms of a sum of two atom operators. Therefore I introduce the projectors $n_i^e = |e\rangle_i \langle e|_i$ and $n_i^g = |g\rangle_i \langle g|_i$.

$$H_{\text{eff},2} = \sum_{\substack{i=1 \\ j=i+1}}^{N-1} V^2(\mathbf{r}_{i,j}) (n_i^e n_j^g + n_i^g n_j^e + n_i^g n_j^g) + \sum_{i=2}^{N-1} [V(\mathbf{r}_{i-1,i}) + V(\mathbf{r}_{i,i+1})]^2 n_{i-1}^g n_i^e n_{i+1}^g \\ + \sum_{i=3}^{N-1} [V(\mathbf{r}_{i-2,i-1}) + V(\mathbf{r}_{i-1,i})]^2 n_{i-2}^g n_{i-1}^g n_i^e + \sum_{i=2}^{N-2} [V(\mathbf{r}_{i,i+1}) + V(\mathbf{r}_{i+1,i+2})]^2 n_i^e n_{i+1}^g n_{i+2}^g \\ + [V(\mathbf{r}_{1,2}) + V(\mathbf{r}_{2,3})]^2 n_1^g n_2^g n_3^g + [V(\mathbf{r}_{N-2,N-1}) + V(\mathbf{r}_{N-1,N})]^2 n_{N-2}^g n_{N-1}^g n_N^e$$

A last step is possible since in the single excitation subspace terms like $n_i^e n_j^g$ simplify to just n_i^e .

$$= \sum_{i=2}^{N-1} \left(V^2(\mathbf{r}_{i-1,i}) + V^2(\mathbf{r}_{i,i+1}) + [V(\mathbf{r}_{i-1,i}) + V(\mathbf{r}_{i,i+1})]^2 \right) n_i^e \\ + \sum_{i=3}^{N-1} [V(\mathbf{r}_{i-2,i-1}) + V(\mathbf{r}_{i-1,i})]^2 n_i^e + \sum_{i=2}^{N-2} [V(\mathbf{r}_{i,i+1}) + V(\mathbf{r}_{i+1,i+2})]^2 n_i^e \\ + \left(V^2(\mathbf{r}_{1,2}) + [V(\mathbf{r}_{1,2}) + V(\mathbf{r}_{2,3})]^2 \right) n_1^e + \left(V^2(\mathbf{r}_{N-1,N}) + [V(\mathbf{r}_{N-2,N-1}) + V(\mathbf{r}_{N-1,N})]^2 \right) n_N^e \\ + \sum_{i=1}^{N-1} V(\mathbf{r}_{i,i+1}) n_i^g n_{i+1}^g$$

Energy scales

The most important energies which determine the dynamics are the trap frequency, which is of the order $\omega \sim 10$ -100 kHz, the distances between atoms who can reliably reach $3 \mu\text{m}$, and the lifetime of the Rydberg states, which is in the order of $100 \mu\text{s}$. From this it already becomes apparent that, in order to see dynamics of the system within the lifetime of the Rydberg states, the coupling strength has to be higher than the trap frequency. This prevents the Lamb-Dicke regime, as the phonon number cannot be assumed as conserved. Thus the first order expansion in the displacements of the atoms has to be taken into account. The relative size of each order of the expansion can be estimated by the factor

$$\varepsilon := \sqrt{\frac{\hbar}{m\omega}} \frac{1}{r} \approx (2-11) \cdot 10^{-3}$$

As the dipole interaction is a function of the angle ϑ between the dipole moment and the axis of the chain, the interaction can be tuned in a way that the first order term vanishes. As can be seen in Fig. 16, the second order term in the expansion is of the order $5 \cdot 10^{-5} - 10^{-3}$, smaller than the leading order.

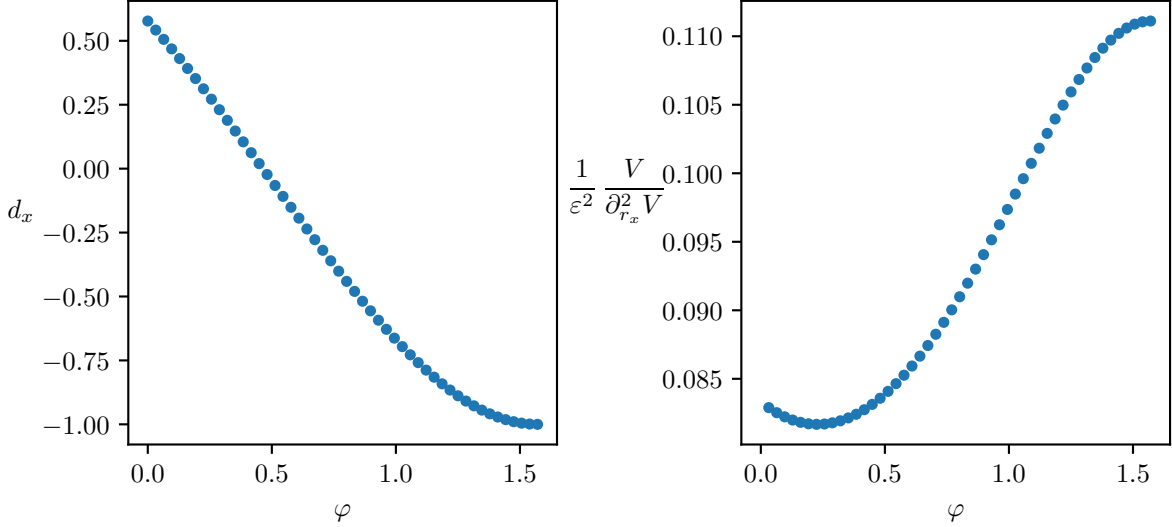


Figure 16: **Dipole orientation and ratio of interactions.** For all angles $\sin \varphi := h/r$ one can find a dipole orientation, shown in the left panel, such that the first order term in the expansion of the interaction vanishes. The ratio between zeroth order and second order term is shown in the right panel.

Spin-Phonon Hamiltonian

Let us first clarify the approximations and choices made that lead to the spin-phonon coupled Hamiltonian that is used in the following. The consideration is restricted to the single-excitation subspace with respect to the electronic degrees of freedom, i.e. the spin degrees of freedom. This is achieved by going into a regime where the energy distance between states is much larger than their coupling. As a second feature the system shall be arranged in a way that the number of motional excitations is conserved as well. This is achieved by tuning the linear phonon creation term, stemming from the first order Taylor expansion of the coupling potential, to zero. This procedure is laid out in the following.

Considering only nearest neighbor interactions the interaction Hamiltonian expanded to the second order in the displacements from the trap center reads

$$\begin{aligned}
 H_{\text{int}} &= \sum_{i,j=1; i < j}^{N-1} \left[V(\mathbf{r}_{i,j}) + \sum_{n \in \{i,j\}} \frac{\partial V(\mathbf{r}_{i,j})}{\partial \mathbf{R}_n} \delta \hat{\mathbf{R}}_n + \frac{1}{2} \sum_{n,m \in \{i,j\}} \frac{\partial^2 V(\mathbf{r}_{i,j})}{\partial \mathbf{R}_n \partial \mathbf{R}_m} \delta \hat{\mathbf{R}}_n \delta \hat{\mathbf{R}}_m \right] \sigma_i^+ \sigma_j^- + \text{h.c.} \\
 &= \sum_{i,j=1; i < j}^{N-1} \left[V(\mathbf{r}_{i,j}) + \frac{\partial V(\mathbf{r}_{i,j})}{\partial \mathbf{r}_{i,j}} (\delta \hat{\mathbf{R}}_j - \delta \hat{\mathbf{R}}_i) \right. \\
 &\quad \left. + \frac{1}{2} \frac{\partial^2 V(\mathbf{r}_{i,j})}{\partial \mathbf{r}_{i,j}^2} (\delta \hat{\mathbf{R}}_i^2 + \delta \hat{\mathbf{R}}_j^2 - \delta \hat{\mathbf{R}}_i \delta \hat{\mathbf{R}}_j - \delta \hat{\mathbf{R}}_j \delta \hat{\mathbf{R}}_i) \right] \sigma_i^+ \sigma_j^- + \text{h.c.}
 \end{aligned}$$

As this step one can introduce a shift in the displacement as was done in Section , i.e. $\{\delta \mathbf{R}_l\} \rightarrow \{\delta \mathbf{R}_l + \mathbf{\Delta}_l\}$. Plugging the shifted displacements in the above equation and considering only the terms linear in

displacement (H_{Lin}), one can collect the following terms.

$$\begin{aligned}
 H_{\text{Lin}} &= \sum_{i,j=1; i < j}^{N-1} \left[\left(-\frac{\partial V(\mathbf{r}_{i,j})}{\partial \mathbf{r}_{i,j}} - \frac{\partial^2 V(\mathbf{r}_{i,j})}{\partial \mathbf{r}_{i,j}^2} (\Delta_j - \Delta_i) \right) \delta \hat{\mathbf{R}}_i \right. \\
 &\quad \left. + \left(\frac{\partial V(\mathbf{r}_{i,j})}{\partial \mathbf{r}_{i,j}} + \frac{\partial^2 V(\mathbf{r}_{i,j})}{\partial \mathbf{r}_{i,j}^2} (\Delta_j - \Delta_i) \right) \delta \hat{\mathbf{R}}_j \right] \sigma_i^+ \sigma_j^- + \text{h.c.} \\
 &= \sum_{i,j=1; i < j}^{N-1} \left[\frac{V_0}{a} \left(\frac{a^4}{r_{i,j}^4} g(\hat{\mathbf{r}}_{i,j}) + \frac{a^5}{r_{i,j}^5} c(\hat{\mathbf{r}}_{i,j}) \left(\frac{\Delta_j}{a} - \frac{\Delta_i}{a} \right) \right) (\delta \hat{\mathbf{R}}_j - \delta \hat{\mathbf{R}}_i) \right] \sigma_i^+ \sigma_j^- + \text{h.c.}
 \end{aligned}$$

This constitutes the condition which the shifts in displacement have to satisfy.

$$G_{i,j} := \frac{V_0}{a} \left(\frac{a^4}{r_{i,j}^4} g(\hat{\mathbf{r}}_{i,j}) + \frac{a^5}{r_{i,j}^5} c(\hat{\mathbf{r}}_{i,j}) \left(\frac{\Delta_j}{a} - \frac{\Delta_i}{a} \right) \right) = 0 \quad \forall i, j \in \{1, N-1\}; i < j$$

With this definition one can write the interaction Hamiltonian as

$$\begin{aligned}
 H_{\text{int}} &= \sum_{i,j=1; i < j}^{N-1} \left[V(\mathbf{r}_{i,j}) + \Delta V_{i,j} + G_{i,j} (\delta \hat{\mathbf{R}}_j - \delta \hat{\mathbf{R}}_i) \right. \\
 &\quad \left. + \frac{1}{2} \frac{\partial^2 V(\mathbf{r}_{i,j})}{\partial \mathbf{r}_{i,j}^2} (\delta \hat{\mathbf{R}}_i^2 + \delta \hat{\mathbf{R}}_j^2 - \delta \hat{\mathbf{R}}_i \delta \hat{\mathbf{R}}_j - \delta \hat{\mathbf{R}}_j \delta \hat{\mathbf{R}}_i) \right] \sigma_i^+ \sigma_j^- + \text{h.c.} \\
 \text{with } \Delta V_{i,j} &= V_0 \frac{a^4}{r_{i,j}^4} g(\hat{\mathbf{r}}_{i,j}) \left(\frac{\Delta_j}{a} - \frac{\Delta_i}{a} \right) + \frac{1}{2} V_0 \frac{a^5}{r_{i,j}^5} c(\hat{\mathbf{r}}_{i,j}) \left(\frac{\Delta_j}{a} - \frac{\Delta_i}{a} \right)^2
 \end{aligned}$$

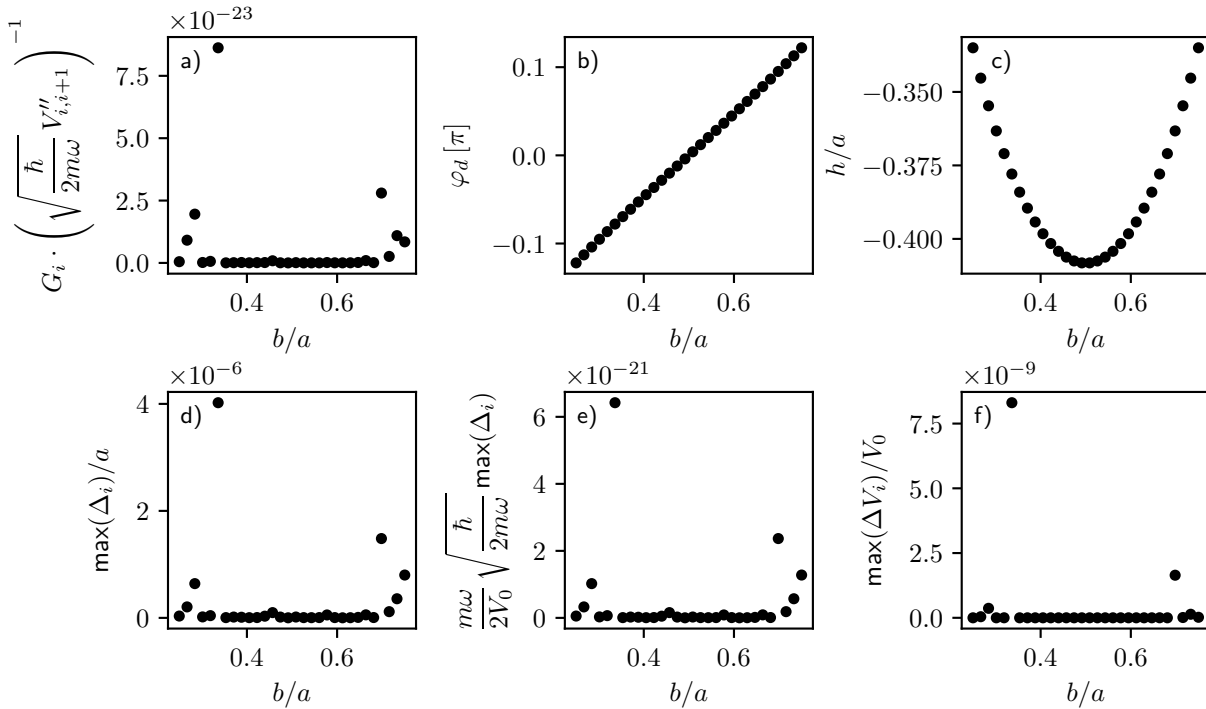


Figure 17: Parameters for negligible first order.

First order approximation in spin degrees

For the first approach I will continue with the Hamiltonian up to first order in the dipole interaction, i.e. without $H_{\text{eff},2}$. The phonons are introduced by again expanding the interaction up to second order.

When we choose the dipole orientation, such that the first order vanishes, the Hamiltonian reads

$$\begin{aligned}
H &= \sum_{i=1}^{N-1} \left[V(\mathbf{r}_{i,i+1}) + \frac{1}{2} \sum_{n,m \in \{i,i+1\}} \frac{\partial^2 V(\mathbf{r}_{i,i+1})}{\partial \mathbf{r}_n \partial \mathbf{r}_m} \delta \hat{\mathbf{r}}_n \delta \hat{\mathbf{r}}_m \right] \sigma_i^+ \sigma_{i+1}^- + \text{h.c.} \\
&= \sum_{i=1}^{N-1} \left[V(\mathbf{r}_{i,i+1}) - \frac{\hbar}{2m\omega} \frac{\partial^2 V(\mathbf{r}_{i,i+1})}{\partial \mathbf{r}_i^2} (p_j^\dagger p_{j+1} + p_{j+1}^\dagger p_j - p_j^\dagger p_j - p_{j+1}^\dagger p_{j+1}) \right] \sigma_i^+ \sigma_{i+1}^- + \text{h.c.}
\end{aligned}$$

The quadrature operators of the phonons are used $\delta \hat{\mathbf{r}}_i = \sqrt{\frac{\hbar}{2m\omega}} (p_i + p_i^\dagger)$. In the following I want to present a brief investigation of the influence of this small second order term on the dynamics of the spins. Ultimately I am interested in the effect of the phonons on the transport of spin excitations via a Rice-Mele scheme. Therefore I will compare the free evolution of the system with and without phonons at conditions mimicking a point in the middle of the Rice-Mele cycle, where the dynamics are the fastest. The results are shown in Fig. 18. As can be seen, for high phonon excitation numbers the difference in the spin excitations during the time period of the whole cycle becomes substantial. Nevertheless, the performance of the Rice-Mele protocol is affected only mildly, and is even slightly improved by the inclusion of the phonons, as can be seen in Fig. 21.

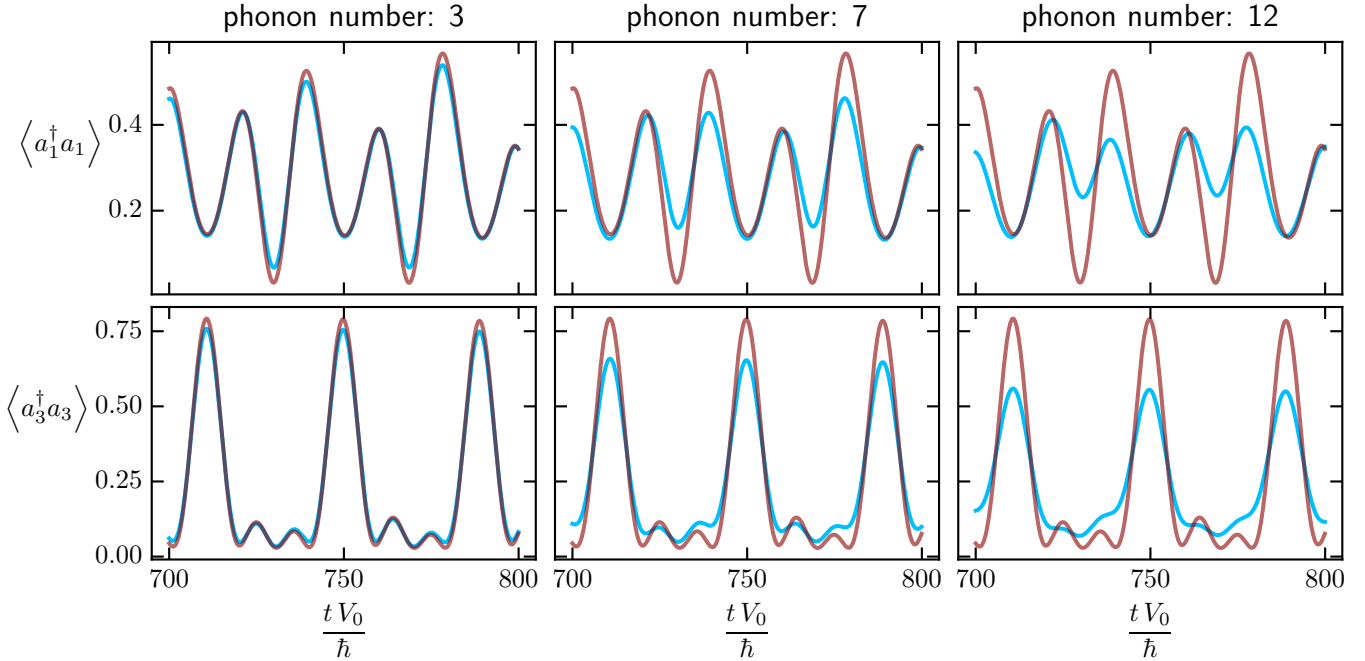


Figure 18: **Comparison of free evolution with and without phonons.** The difference in spin excitation probability between the free evolution with and without phonons is shown for a chain of length $N = 4$. As initial conditions I chose a quadratically decaying excitation profile from the right edge and parameters $b = 1.2r_0$, $h = 0.4r_0$, $a = 3r_0$. The red line shows the trajectory for the case without motional degrees of freedom and the blue one the case with phonons present. For an atom with 12 motional excitations the mean displacement is of the order of $0.15 \mu\text{m}$, which is already around the limit of the expansion to second order being accurate.

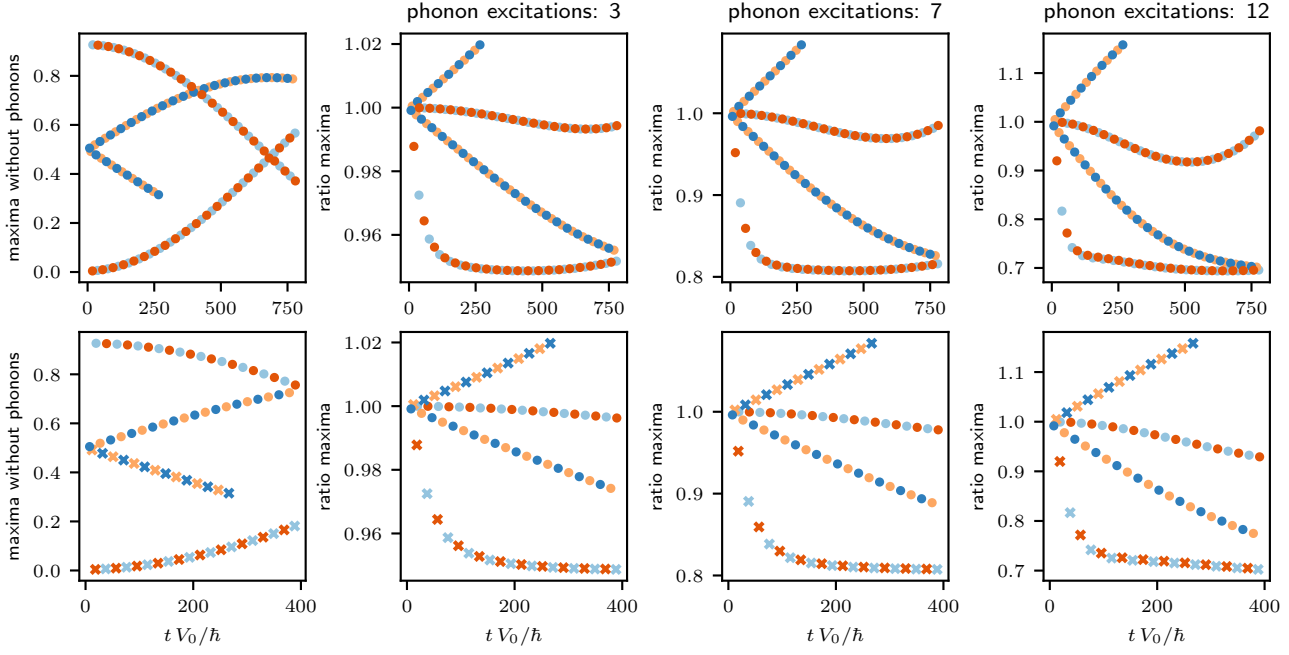


Figure 19: **Comparison of the maxima of the free evolution with and without phonons.** For the same parameters and initial conditions as in Fig. 18 the maxima of the spin excitation probability at the individual sites are compared for the free evolution with and without phonons.

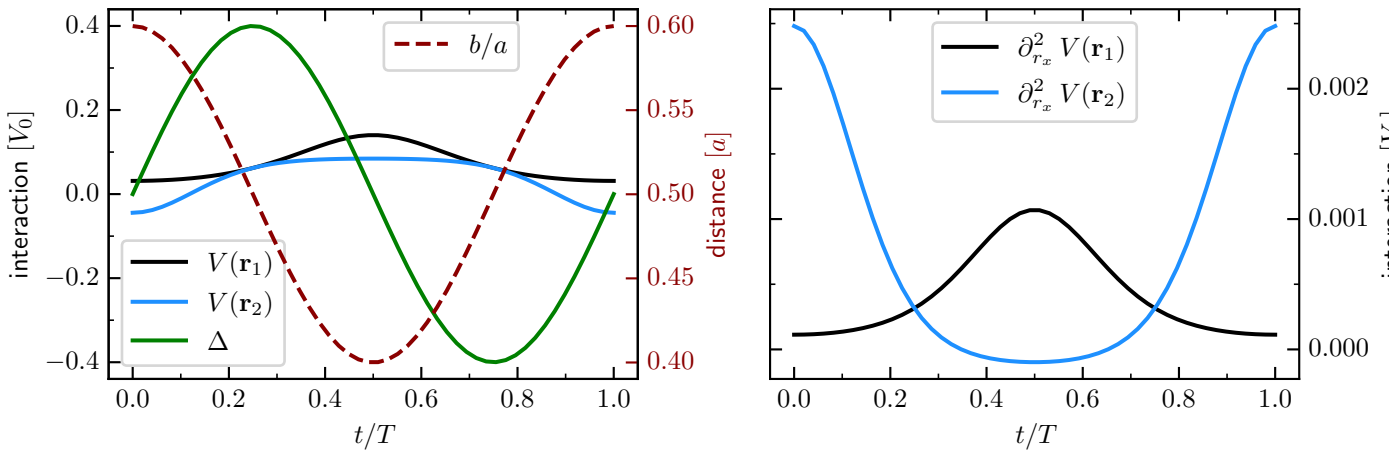


Figure 20: **Rice-Mele scheme.** Depicted is the course of the decisive parameters during the Rice-Mele cycle, i.e. the axial distance between atoms in the unit cell $b = 1.8 - 1.2 - 1.8 r_0$ (the distance orthogonal to the chain axis is kept constant $h = 0.4 r_0$ as well as the distance inside a sublattice $a = 3 r_0$). The detuning takes its maximal value at $\Delta_{\max} = 0.4 V_0$.

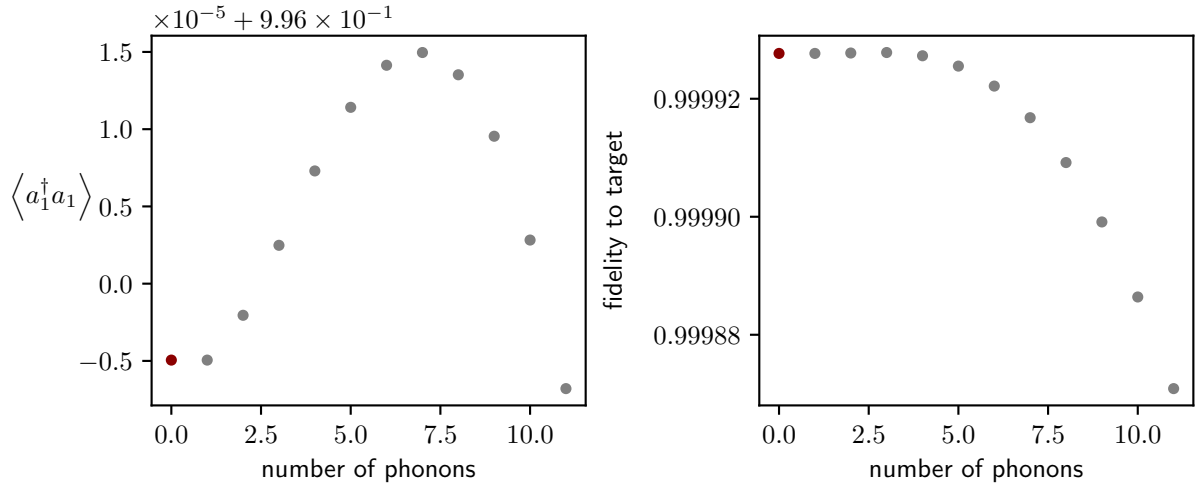


Figure 21: **Transferred excitation in dependence of phonon number.** I show the excitation probability at the left edge at the end of the cycle, when the system is initialized in an eigenstate of the pure spin Hamiltonian localized at the right edge. And parameters of the cycle are $b = 1.8 - 1.2 - 1.8 r_0$, $h = 0.4 r_0$, $a = 3 r_0$, $\Delta_{\max} = 0.4 V_0$.

Spin-phonon coupling

Transport

I want to allow for the propagation of excitations through the chain. The first step towards this is constructing a Hamiltonian which describes the dynamics of a system of Rydberg atoms, where one atom is excited, hence restricting ourselves to the single excitation subspace. The general coupling between the atoms arises from the interaction between the dipole moments of the atoms,

$$V = \sum_{\mathbf{r}} V(\mathbf{r})$$

$$\text{with } V(\mathbf{r}) = V_0 \left(\frac{\hat{\mathbf{d}}_1 \cdot \hat{\mathbf{d}}_2}{r^3} - 3 \frac{(\mathbf{r} \cdot \hat{\mathbf{d}}_1)(\mathbf{r} \cdot \hat{\mathbf{d}}_2)}{r^5} \right).$$

Each individual atom is characterized by its position in the lattice and its atomic energy level which is captured by the non-interacting Hamiltonian H_0 . The total Hamiltonian thus reads

$$H = H_0 + V$$

We will investigate the system in a regime, where the coupling due to the dipole forces is much smaller than the energy gap Δ , which separates the energy of the atoms from their next higher and lower states, i.e. $\|V\| \ll \Delta$. In order to gain an approximate effective Hamiltonian, which acts in a closed way on the one excitation subspace, I will perform a Schrieffer-Wolff transform up to second order [bravyi_schrieffer-wolff_2011]. Let P_0 be the projector on the one excitation subspace. Then an exact alternative Hamiltonian H' can be constructed by the transformation

$$H' = e^S H e^{-S}$$

for which holds, that it is block-diagonal with respect to the subspaces defined by the projections P_0 and $1 - P_0$. Thus the alternative Hamiltonian H' is closed with respect to the one excitation subspace and one can define the exact effective Hamiltonian as

$$H_{\text{eff}} = P_0 e^S H e^{-S} P_0$$

Let ε be of the order of the coupling $\varepsilon = \|V\|$. The transformation generating operator S can be expanded in orders of V .

$$S = \sum_{n=1}^{\infty} \varepsilon^n S_n$$

I will consider the effective Hamiltonian up to second order in the coupling. This lets H_{eff} read

$$H_{\text{eff}} = H_0 P_0 + P_0 V P_0 + \frac{1}{2} P_0 [S_1, V_{\text{od}}] P_0,$$

with $V_{\text{od}} = P_0 V (1 - P_0) + (1 - P_0) V P_0$ being the off-diagonal part of the interaction. The first term of the generator S has the form

$$S_1 = \mathcal{L}(V_{\text{od}})$$

$$\text{with } \mathcal{L}(X) = \sum_{\substack{i,j \in E \\ E_i \neq E_j}} \frac{\langle i | V | j \rangle}{E_i - E_j} |i\rangle \langle j|$$

The sum in the superoperator $\mathcal{L}$ goes over all energy eigenstates, where the i, j combination of states with equal energy are omitted. It can be written via a sum over all energies in the single excitations subspace E_1 and all the other eigenenergies $\tilde{E}$.

$$\mathcal{L}(X) = \sum_{\substack{i \in E_1 \\ j \in \tilde{E}}} \frac{\langle i|V|j\rangle}{E_i - E_j} |i\rangle \langle j| - \text{h.c.}$$

From this the commutator term in the effective Hamiltonian can be calculated.

$$P_0 [S_1, V_{\text{od}}] P_0 = 2 \sum_{\substack{i \in E_1 \\ j \in \tilde{E}}} \frac{|\langle i|V|j\rangle|^2}{E_i - E_j} |i\rangle \langle i| =: H_{\text{eff},2}$$

Thus finally the effective Hamiltonian reads

$$H_{\text{eff}} = H_0 P_0 + P_0 V P_0 + \sum_{\substack{i \in E_1 \\ j \in \tilde{E}}} \frac{|\langle i|V|j\rangle|^2}{E_i - E_j} |i\rangle \langle i|$$

For the effective Hamiltonian the contribution $H_0 P_0$ is just a constant term and can be neglected. Further because of the large relative size of Δ over the interaction strength, I will only include energies into the above sum which differ from the single excitation subspace by Δ . Further the zero excitations subspace does not couple to the one with one excitation in the considered order of expansion. Thus only eigenstates in the double excitation subspace have to be taken into account for the sum.

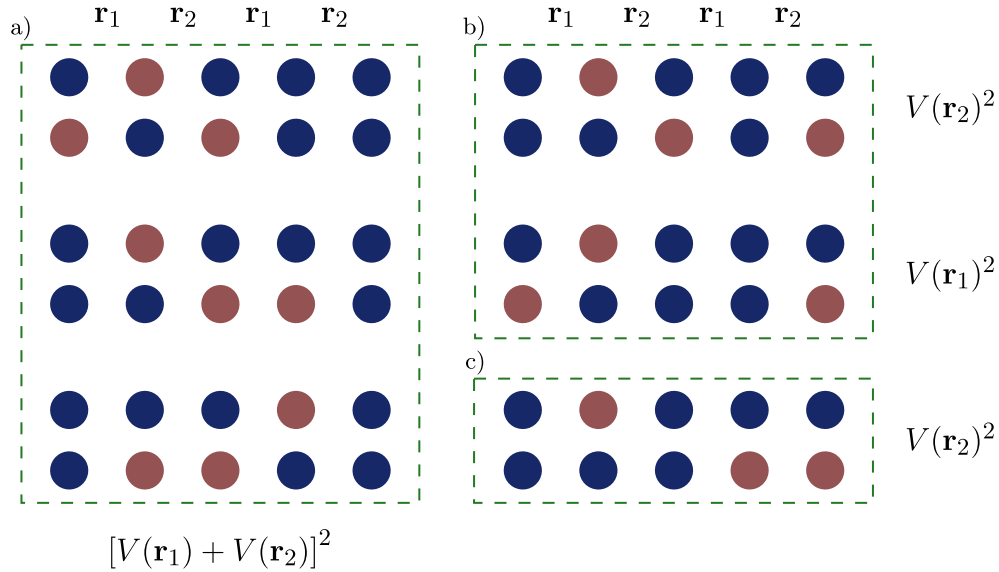


Figure 22: **Coupling schemes.** There are different ways double excited states couple to the single excited ones. The top row always depicts a state of the single, while the bottom row depicts a state of the double excited subspace. Generally configurations of two different subspaces are coupled to each other, when for two neighboring atoms one configuration has the inverted excitation scheme with respect to the other. I assign them to three categories. The first category is depicted in a). It couples two neighboring atoms twice. In the picture all possible cases are shown. For b) only two atoms couple two each other. Here the coupling takes place at a pair of $|e\rangle$ and $|g\rangle$ in both configurations. If the excitation in the single excitation subspace is at position $2 < i < N - 1$, there are $N - 4$ different states in the double excited subspace that contribute a term $V(\mathbf{r}_1)^2$ and the same number for $V(\mathbf{r}_2)^2$. Category c) couples two unexcited sites in the single excitations subspace to two excited sites in the double excited subspace. For this case there are $N/2 - 2$ states that contribute $V(\mathbf{r}_1)^2$ and $N/2 - 2$ states that contribute $V(\mathbf{r}_2)^2$, if the single excited state is excited at atom $2 < i < N - 1$.

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$$H_{\text{eff},2} = \sum_{i=1}^N \tilde{V}_i |i\rangle \langle i|$$

$$\tilde{V}_i = -\frac{1}{\Delta} \begin{cases} 3 [V(\mathbf{r}_1) + V(\mathbf{r}_2)]^2 + (N-4 + \frac{N}{2} - 2) V^2(\mathbf{r}_1) + (N-4 + \frac{N}{2} - 2) V^2(\mathbf{r}_2) & \text{if } 1 \neq i \neq N \\ [V(\mathbf{r}_1) + V(\mathbf{r}_2)]^2 + (N-3 + \frac{N}{2} - 1) V^2(\mathbf{r}_1) + (\frac{N}{2} - 1) V^2(\mathbf{r}_2) & \text{if } i = 1 \\ 2 [V(\mathbf{r}_1) + V(\mathbf{r}_2)]^2 + (N-3 + \frac{N}{2} - 2) V^2(\mathbf{r}_1) + (N-4 + \frac{N}{2} - 1) V^2(\mathbf{r}_2) & \text{if } i = 2 \\ 2 [V(\mathbf{r}_1) + V(\mathbf{r}_2)]^2 + (N-4 + \frac{N}{2} - 1) V^2(\mathbf{r}_1) + (N-3 + \frac{N}{2} - 2) V^2(\mathbf{r}_2) & \text{if } i = N-1 \\ [V(\mathbf{r}_1) + V(\mathbf{r}_2)]^2 + (\frac{N}{2} - 1) V^2(\mathbf{r}_1) + (N-3 + \frac{N}{2} - 1) V^2(\mathbf{r}_2) & \text{if } i = N \end{cases}$$

The part involving interactions within the same subspace can be written as

$$P_0 V P_0 = \sum_{\substack{i=1 \\ j=i+1}}^{N-1} V(\mathbf{r}_{i,j}) \sigma_i^+ \sigma_j^- + \text{h.c.},$$

where $\mathbf{r}_{i,j}$ is the distance between the i th and j th site. I now also want to write $H_{\text{eff},2}$ in terms of a sum of two atom operators. Therefore I introduce the projectors $n_i^e = |e\rangle_i \langle e|_i$ and $n_i^g = |g\rangle_i \langle g|_i$.

$$H_{\text{eff},2} = \sum_{\substack{i=1 \\ j=i+1}}^{N-1} V^2(\mathbf{r}_{i,j}) (n_i^e n_j^g + n_i^g n_j^e + n_i^g n_j^g) + \sum_{i=2}^{N-1} [V(\mathbf{r}_{i-1,i}) + V(\mathbf{r}_{i,i+1})]^2 n_{i-1}^g n_i^e n_{i+1}^g \\ + \sum_{i=3}^{N-1} [V(\mathbf{r}_{i-2,i-1}) + V(\mathbf{r}_{i-1,i})]^2 n_{i-2}^g n_{i-1}^g n_i^e + \sum_{i=2}^{N-2} [V(\mathbf{r}_{i,i+1}) + V(\mathbf{r}_{i+1,i+2})]^2 n_i^e n_{i+1}^g n_{i+2}^g \\ + [V(\mathbf{r}_{1,2}) + V(\mathbf{r}_{2,3})]^2 n_1^e n_2^g n_3^g + [V(\mathbf{r}_{N-2,N-1}) + V(\mathbf{r}_{N-1,N})]^2 n_{N-2}^g n_{N-1}^g n_N^e$$

A last step is possible since in the single excitation subspace terms like $n_i^e n_j^g$ simplify to just n_i^e .

$$= \sum_{i=2}^{N-1} \left(V^2(\mathbf{r}_{i-1,i}) + V^2(\mathbf{r}_{i,i+1}) + [V(\mathbf{r}_{i-1,i}) + V(\mathbf{r}_{i,i+1})]^2 \right) n_i^e \\ + \sum_{i=3}^{N-1} [V(\mathbf{r}_{i-2,i-1}) + V(\mathbf{r}_{i-1,i})]^2 n_i^e + \sum_{i=2}^{N-2} [V(\mathbf{r}_{i,i+1}) + V(\mathbf{r}_{i+1,i+2})]^2 n_i^e \\ + \left(V^2(\mathbf{r}_{1,2}) + [V(\mathbf{r}_{1,2}) + V(\mathbf{r}_{2,3})]^2 \right) n_1^e + \left(V^2(\mathbf{r}_{N-1,N}) + [V(\mathbf{r}_{N-2,N-1}) + V(\mathbf{r}_{N-1,N})]^2 \right) n_N^e \\ + \sum_{i=1}^{N-1} V(\mathbf{r}_{i,i+1}) n_i^g n_{i+1}^g$$

First order approximation

For the first approach I will continue with the Hamiltonian up to first order in the dipole interaction, i.e. without $H_{\text{eff},2}$. The phonons are introduced by again expanding the interaction up to second order. I will restrict the consideration also to the constant phonon excitation subspace. For this reason the first order expansion of the dipole interaction will be ignored, as it couples subspaces with different phonon numbers. The Hamiltonian thus reads

$$H = \sum_{i=1}^{N-1} \left[V(\mathbf{r}_{i,i+1}) + \sum_{n,m \in \{i,i+1\}} \frac{\partial^2 V(\mathbf{r}_{i,i+1})}{\partial \mathbf{r}_n \partial \mathbf{r}_m} \delta \hat{\mathbf{r}}_n \delta \hat{\mathbf{r}}_m \right] \sigma_i^+ \sigma_{i+1}^- + \text{h.c.}$$

The hats in $\delta\hat{\mathbf{r}}$ indicates operators. Here also the operators are projected on the constant phonon subspace. What has to be accounted for is that this second order expansion is only valid if the displacement is small, which means $1/\sqrt{m\omega} \ll 1$ as $\|a\| \sim 1$.

$$\left| 1 - \frac{V(\mathbf{r} + \delta\mathbf{r})}{V(\mathbf{r})} \right| \ll 1$$

$$\left| \frac{\partial^2 V(\mathbf{r})}{\partial \mathbf{r}^2} \delta\mathbf{r}^2 \right| \ll |V(\mathbf{r})|$$

When quantizing the displacements to phonon operators $\delta\mathbf{r}^2 \rightarrow \frac{2}{m\omega} a^\dagger a$, one finds the condition

$$\frac{2}{m\omega} \left| \frac{\partial^2 V(\mathbf{r})}{\partial \mathbf{r}^2} \right| \ll |V(\mathbf{r})|,$$

as $a^\dagger a$ is of the order of 1.

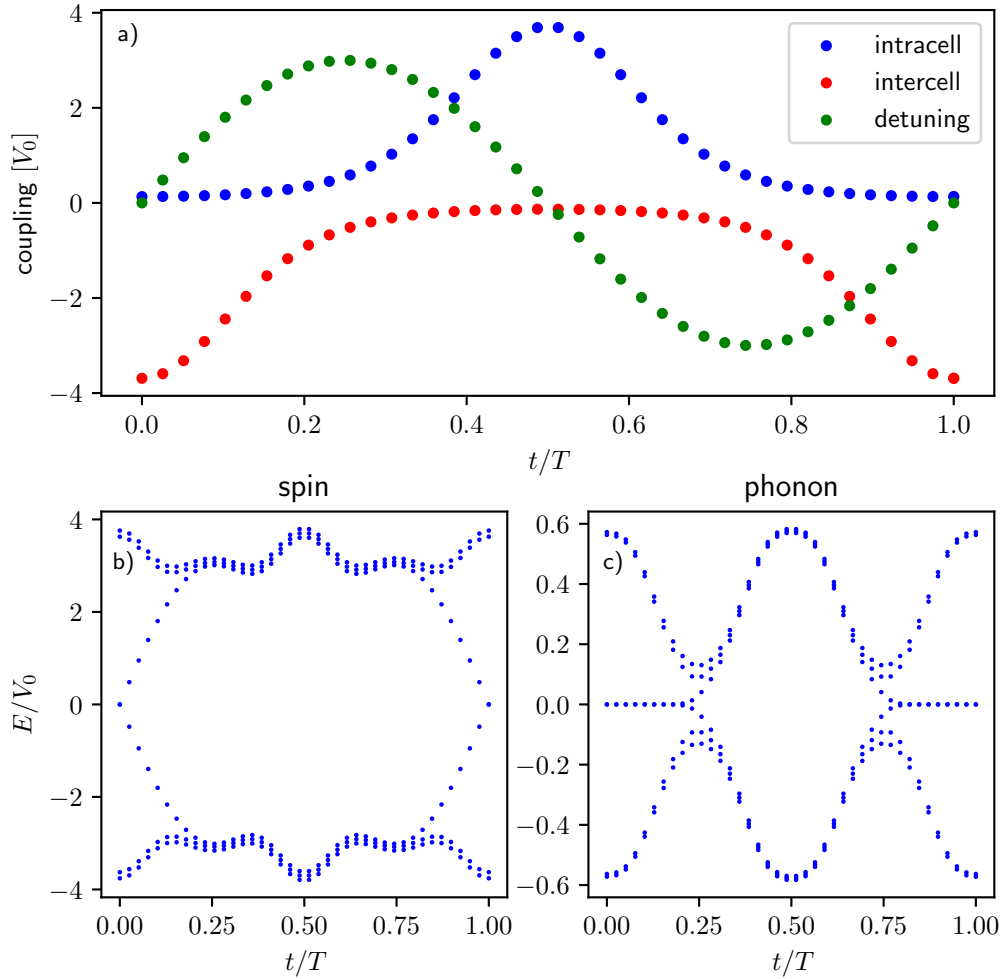


Figure 23: **Spin transfer protocol.** In a the course of the interaction during the transfer protocol is shown. b-c show the spectrum of the spin and phonon Hamiltonian respectively during the pump cycle each for one excitation in the respective degrees of freedom. For the spin Hamiltonian a pure dipole dipole interaction is used, where the dipole moments of the atoms are orientated in way that puts the interactions within the same sublattice to zero. This choice improves the accuracy of the nearest neighbor assumption. The phonon spectrum is received by the methods used in the previous chapters. It therefore depicts a scenario without spin degrees of freedom.

As a next step I want to first discard the phonon dynamics and realize a transfer protocol for a spin excitation through the chain. For this purpose I implement a Rice-Mele model configuration for the

chain of Rydberg atoms. This is achieved by including an onsite potential with alternating sign of the form

$$H_{\text{OS}} = \sum_{i=1}^N (-1)^i \Delta \sigma_i^+ \sigma_i^-.$$

Note that the detuning Δ is now a different parameter from above. By changing the position and detuning in an adiabatic cyclic way, one can implement a transfer protocol, which is depicted in Fig. 23. The system is initialized with the eigenstate of the spin Hamiltonian located at the edge. This is possible, as I start the transfer protocol in the topological phase. After the the transfer period I compare the evolved state with the target state, which is the eigenstate located at the other side. I use a chain of six atoms in total. The resulting fidelity between final and target state for this protocol is 0.9996.

I can now compare which influence the introduction of photons has on the transfer quality. Therefore I test the effect of four different phonon initial states on the transfer process. The different initial phonon states can be read of Fig. 24. Fig. 25 shows that the evolvment of the phonon degrees of freedom does not change the quality of the transfer to a very considerable portion, where I investigated how the strength of the phonon-phonon interactions effect the fidelity of final and target state. In Fig. 26 I show the time evolution of the phononic degrees of freedom for two initial phonon excitations.

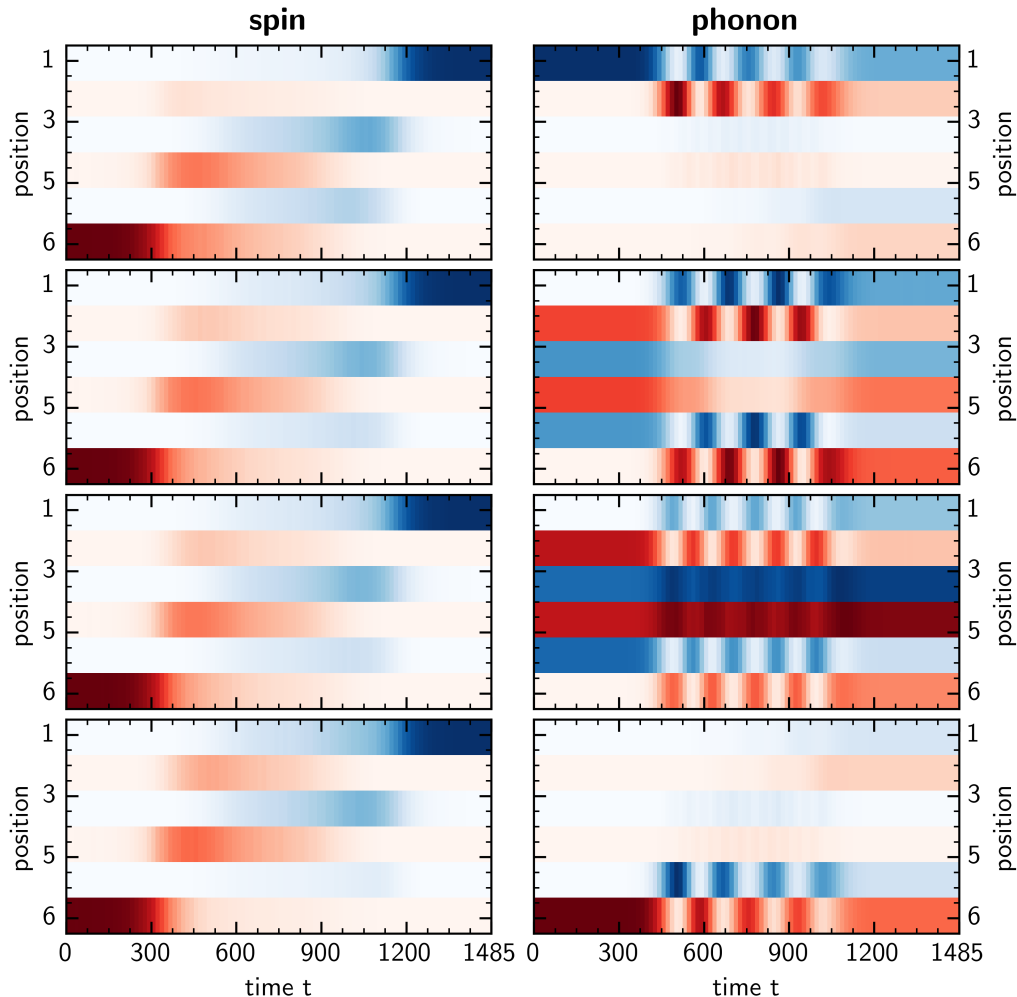


Figure 24: **Spin transfer process.** The time evolution of the spin and phonon degrees of freedom is depicted during the adiabatic transfer protocol. For the same transfer protocol the system was prepared in different initial phonon configurations, which stored one excitation in total.

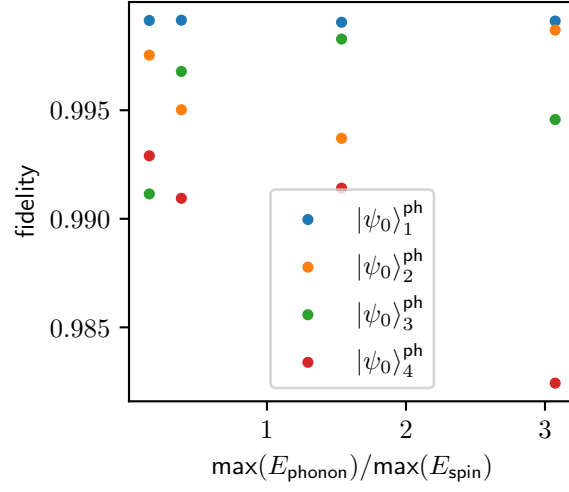


Figure 25: **Fidelity in dependence of phonon interaction strength.** For the 4 initial phonon configurations introduced in Fig. 24 (the labeling runs from top to bottom) different ratios of spin dipole interactions to phonon interactions are chosen. The fidelity between the final and target state is shown in dependence of interaction ratio and initial phonon state.

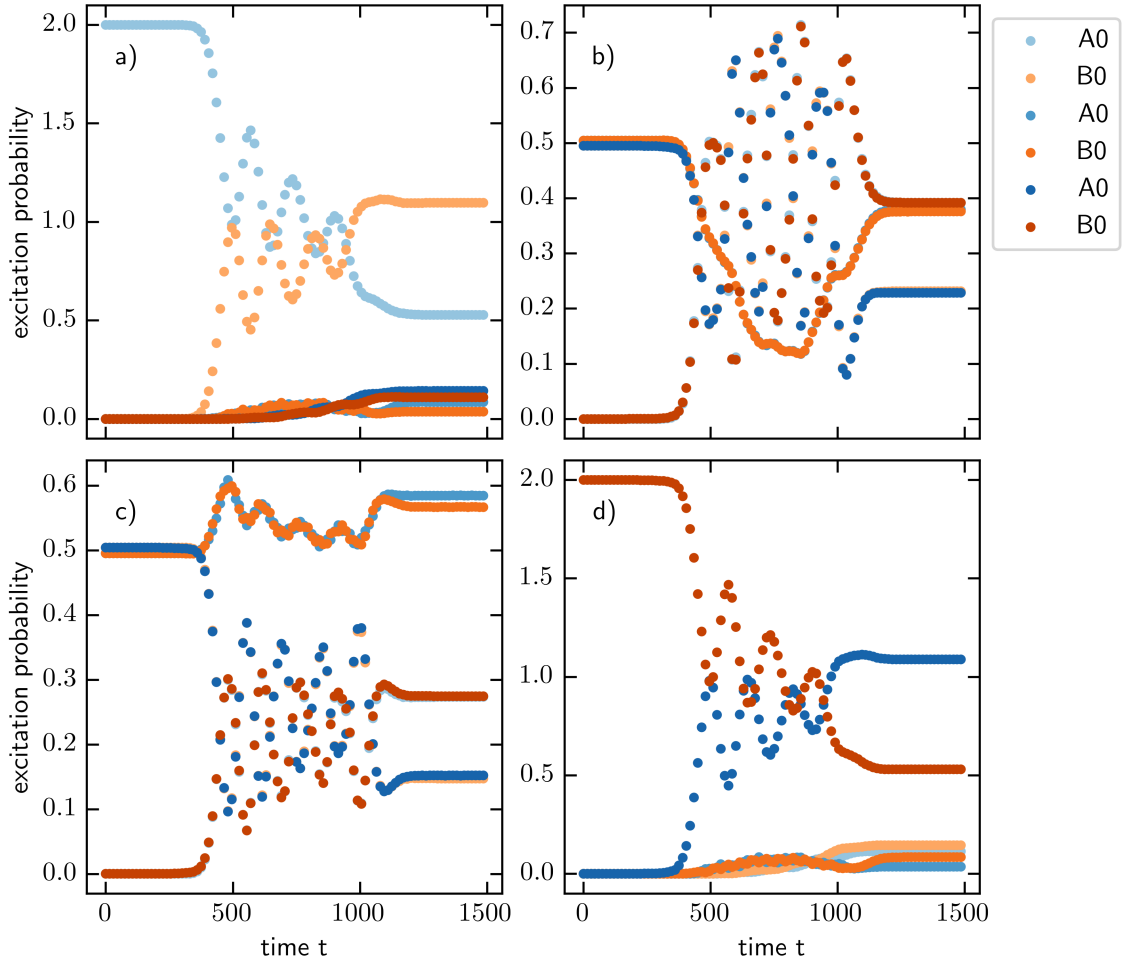


Figure 26: **Phonon evolution for two excitations.** For two phonon excitations the evolution of the phonons during the transfer period is depicted.

Winding number of the spin-phonon coupled Hamiltonian

Next I want to determine the topological phase of the Hamiltonian where spin and motional degrees of freedom are coupled in first order Schrieffer-Wolff approximation. I will again consider only nearest neighbor interaction and will also constrain the dynamics to the single excitation subspace, both in spin and phonon degrees of freedom. For this purpose I will write the interactions in terms of fermionic spin and phonon operators. a_i and b_i ($p_{A,i}$ and $p_{B,i}$) are spin (phonon) annihilation operators acting, respectively, on the sublattice A and B sites in unit cell i . The Hamiltonian then reads

$$\begin{aligned}
 H &= H_{\text{sp}} + H_{\text{spph}} \\
 \text{with } H_{\text{sp}} &= \sum_i V_1 a_i^\dagger b_i + V_2 b_i^\dagger a_{i+1} + \text{h.c.} \\
 H_{\text{spph}} &= \sum_i V_3 \left[p_{A,i}^\dagger p_{B,i} + p_{B,i}^\dagger p_{A,i} - p_{A,i}^\dagger p_{A,i} - p_{B,i}^\dagger p_{B,i} \right] a_i^\dagger b_i + \text{h.c.} \\
 &\quad + \sum_i V_4 \left[p_{A,i+1}^\dagger p_{B,i} + p_{B,i}^\dagger p_{A,i+1} - p_{A,i+1}^\dagger p_{A,i+1} - p_{B,i}^\dagger p_{B,i} \right] a_{i+1}^\dagger b_i + \text{h.c.}
 \end{aligned}$$

Here the intracell couplings V_1 , V_3 , and the intercell couplings V_2 , V_4 are constants which depend on the parameters distance and dipole moment. Transforming this Hamiltonian into momentum space by simply performing a Fourier transform of every operator does not simplify the description and does not allow for an analysis of the winding number.

But by building collective operators of spin and phonon degrees of freedom one is able to make use of the one dimensional character of the chain and achieve a momentum space Hamiltonian with decoupled momenta. First I restrict myself to the spin phonon coupled term of the Hamiltonian. I group all operators in H_{spph} acting on the same sublattice together and combine them into one operator, defined on an adapted space. By this procedure I get a couple of new operators. All of them act on the same space, which is defined on one site.

$$\begin{pmatrix} s_i^\uparrow p_i^\uparrow \\ s_i^\uparrow p_i^\downarrow \\ s_i^\downarrow p_i^\uparrow \\ s_i^\downarrow p_i^\downarrow \end{pmatrix} =: \begin{pmatrix} |1\rangle \\ |2\rangle \\ |3\rangle \\ |4\rangle \end{pmatrix}$$

For $D \in \{A, B\}$ and $d \in \{a, b\}$ I define

$$\begin{aligned}
 c_i^{\bar{D}, D} &:= p_{D,i}^\dagger d_i = |3\rangle \langle 2| \\
 c_i^{D, \bar{D}} &:= p_{D,i} d_i^\dagger = |2\rangle \langle 3| \\
 c_i^{\bar{D}, \bar{D}} &:= p_{D,i}^\dagger d_i^\dagger = |1\rangle \langle 4| \\
 c_i^{D, D} &:= p_{D,i} d_i = |4\rangle \langle 1| \\
 c_i^{\bar{D}} &:= d_i^\dagger = |1\rangle \langle 3| + |2\rangle \langle 4| \\
 c_i^D &:= d_i = |3\rangle \langle 1| + |4\rangle \langle 2| \\
 c_i^{\bar{D}, D, D} &:= p_{D,i}^\dagger p_{D,i} d_i = |3\rangle \langle 1| \\
 c_i^{\bar{D}, D, \bar{D}} &:= p_{D,i}^\dagger p_{D,i} d_i^\dagger = |1\rangle \langle 3|
 \end{aligned}$$

what lets the Hamiltonian read

$$\begin{aligned}
 H_{\text{spph}} &= \sum_i V_3 \left[c_i^{\bar{A}, \bar{A}} c_i^{B, B} + c_i^{A, \bar{A}} c_i^{\bar{B}, B} - c_i^{\bar{A}, A, \bar{A}} c_i^B - c_i^{\bar{A}} c_i^{\bar{B}, B, B} \right] + \text{h.c.} \\
 &\quad + \sum_i V_4 \left[c_{i+1}^{\bar{A}, \bar{A}} c_i^{B, B} + c_{i+1}^{A, \bar{A}} c_i^{\bar{B}, B} - c_{i+1}^{\bar{A}, A, \bar{A}} c_i^B - c_{i+1}^{\bar{A}} c_i^{\bar{B}, B, B} \right] + \text{h.c.}
 \end{aligned}$$

Because of the single excitation restriction, exactly one phonon and spin has to be present at the two neighboring sites for a term acting on those two neighboring sites to contribute. Therefore the combined space of the two neighboring sites collapses to the dimension of the one site space, represented by the

basis

$$\begin{pmatrix} s_1^\uparrow p_1^\uparrow & s_2^\downarrow p_2^\downarrow \\ s_1^\uparrow p_1^\downarrow & s_2^\downarrow p_2^\uparrow \\ s_1^\downarrow p_1^\uparrow & s_2^\uparrow p_2^\downarrow \\ s_1^\downarrow p_1^\downarrow & s_2^\uparrow p_2^\uparrow \end{pmatrix}$$

The Hamiltonian in k -space then reads

$$\begin{aligned} H_{\text{spph}} &= \sum_k V_3 \left[c_k^{\bar{A},\bar{A}} c_k^{B,B} + c_k^{A,\bar{A}} c_k^{\bar{B},B} - c_k^{\bar{A},A,\bar{A}} c_k^B - c_k^{\bar{A}} c_k^{\bar{B},B,B} \right] + \text{h.c.} \\ &+ \sum_k V_4 e^{ika} \left[c_k^{\bar{A},\bar{A}} c_k^{B,B} + c_k^{A,\bar{A}} c_k^{\bar{B},B} - c_k^{\bar{A},A,\bar{A}} c_k^B - c_k^{\bar{A}} c_k^{\bar{B},B,B} \right] + \text{h.c.} \\ &= \sum_k (V_3 + V_4 e^{ika}) \left[c_k^{\bar{A},\bar{A}} c_k^{B,B} + c_k^{A,\bar{A}} c_k^{\bar{B},B} - c_k^{\bar{A},A,\bar{A}} c_k^B - c_k^{\bar{A}} c_k^{\bar{B},B,B} \right] + \text{h.c.} \\ &=: \sum_k H_{\text{spph}}^k \end{aligned}$$

The blocks in k -space then read

$$H_{\text{spph}} = \begin{pmatrix} 0 & 0 & -(V_3 + V_4 e^{ika}) & V_3 + V_4 e^{ika} \\ 0 & 0 & V_3 + V_4 e^{ika} & -(V_3 + V_4 e^{ika}) \\ -(V_3 + V_4 e^{-ika}) & V_3 + V_4 e^{-ika} & 0 & 0 \\ V_3 + V_4 e^{-ika} & -(V_3 + V_4 e^{-ika}) & 0 & 0 \end{pmatrix}$$

When considering the total Hamiltonian including H_{sp} one has to expand the space for the collective operators as now the Hamiltonian does not only couple states where spin and phonon excitations are next to each other. But only two new states have to be introduced, $s_1^\uparrow p_1^\downarrow s_2^\downarrow p_2^\downarrow$ and $s_1^\downarrow p_1^\downarrow s_2^\uparrow p_2^\downarrow$. The basis is now family

$$\begin{pmatrix} s_1^\uparrow p_1^\uparrow & s_2^\downarrow p_2^\downarrow \\ s_1^\uparrow p_1^\downarrow & s_2^\downarrow p_2^\uparrow \\ s_1^\downarrow p_1^\uparrow & s_2^\downarrow p_2^\downarrow \\ s_1^\downarrow p_1^\downarrow & s_2^\uparrow p_2^\downarrow \\ s_1^\uparrow p_1^\downarrow & s_2^\uparrow p_2^\uparrow \\ s_1^\downarrow p_1^\downarrow & s_2^\uparrow p_2^\downarrow \end{pmatrix} =: \begin{pmatrix} |1\rangle \\ |2\rangle \\ |3\rangle \\ |4\rangle \\ |5\rangle \\ |6\rangle \end{pmatrix}$$

In this basis the spin Hamiltonian in momentum space H_{sp}^k reads

$$H_{\text{sp}}^k = \begin{pmatrix} 0 & \alpha \\ \alpha^\dagger & 0 \end{pmatrix}$$

with $\alpha = (V_1 + V_2 e^{ika}) \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

The total Hamiltonian in momentum space thus reads

$$H^k = \begin{pmatrix} 0 & \beta \\ \beta^\dagger & 0 \end{pmatrix}$$

with $\beta = \begin{pmatrix} V_1 - V_3 + (V_2 - V_4) e^{ika} & V_3 + V_4 e^{ika} & 0 \\ V_3 + V_4 e^{ika} & V_1 - V_3 + (V_2 - V_4) e^{ika} & 0 \\ 0 & 0 & V_1 + V_2 e^{ika} \end{pmatrix}$

The eigenvalues of β are

$$\begin{aligned} \beta_1 &= V_1 - 2V_3 + (V_2 - 2V_4) e^{ika} \\ \beta_2 = \beta_3 &= V_1 + V_2 e^{ika} \end{aligned}$$

Problems

The main practical problem is that the total Hamiltonian is heavily degenerate $\sim N$. The winding number can however only take values between 0 and 3. Which eigenvalues are the topological protected ones?

This might be not due to an erroneous derivation of the momentum space Hamiltonian. But I don't know. At first glance there is a problem with the kernel of the Hamiltonian. In real space it has a finite dimension of the magnitude of N . But in momentum space the kernel seems to vanish. I think the reason is that I only considered a basis for the combined coupling operators, which actually couple, hence discarding the kernel. But I think this is no unusual proceed. I don't know if the winding number is valid for degenerate Hamiltonians.

- Go to higher dimensions, e.g. transport on the closed surface of a 2D-lattice
- Spin-phonon coupling:
 - What effect has the spin-phonon coupling on transport performance?
 - Is it possible to transport both spins and phonons simultaneously?

Spin-phonon coupling

$$H = \sum_j H_{\text{ph}}^{j,j+1} a_j^\dagger a_{j+1} + \text{h.c.}$$

with

$$H_{\text{ph}}^{j,j+1} = V_{j,j+1}^{\text{der}} (p_j^\dagger p_{j+1} + p_{j+1}^\dagger p_j - p_j^\dagger p_j - p_{j+1}^\dagger p_{j+1})$$

a_j and p_j are spin and phonon annihilation operators respectively.

Attempt of analytic solution for the time evolution (3 atoms)

I'm considering 3 atoms with the Hamiltonian

$$H = H_{12} a^\dagger b + H_{23} b^\dagger c + \text{h.c.}$$

with $H_{ij} = p_i^\dagger p_j + p_j^\dagger p_i - p_i^\dagger p_i - p_j^\dagger p_j$

I want to calculate the time evolution of the occupation operator $a^\dagger a$.

$$a^\dagger a(t) = e^{iHt} a^\dagger a e^{-iHt} = \sum_n \frac{(it)^n}{n!} [H, a^\dagger a]_n =: \sum_n \frac{(it)^n}{n!} C_n$$

with $[H, a^\dagger a]_n = [H, [H, a^\dagger a]_{n-1}]$.

$$C_2 = 2H_{12}^2 (a^\dagger a - b^\dagger b) + H_{23} H_{12} c^\dagger a + H_{12} H_{23} a^\dagger c$$

$$C_3 = (4H_{12}^3 + H_{23}^2 H_{12}) b^\dagger a - (4H_{12}^3 + H_{12} H_{23}^2) a^\dagger b - 3H_{23} H_{12}^2 c^\dagger b + 3H_{12}^2 H_{23} b^\dagger c$$

$$C_4 = (8H_{12}^4 + 2H_{12} H_{23}^2 H_{12}) a^\dagger a - (8H_{12}^4 + 4H_{12}^2 H_{23}^2 + 4H_{23}^2 H_{12}^2) b^\dagger b + 6H_{23} H_{12}^2 H_{23} c^\dagger c$$

$$+ (7H_{23} H_{12}^3 + H_{23}^3 H_{12}) c^\dagger a + (7H_{12}^3 H_{23} + H_{12} H_{23}^3) a^\dagger c$$

Concentrating on the odd indices one can determine a recursion relation for the coefficients of $a^\dagger b$ and $b^\dagger c$.

$$c'_{ab} = H_{12}^2 c_{ab} + c_{ab} (H_{12}^2 + H_{23}^2) + 2H_{12} c_{ab}^\dagger H_{12} - 2H_{12} c_{bc} H_{23} - H_{12} H_{23} c_{bc}^\dagger$$

$$c'_{bc} = (H_{12}^2 + H_{23}^2) c_{bc} + c_{bc} H_{23}^2 + 2H_{23} c_{bc}^\dagger H_{23} - 2H_{12} c_{ab} H_{23} - c_{ab}^\dagger H_{12} H_{23}$$

By redefining the coefficients as

$$c_{ab} =: H_{12} \alpha, \quad c_{bc} =: \beta H_{23}$$

one can reach at a slightly simpler recursion relation

$$\alpha_{n+1} = H_{12}^2 \alpha_n + \alpha_n (H_{12}^2 + H_{23}^2) + 2\alpha_n^\dagger H_{12}^2 + 2\beta_n H_{23}^2 + H_{23}^2 \beta_n^\dagger$$

$$\beta_{n+1} = (H_{12}^2 + H_{23}^2) \beta_n + \beta_n H_{23}^2 + 2H_{23}^2 \beta_n^\dagger + 2H_{12}^2 \alpha_n + \alpha_n^\dagger H_{12}^2$$

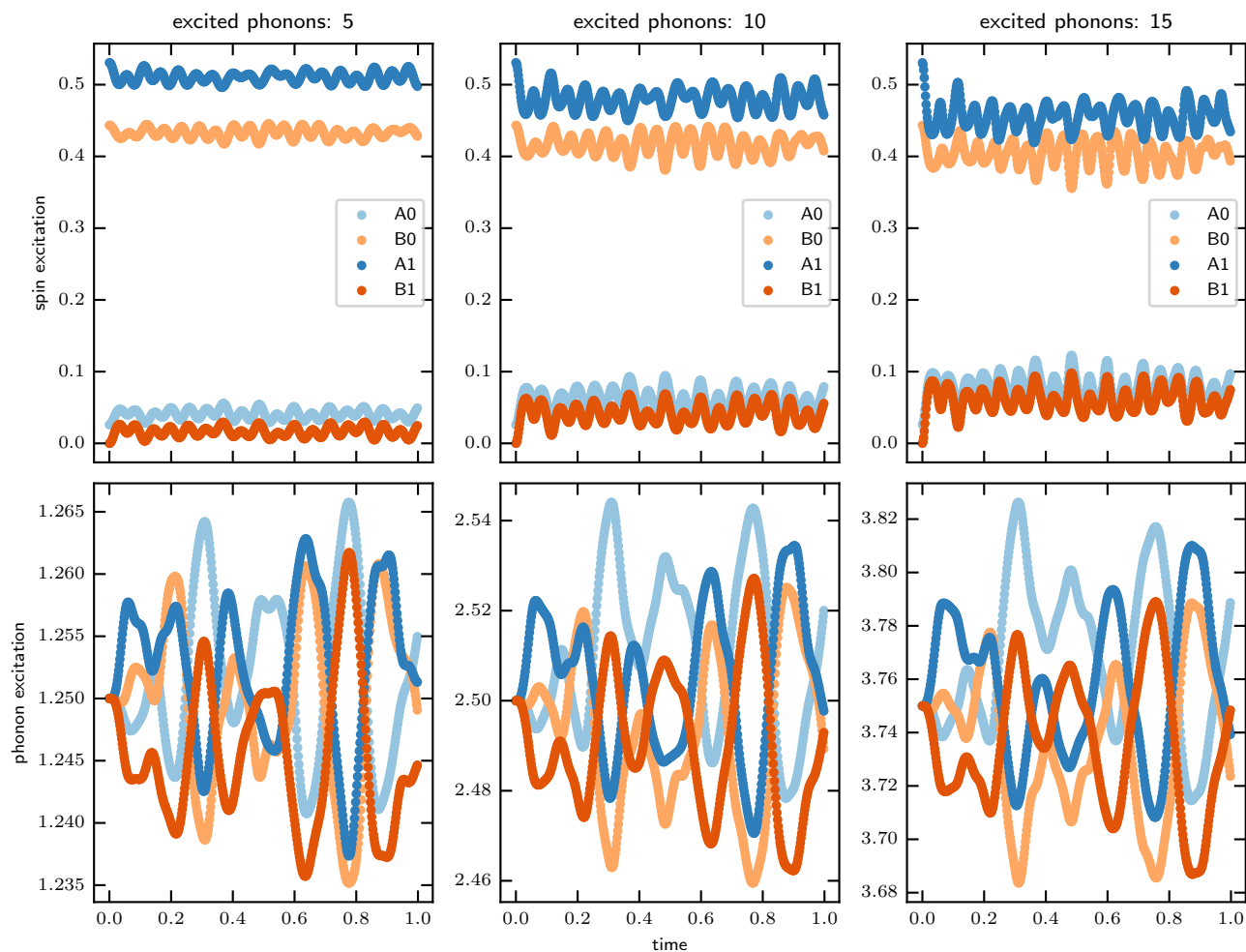


Figure 27: Evolution with only spin-phonon coupling term.

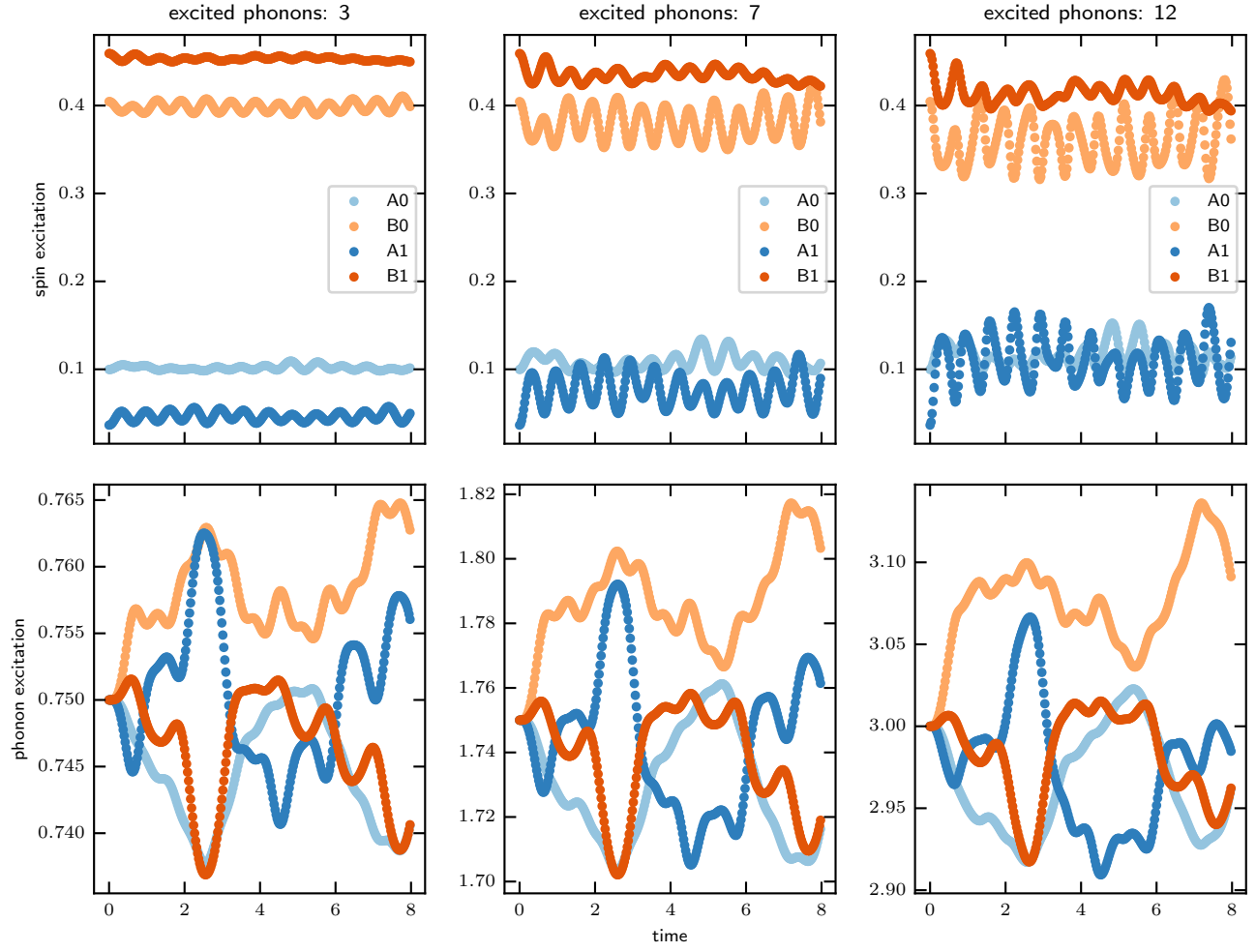


Figure 28: **Evolution with only spin-phonon coupling term.**

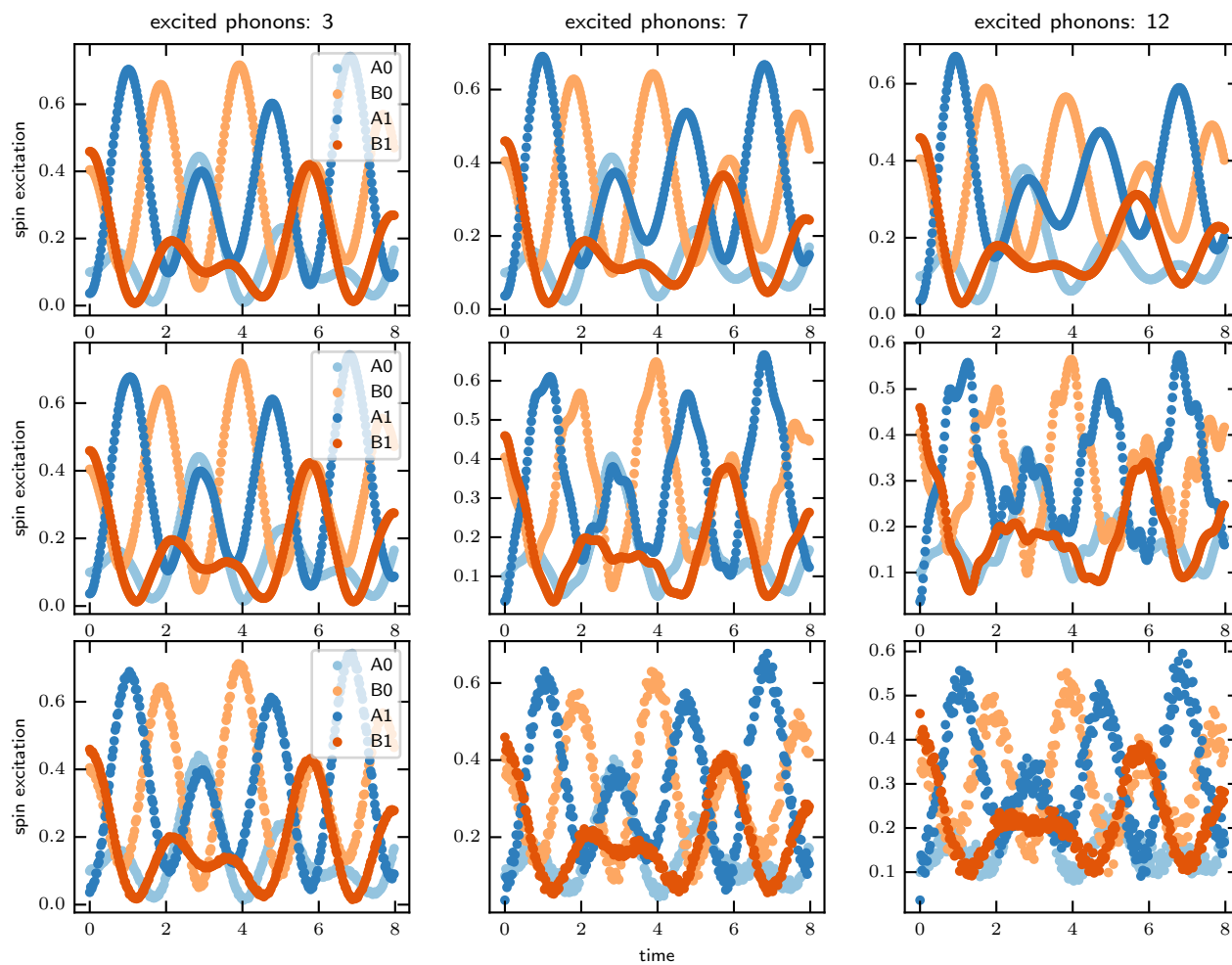


Figure 29: Summary spin evolution

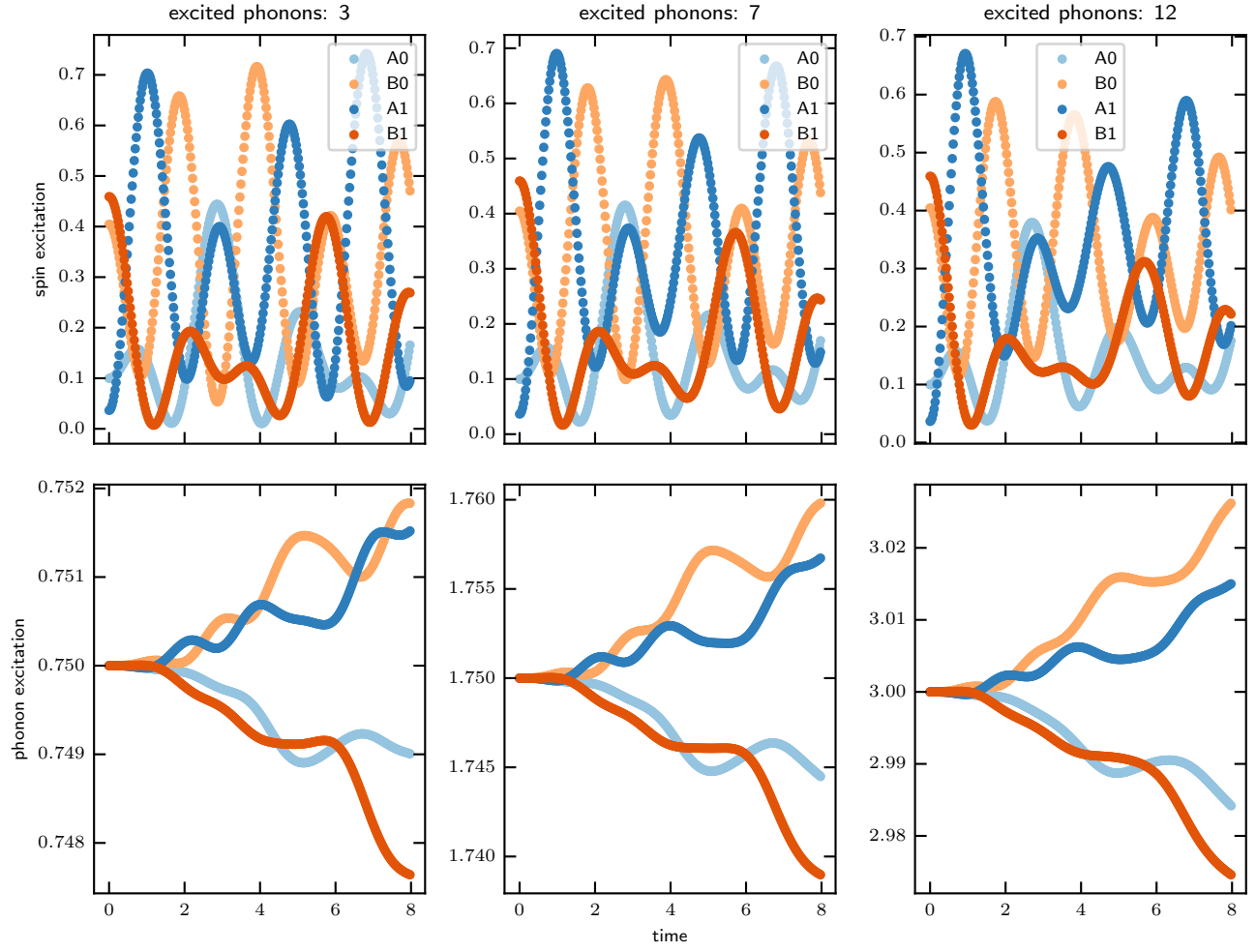


Figure 30: Evolution with full Hamiltonian ratio spin only: coupling 10:1.

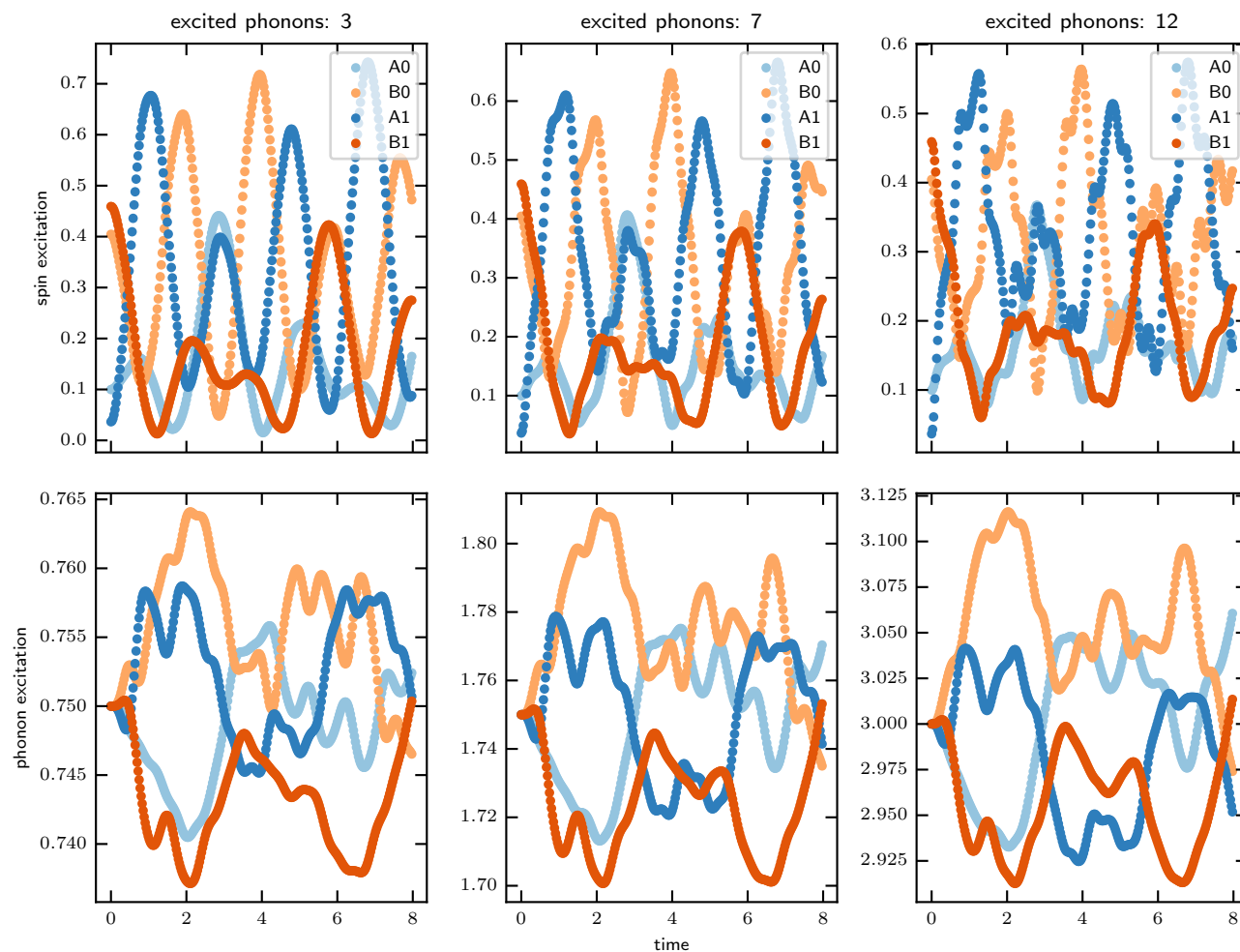


Figure 31: Evolution with full Hamiltonian ratio spin only: coupling 1:1.

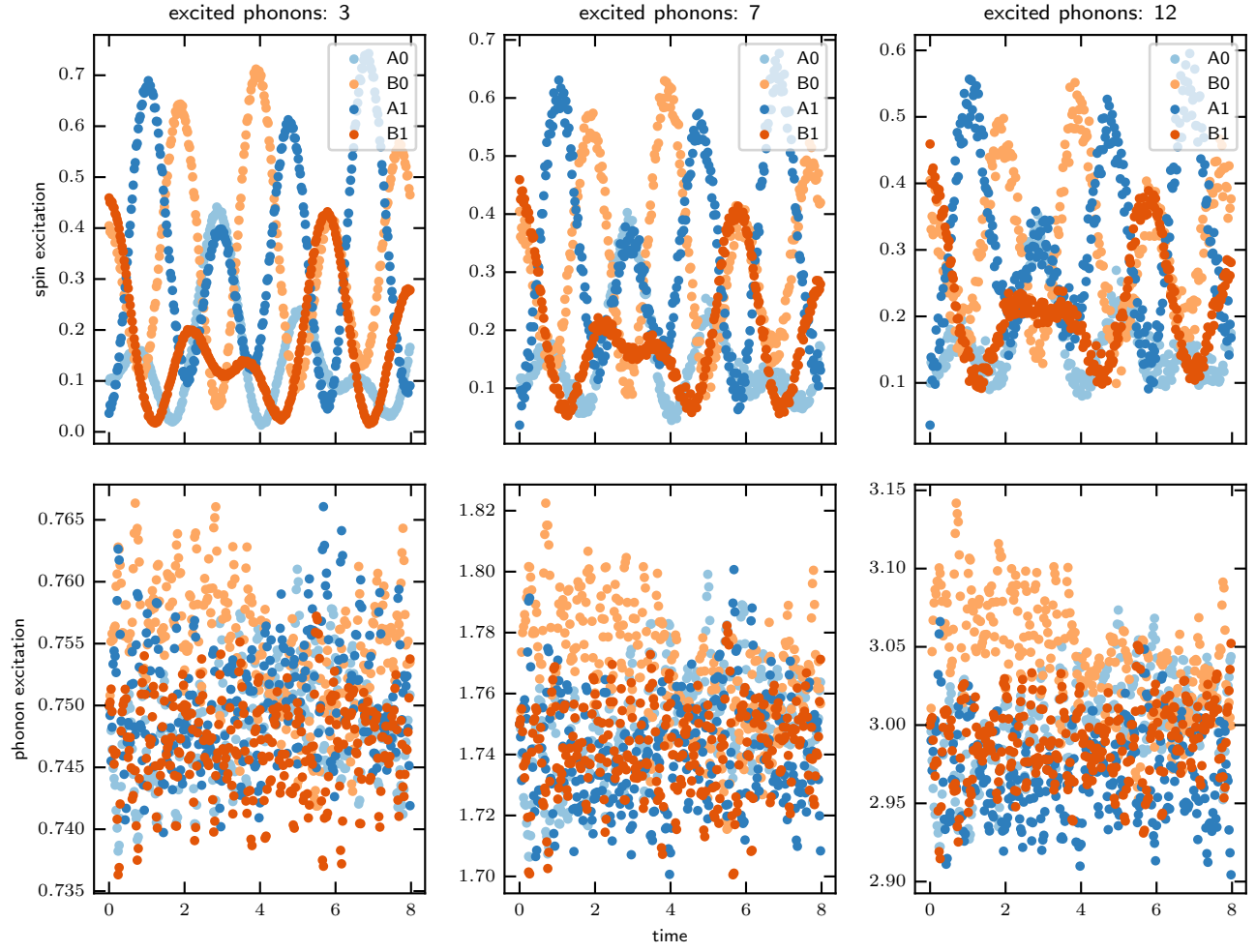


Figure 32: Evolution with full Hamiltonian ratio spin only: coupling 1:10.

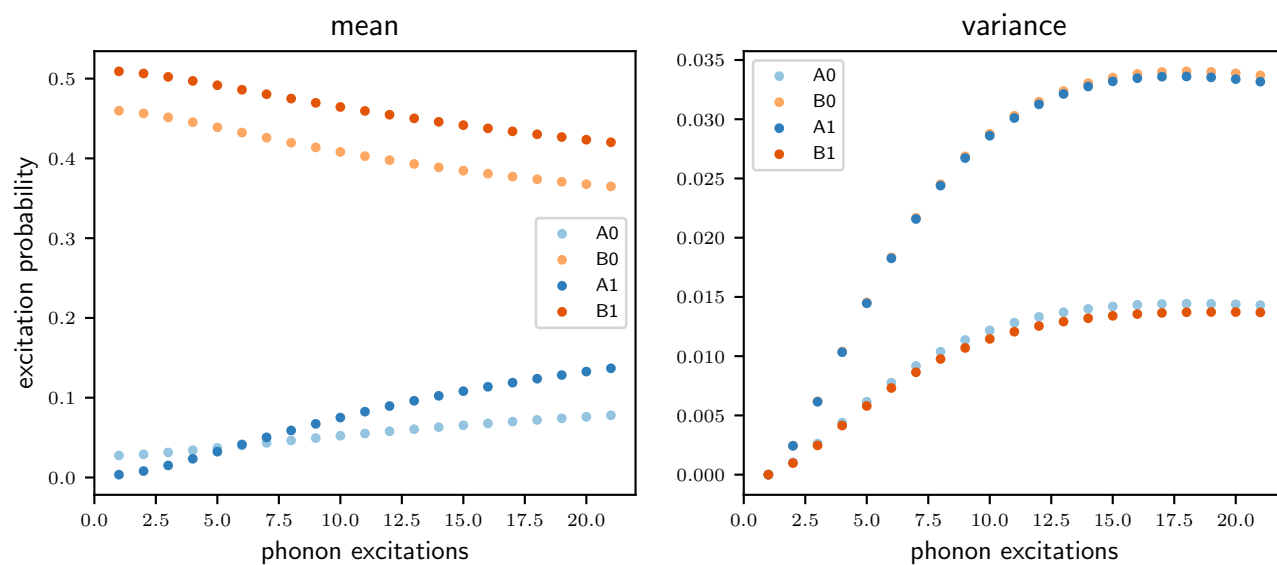
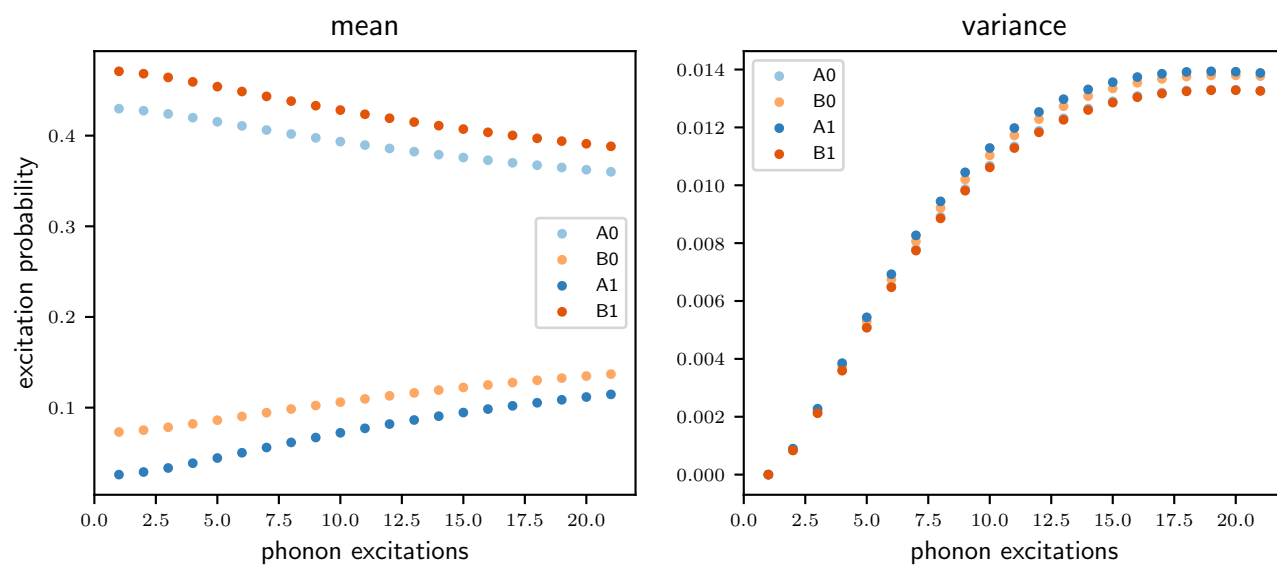
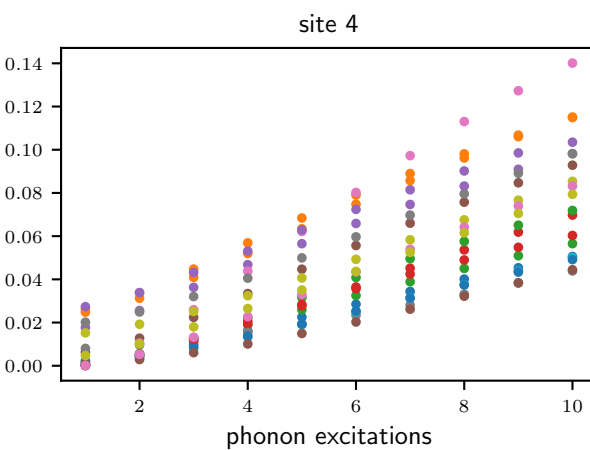
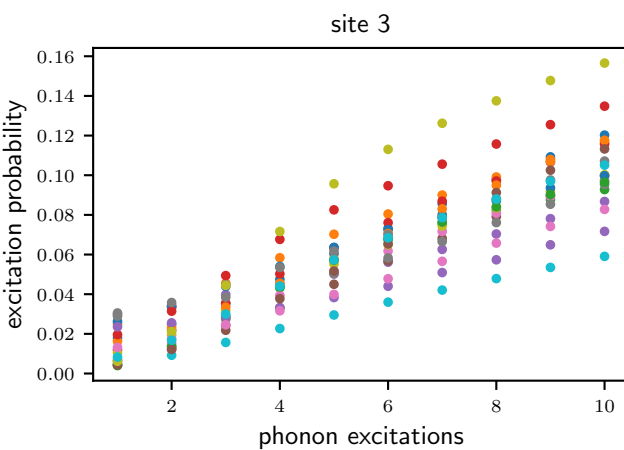
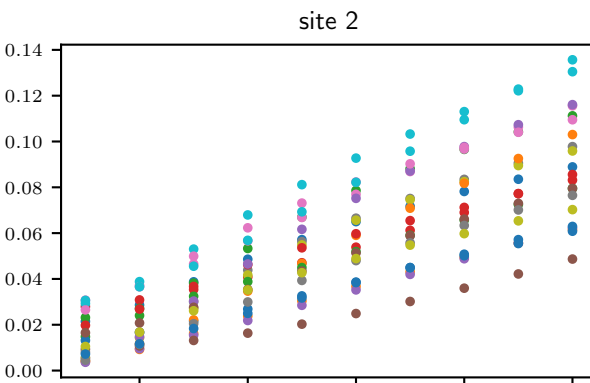
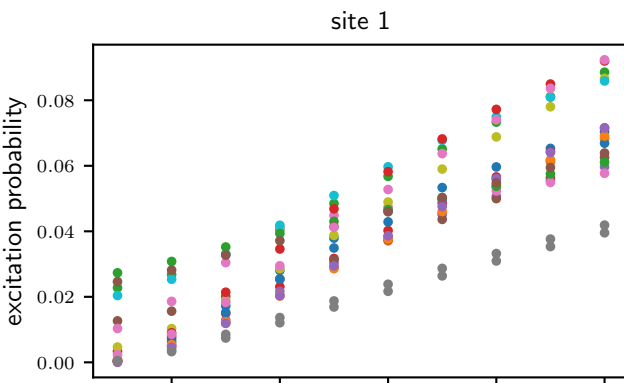
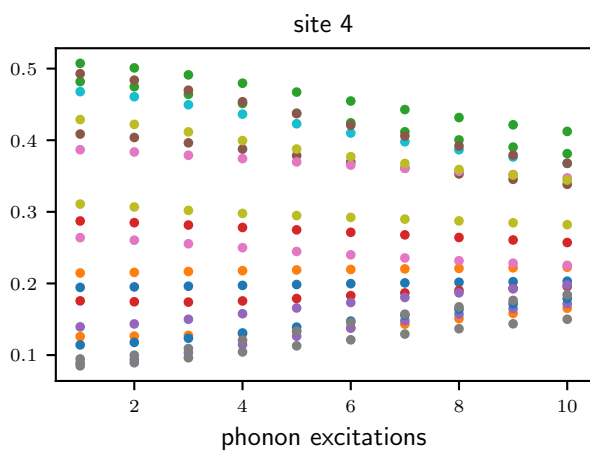
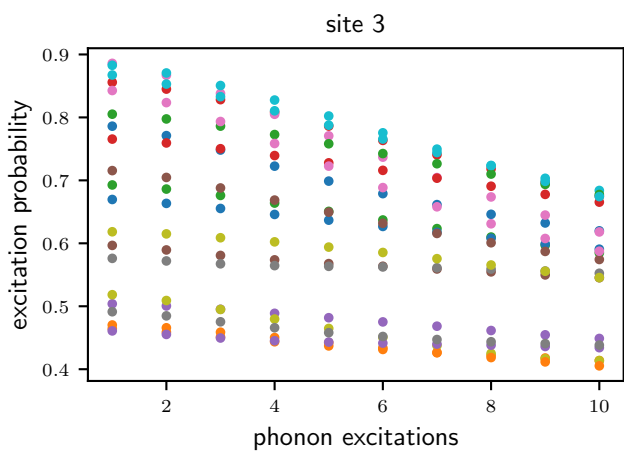
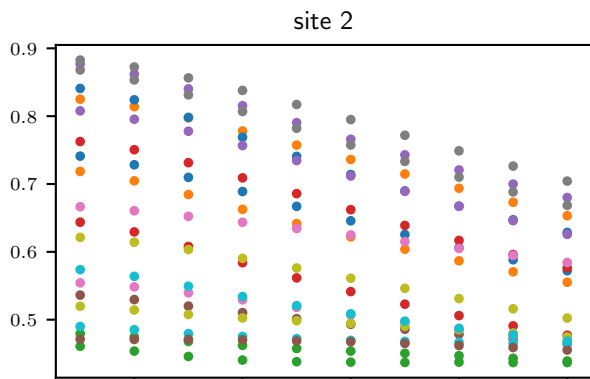
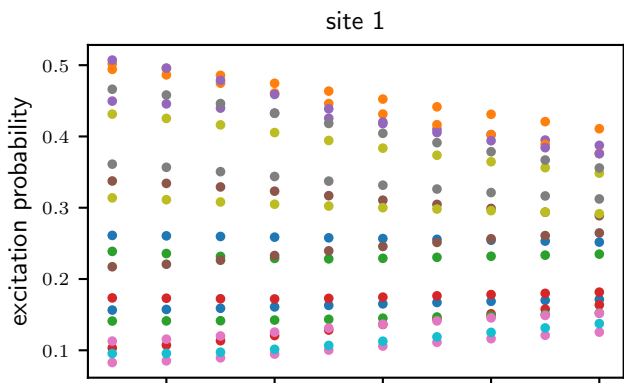
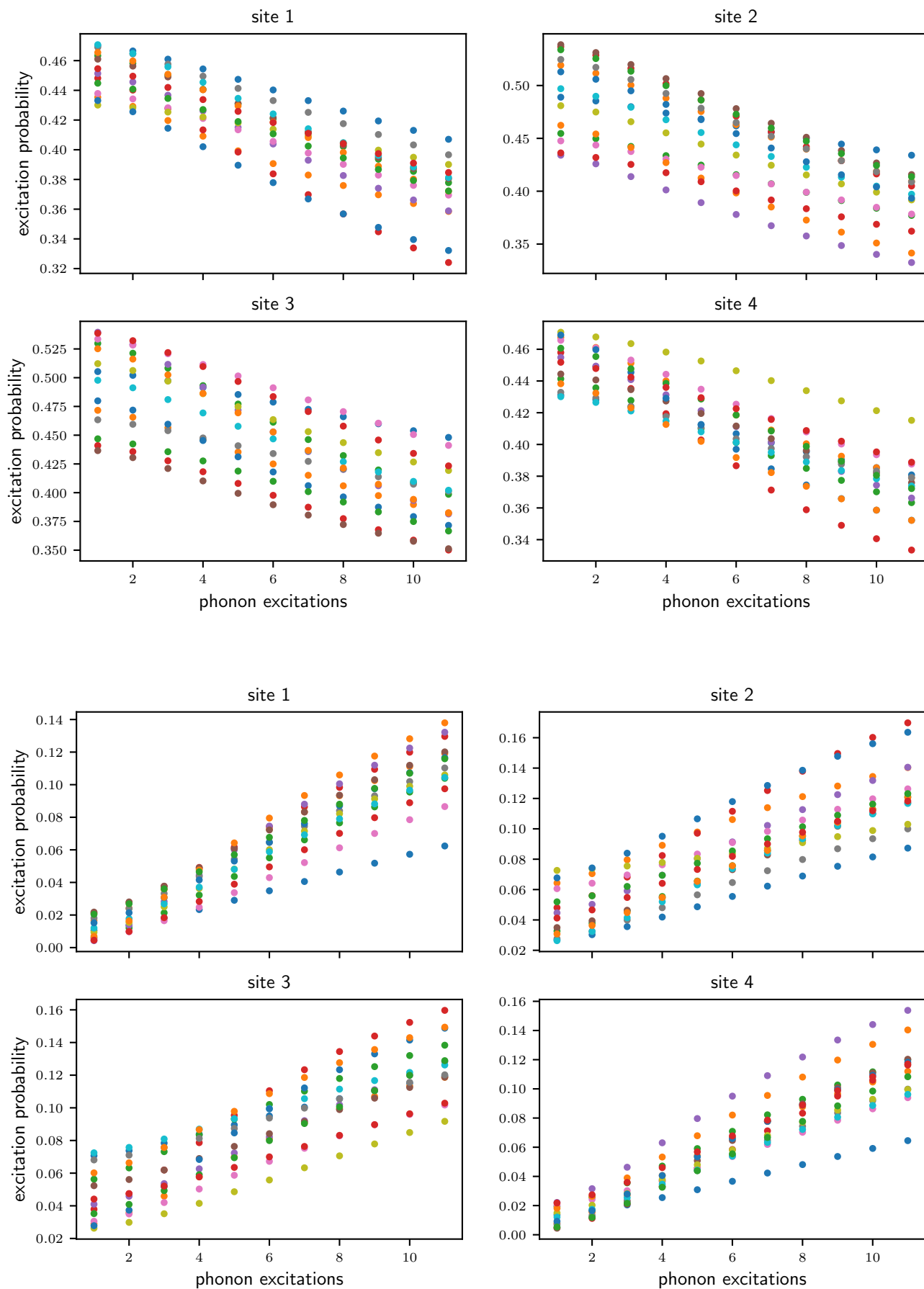


Figure 33: spin-phonon coupling only.







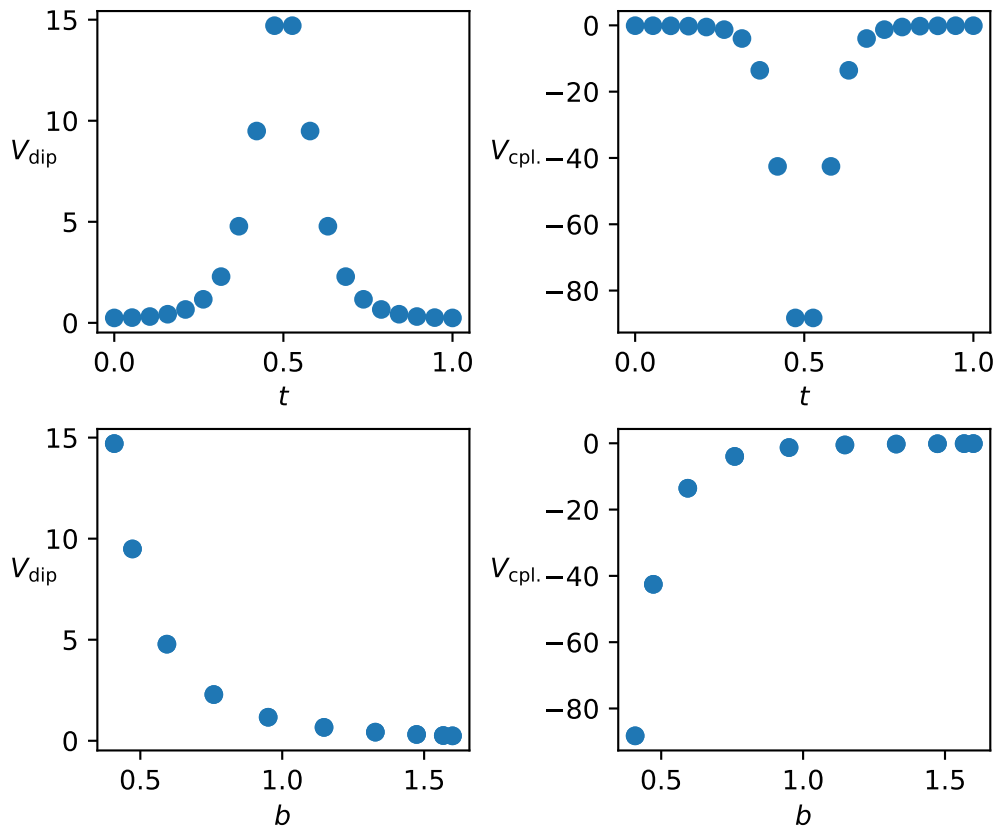


Figure 34: **Interaction strengths during Rice-Mele scheme.**

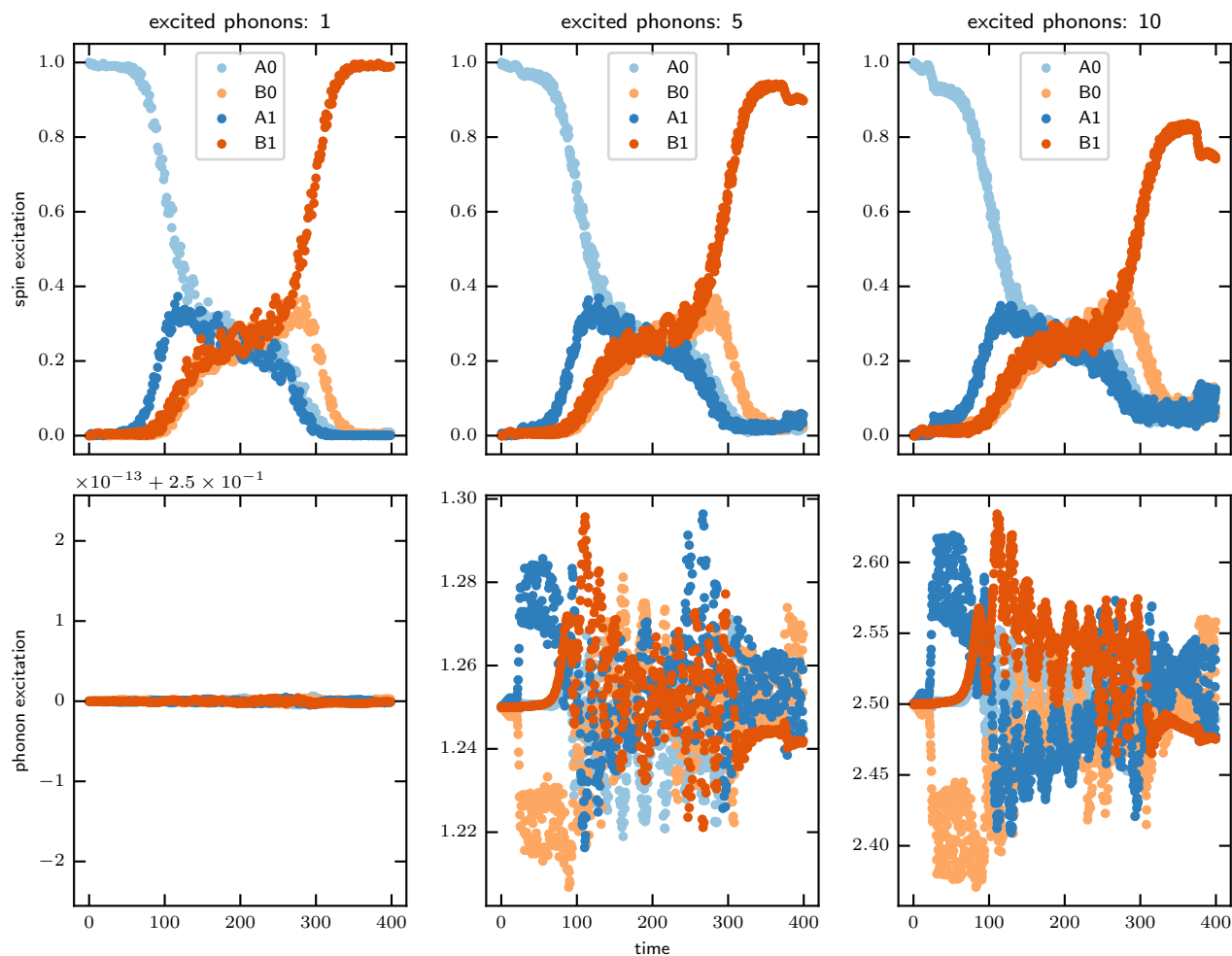


Figure 35: Rice-Mele scheme.

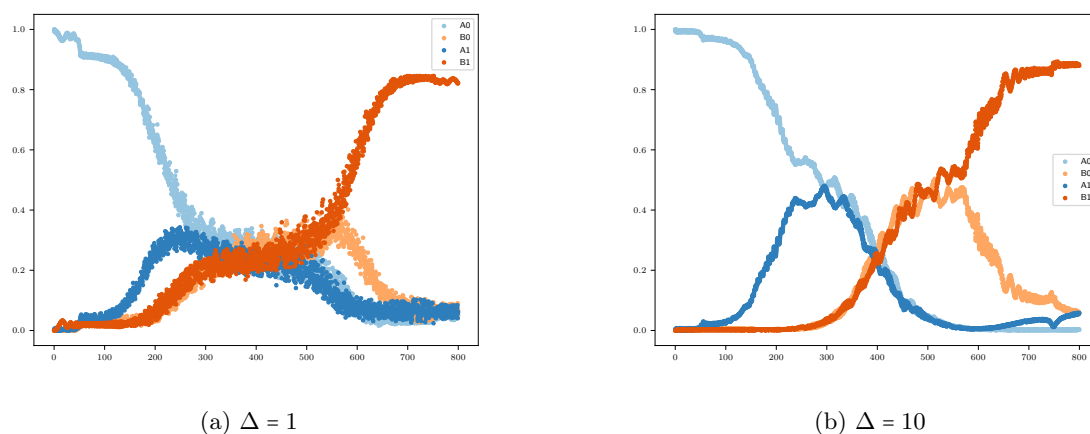


Figure 36: longer time evolution for phonon number 10.

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